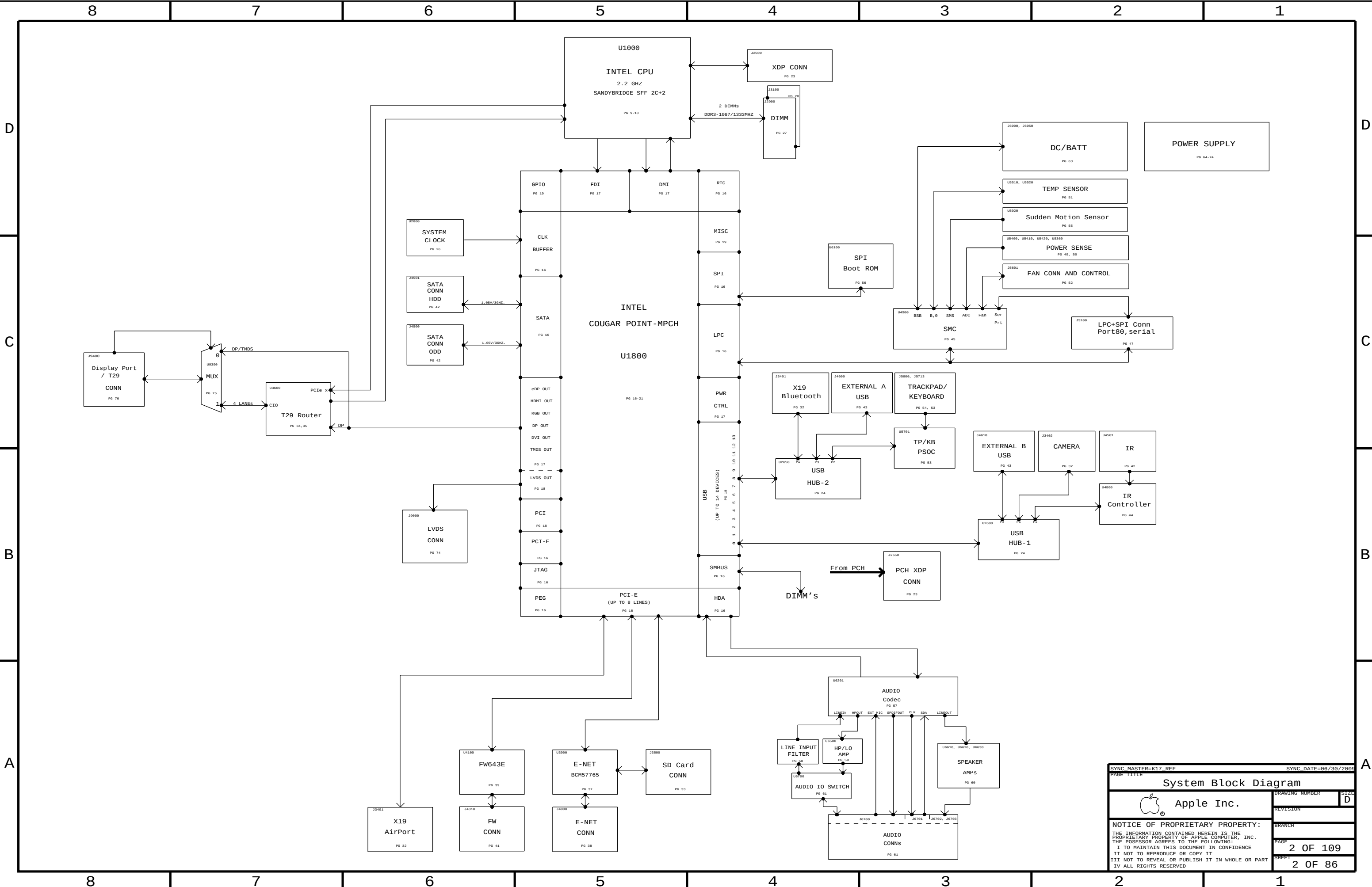




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SYNC DATE=06/30/2009

System Block Diagram

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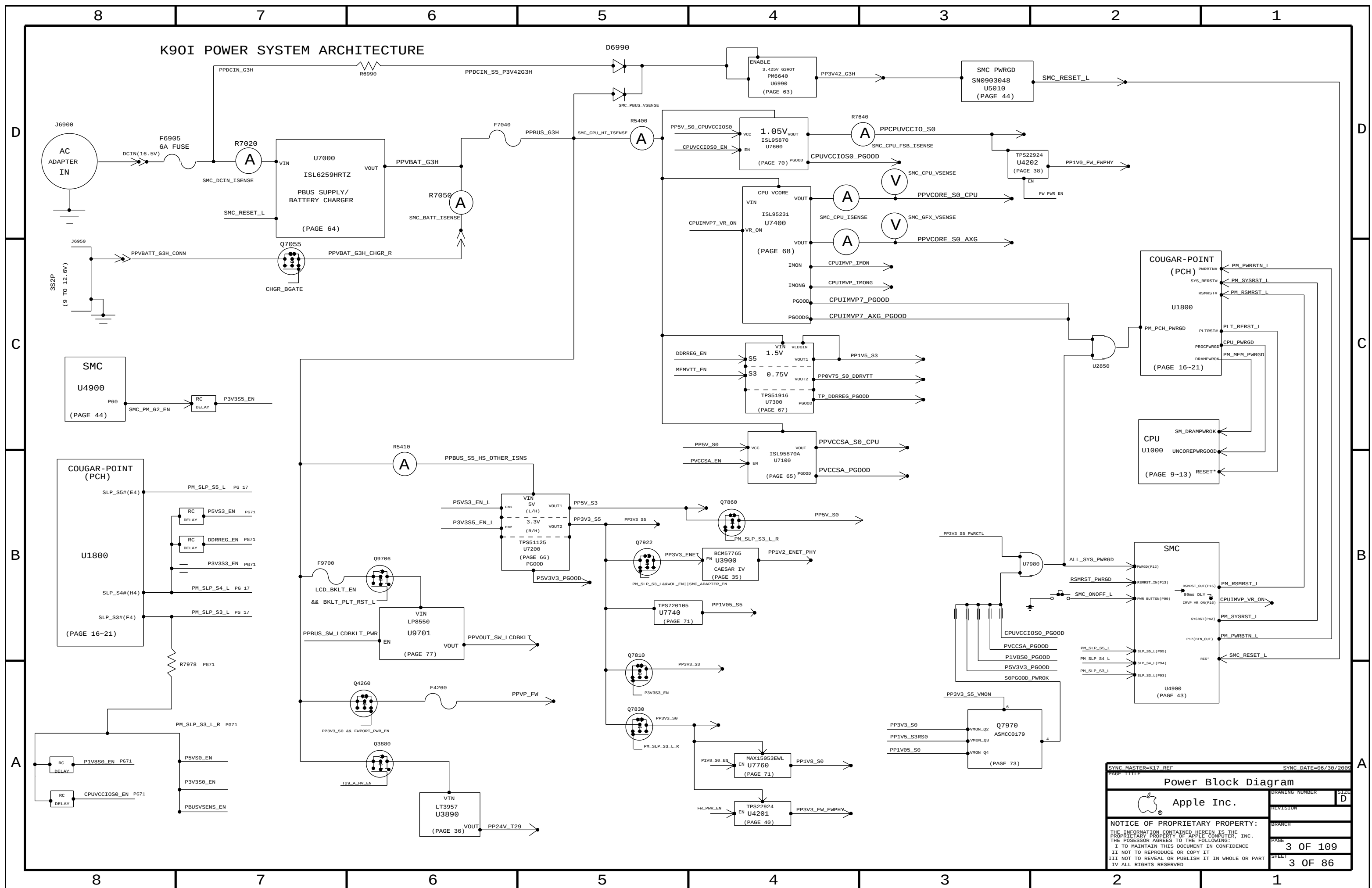
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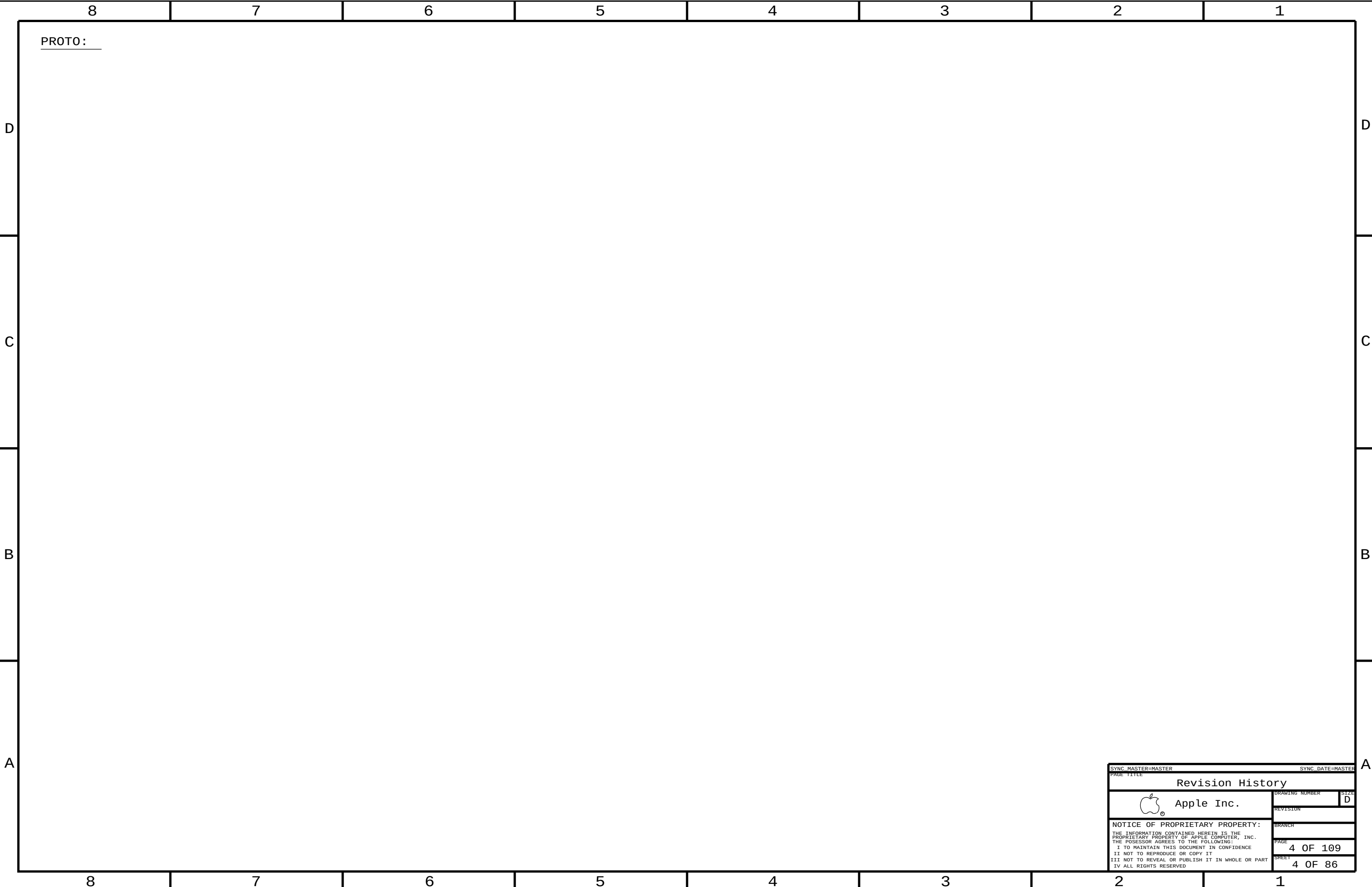
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Revision History

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-1294	PCBA, 2.5G, K90i	K90i_COMMON, CPU_2_5GHZ, EEEE_DDRQ
639-1581	PCBA, 2.7G, K90i	K90i_COMMON, CPU_2_7GHZ, EEEE_DH78
639-1698	PCBA, 2.6G, K90i	K90i_COMMON, CPU_2_6GHZ, EEEE_DH8F
639-1699	PCBA, 2.3G, K90i	K90i_COMMON, CPU_2_3GHZ, EEEE_DH8G
085-1998	K90i MLB DEVELOPMENT BOM	K90i_DEVEL: ENG

K90i BOM GROUPS

BOM GROUP	BOM OPTIONS
K90i_COMMON	ALTERNATE_COMMON, K90i_COMMON1, K90i_COMMON2, K90i_DEBUG:ENG, K90i_PROGPARTS, USBHUB_2513B, T29BST:Y
K90i_COMMON1	BATT_3S, CPUMEN_S0, SMC_DEBUG_YES, HUB1_2NONREM, HUB2_3NONREM, T29:YES, DP_SDRV: A2, SDRV_PD, SDRV12C:MCU
K90i_COMMON2	MIKEY, KB_BL
K90i_PROGPARTS	BOOTROM_PROG, SMC_PROG, TPAD_PROG, ENET_PROG, T29ROM:PROG, T29MCU:PROG
K90i_DEVEL:ENG	BKLT:ENG, BMON:ENG, XDP_CONN, XDP_CPU: BPM, XDP_PCH, LCPPLUS, VREFMRGN, S0PGOOD_ISL, INVPISNS_ENG
K90i_DEVEL:PVT	LCPPLUS, XDP_CONN, XDP_PCH
K90i_DEBUG:ENG	DEVEL_BOM, SMC_DEBUG_YES, XDP
K90i_DEBUG:PVT	DEVEL_BOM, BKLT:PROD, BMON:PROD, SMC_DEBUG_YES, XDP, VREFMRGN_NOT
K90i_DEBUG:PROD	BKLT:PROD, BMON:PROD, SMC_DEBUG_YES, XDP, VREFMRGN_NOT, LCPPLUS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3934	1	SNB, 2C, QXXX, ES1, 2, 2, 35W, B2, 3M, GT1, BGA	U1000	CRITICAL	CPU_2_2GHZ
337S4058	1	SNB, Q1RA, QS, J1, 2, 5, 35W, 2+2, 1.30, 3M, BGA	U1000	CRITICAL	CPU_2_5GHZ
337S4057	1	SNB, Q1R3, QS, J1, 2, 7, 35W, 2+2, 1.30, 4M, BGA	U1000	CRITICAL	CPU_2_7GHZ
337S4024	1	SNB, Q1R9, QS, J1, 2, 3, 35W, 2+2, 1.30, 3M, BGA	U1000	CRITICAL	CPU_2_3GHZ
337S4064	1	SNB, Q1R7, QS, J1, 2, 6, 35W, 2+2, 1.30, 3M, BGA	U1000	CRITICAL	CPU_2_6GHZ
337S4029	1	IC, PCH, COUGARPOINT, SLH90, PRQ, BD82HM65	U1800	CRITICAL	
343S0534	1	IC, BCM5776580, ENET&S0, 8XB	U3900	CRITICAL	
338S0753	1	IC, FN643-E2, 1394B PHY/ONC1 LINC/PCI-E, S2	U4100	CRITICAL	
338S0921	1	IC, T29-C0, 220 FCBG6A, 15x15MM	U3600	CRITICAL	T29: YES
353S3055	1	IC, PI3VEDP212, X2 DISPLAYPORT 2:1 MUX, QFN	U9390	CRITICAL	

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
33550663	1	IC, FLASH, SERIAL, SPI, 1MBIT, 2V7, BP, S01C	U3990	CRITICAL	ENET_BLANK
341S3026	1	IC ENET, 11MBITFLAH, CIV REV01, K60/K62	U3990	CRITICAL	ENET_PROG
33550777	1	IC, EEPROM, SERIAL, SPI, 1KX8, 1.8V, MLP8, LF	U3690	CRITICAL	T29ROM:BLANK
341T0317	1	IC, T29 ASSY	U3690	CRITICAL	T29ROM: PROG
337S3997	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29MCU:BLANK
341S2939	1	IC, PROGMD, LPC1112A, T29 PORT MCU, HVQFN25	U9330	CRITICAL	T29MCU: PROG
338S0895	1	IC, SMC, HS8/2117/9MMx9MM, TLP	U4900	CRITICAL	SMC_BLANK
341T0300	1	IC, SMC, K901	U4900	CRITICAL	SMC_PROG
335S0770	1	64 MBIT SPI SERIAL DUAL I/O FLASH	U6100	CRITICAL	BOOTROM_BLANK
335S0769	1	64 MBIT SPI SERIAL DUAL I/O FLASH	U6100	CRITICAL	BOOTROM_BLANK
341T0299	1	IC, EFI ROM, K901	U6100	CRITICAL	BOOTROM_PROG
341S2384	1	IR, ENCODE II, CY7C63803-LQXC	U4800	CRITICAL	
341S3024	1	IC, TP PSOC, K90, K901, K91, K91F, K92	U5701	CRITICAL	TPAD_PROG

Development BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM_OPTION
085-1998	1	K901 MLB DEVELOPMENT	DEVEL	CRITICAL	DEVEL_BOM

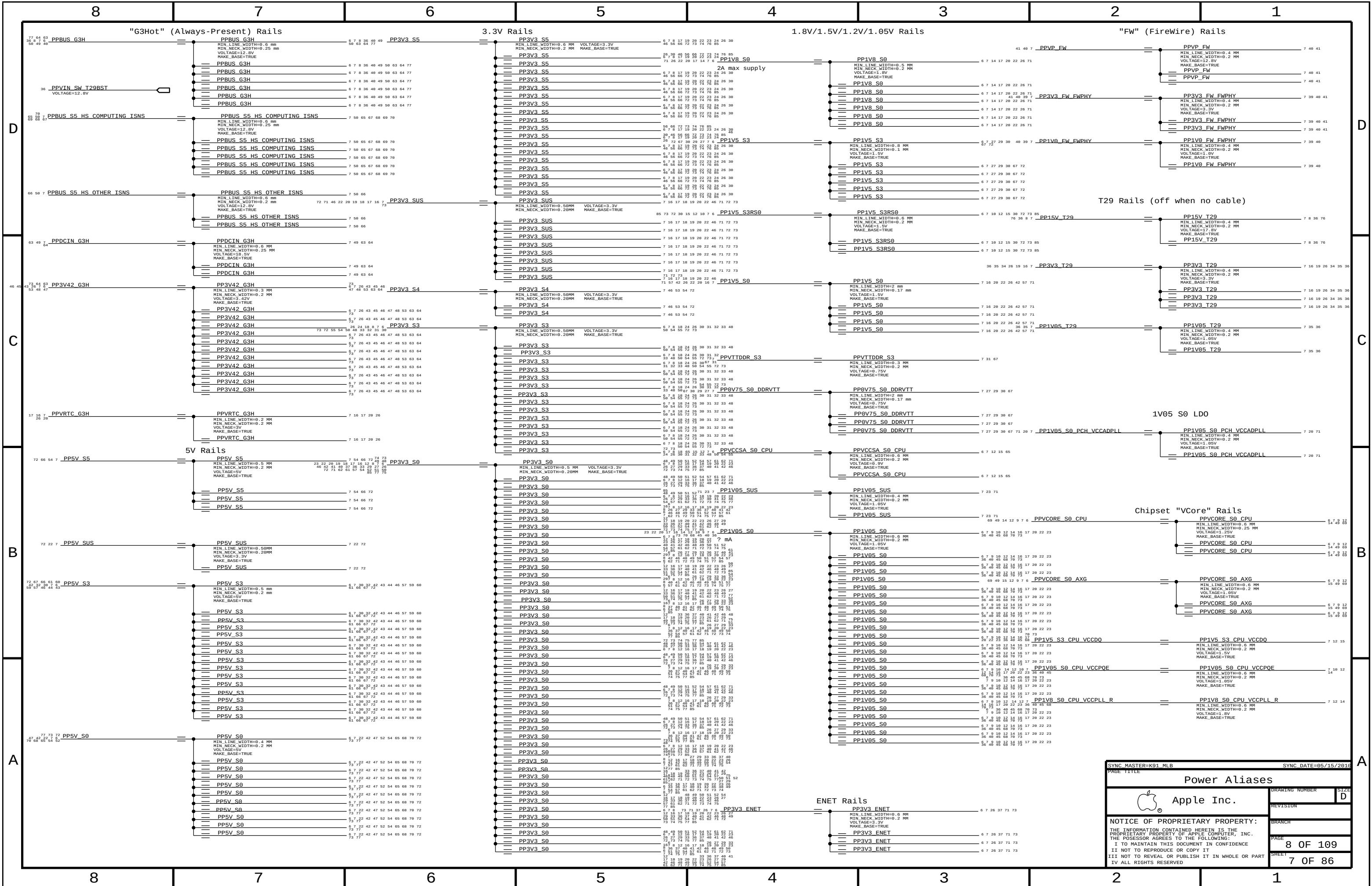
Bar Code Labels / EEEE #'s

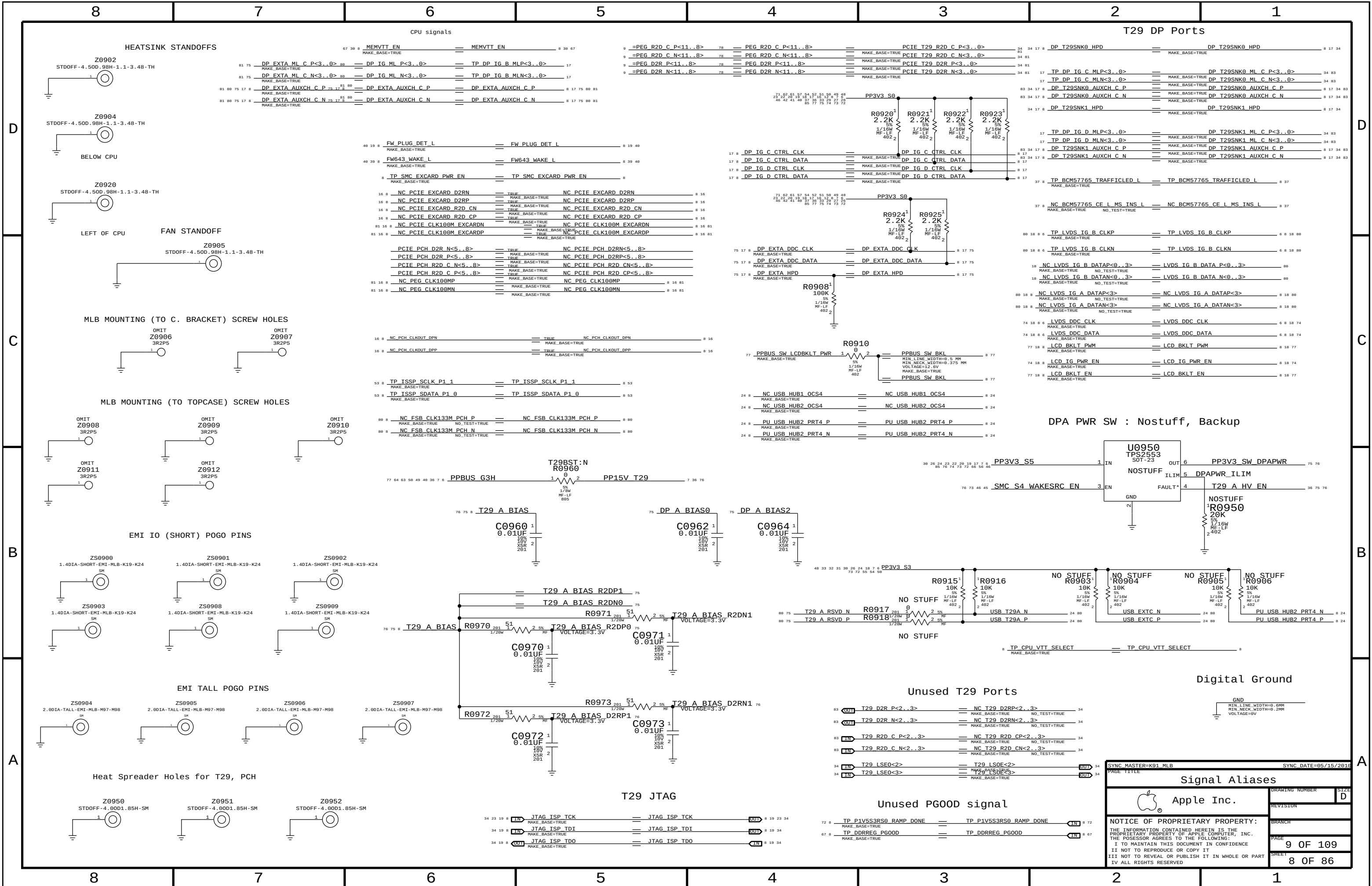
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DDRQ]	CRITICAL	EEEE_DDRQ
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DH78]	CRITICAL	EEEE_DH78
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DH8F]	CRITICAL	EEEE_DH8F
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:DH8G]	CRITICAL	EEEE_DH8G

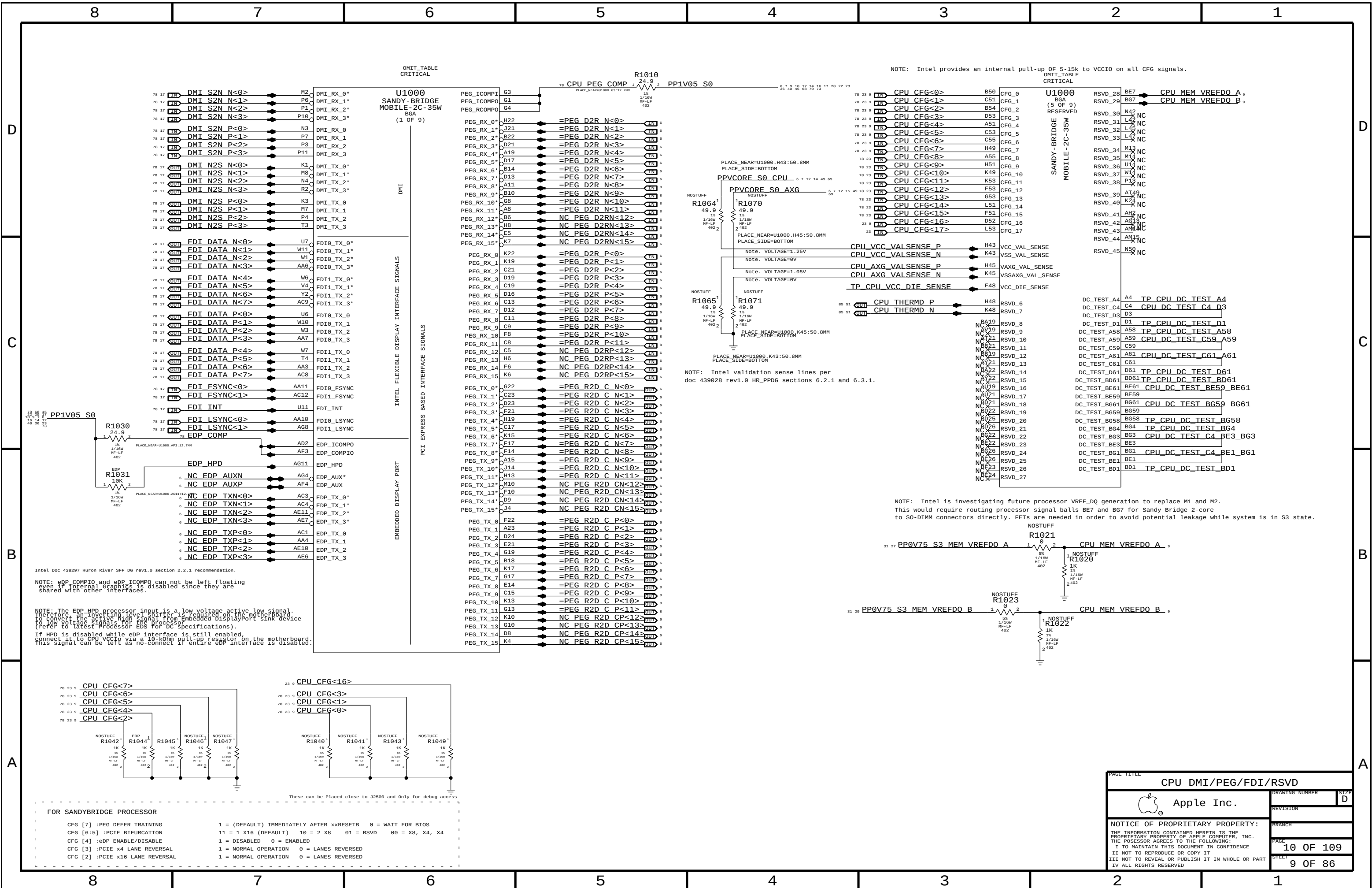
Alternate Parts

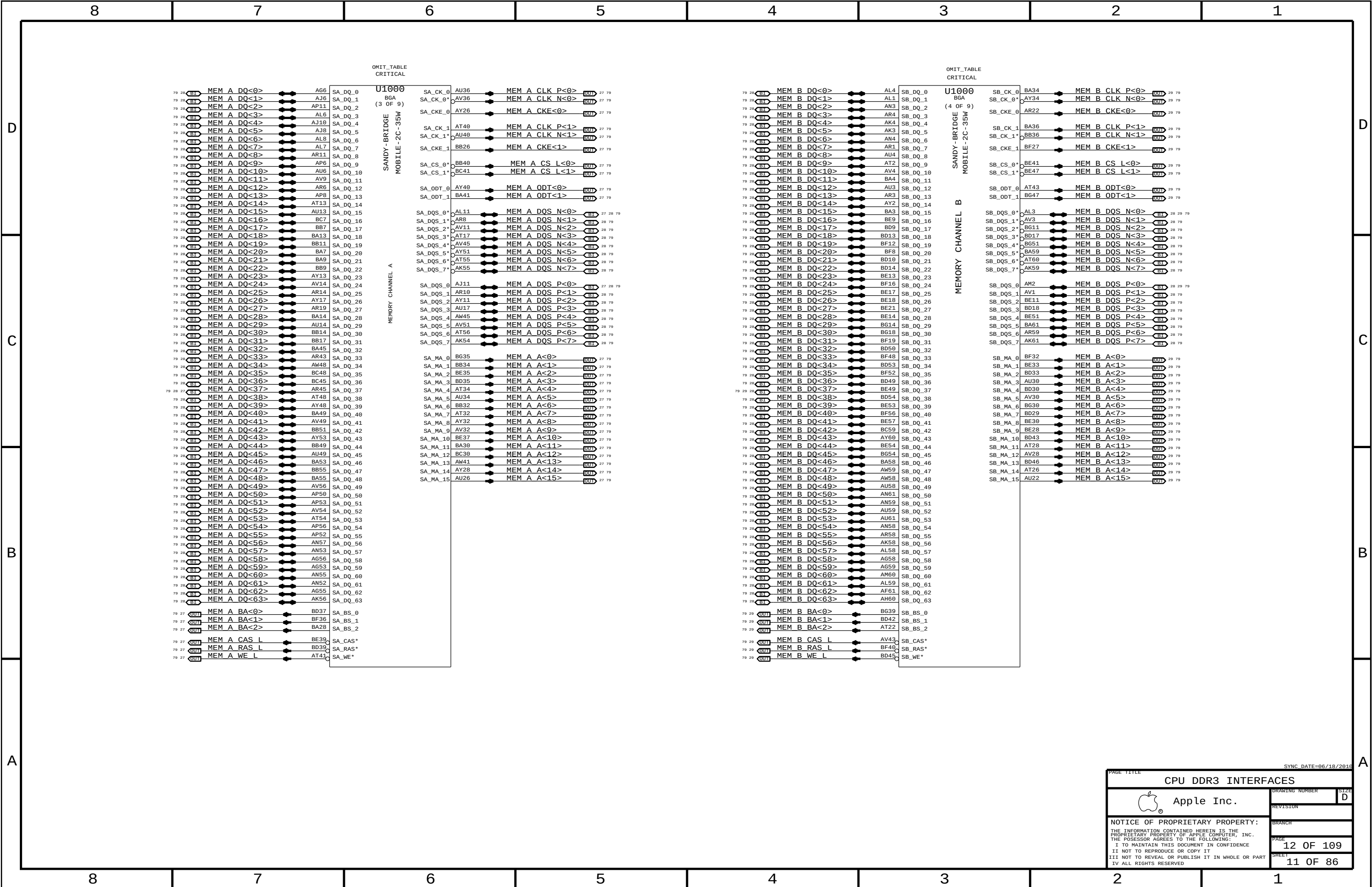
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0603	138S0602		ALL	Murata alt to Samsung
15750058	15750055		ALL	Delta alt to TDK Magnetics
516S0805	516S0806		ALL	Molex alt to Foxconn
128S0303	128S0282		ALL	Panasonic alt to Sanyo
138S0676	138S0691		ALL	Murata alt to Samsung
152S0778	152S0693		ALL	Cyntec alt to Vishay
376S0855	376S0613		ALL	Diodes alt to Toshiba
376S0977	376S0859		ALL	Diodes alt to Toshiba
376S0972	376S0612		ALL	Rohm alt to Toshiba
376S0927	376S0966		ALL	Fairchild alt to Renesas
376S0927	376S0790		ALL	Fairchild alt to CICOLO
376S0960	376S0801		ALL	Renesas alt to Renesas
376S0790	376S0928		ALL	CICOLO alt to Fairchild
376S0928	376S0895		ALL	Fairchild alt to Renesas
376S0937	376S0845		ALL	Fairchild alt to Renesas
376S0777	376S0761		ALL	ADN alt to Siliconix
376S0957	376S0958		ALL	Fairchild alt to Fairchild
376S0953	376S0958		ALL	Fairchild alt to Renesas
353S3085	353S1658		ALL	STmicro alt to LT

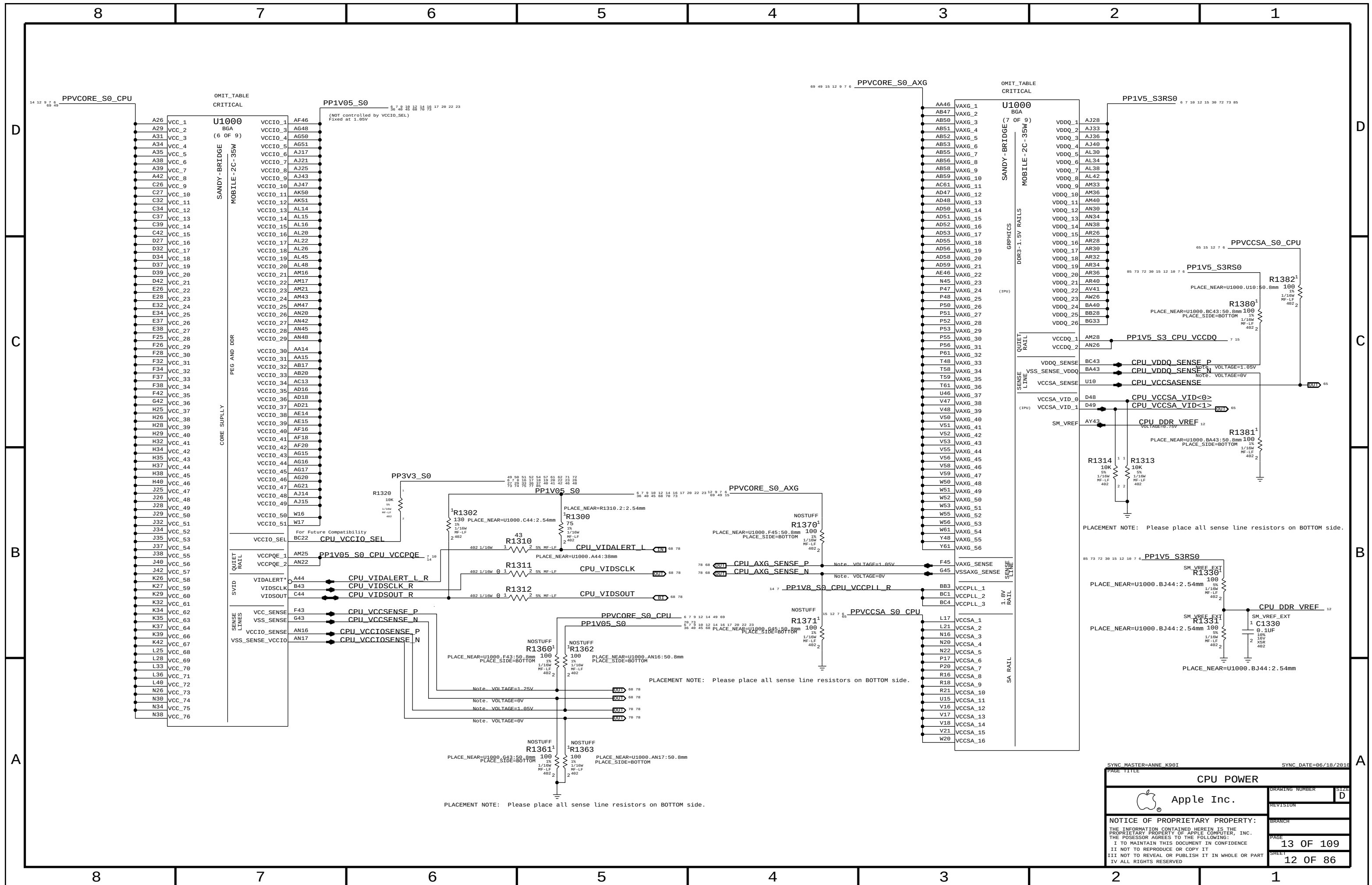
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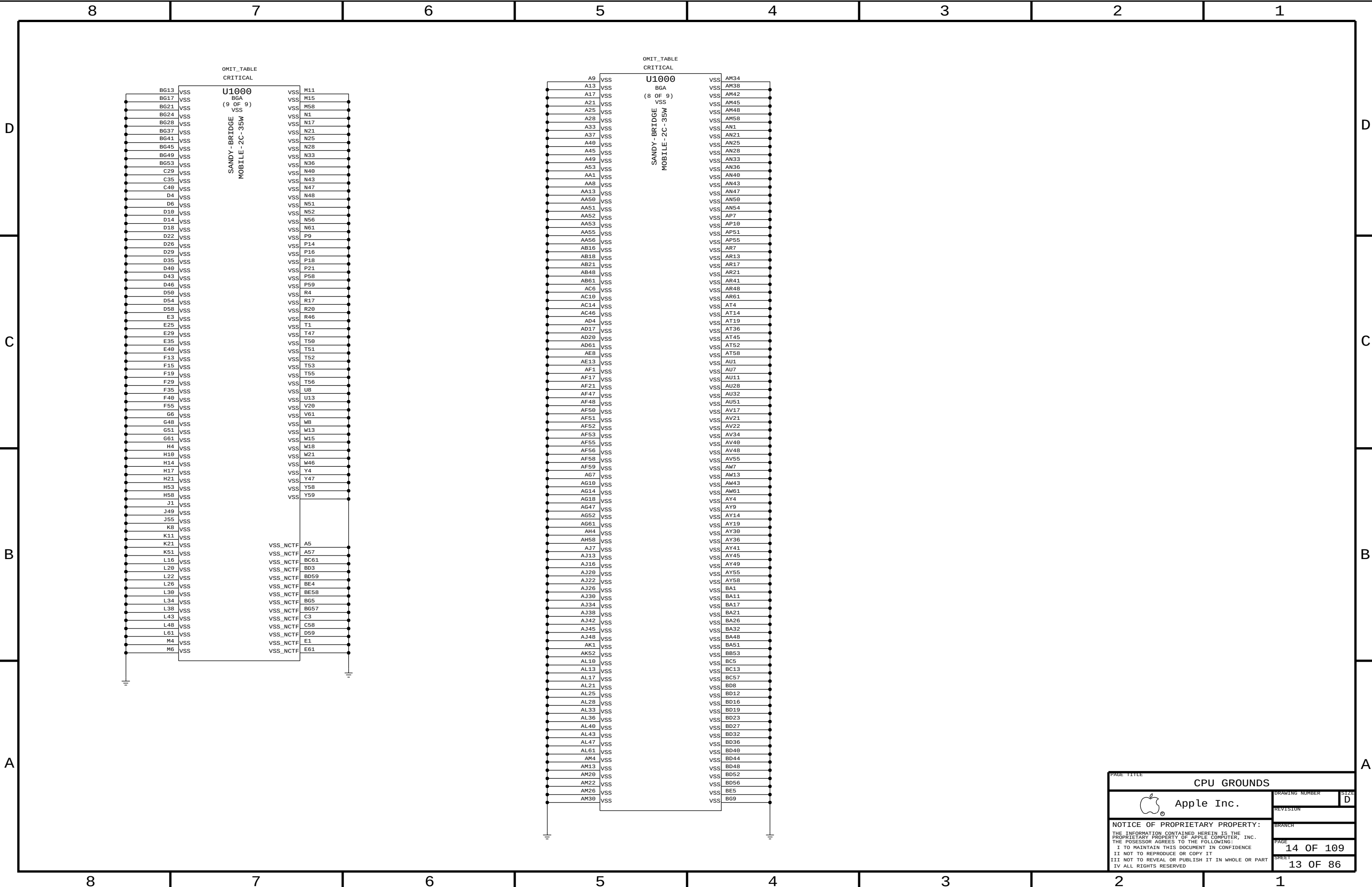


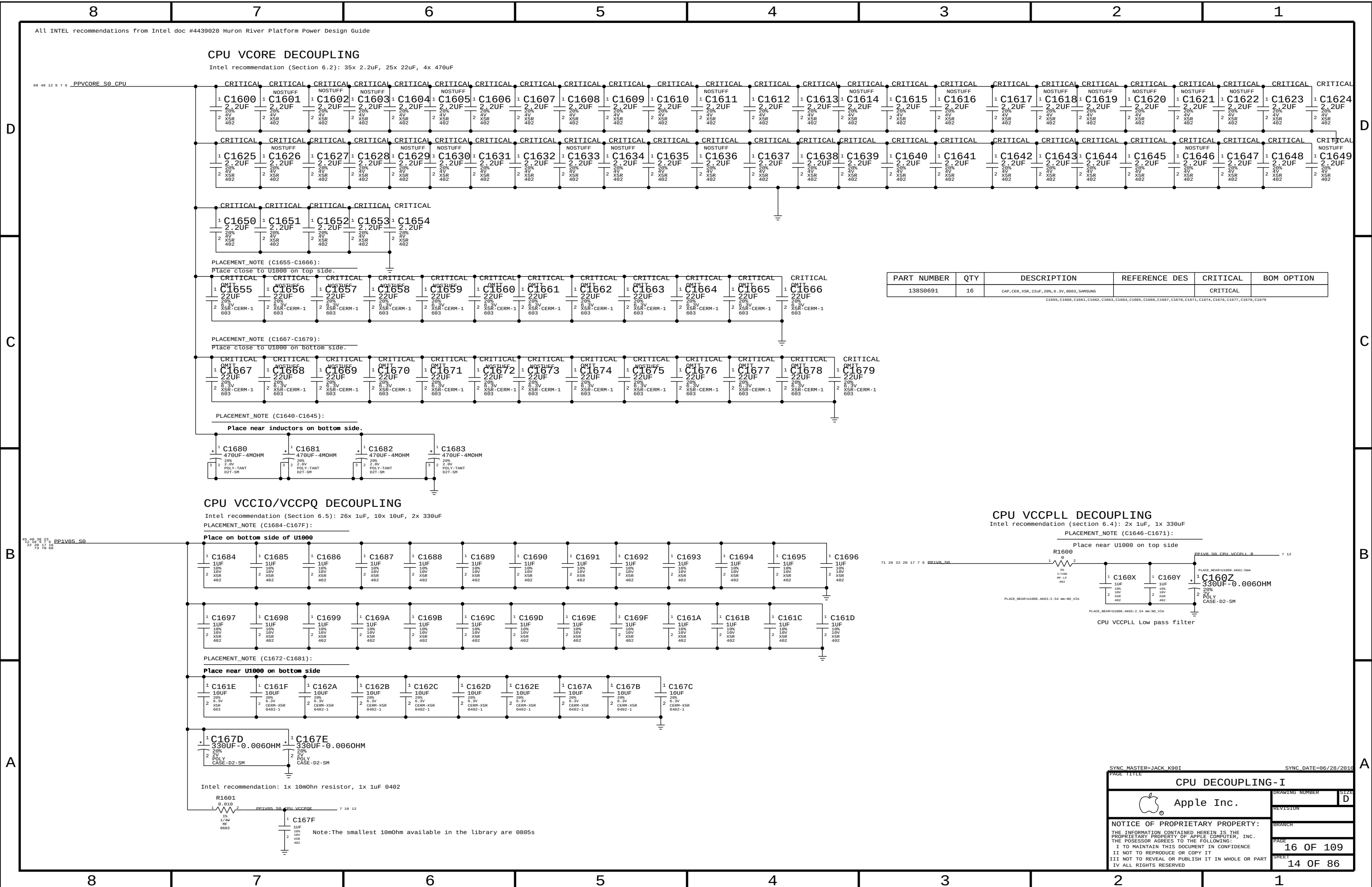
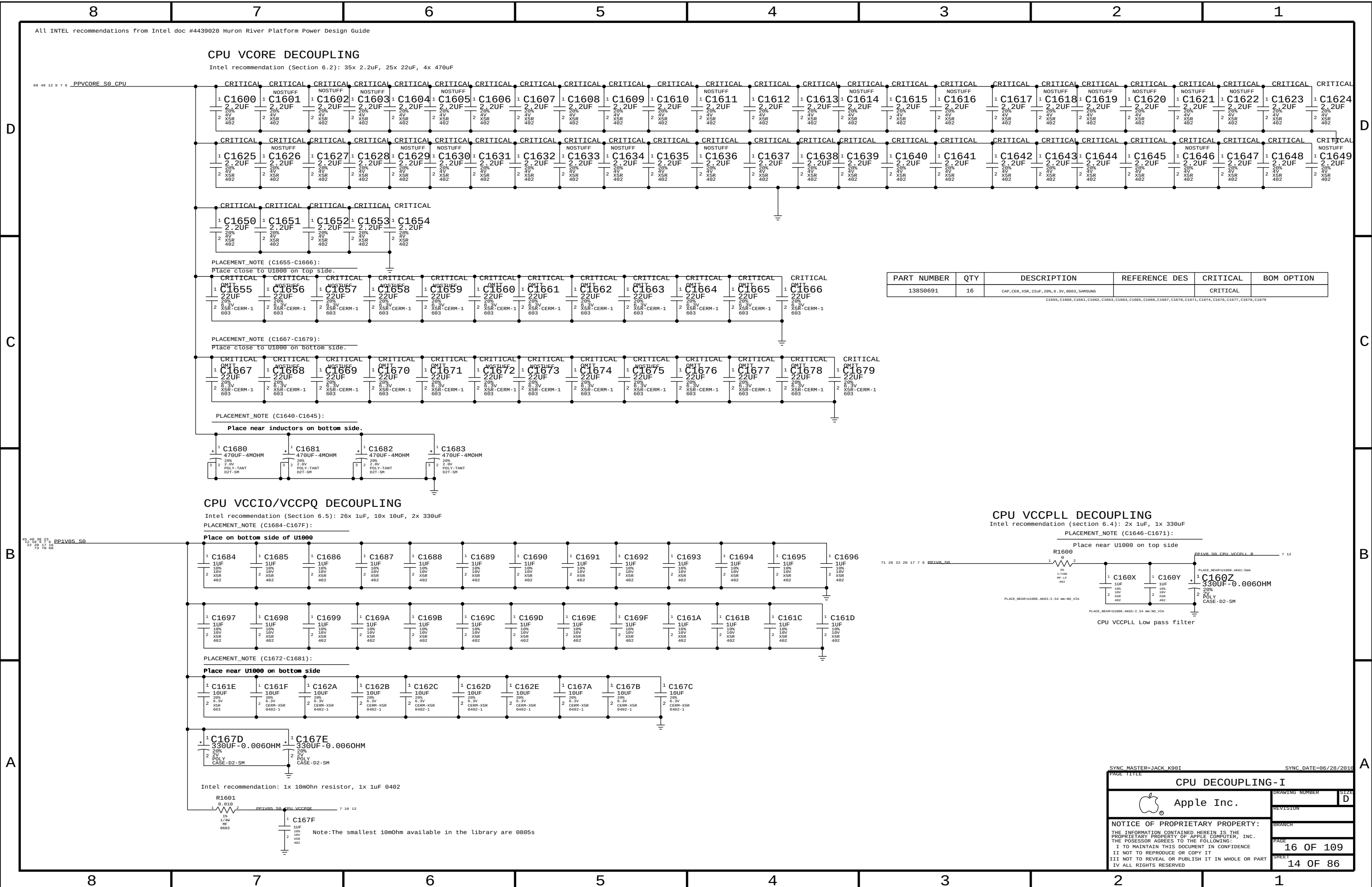










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This figure displays three detailed circuit board layouts for CPU decoupling components, organized by column headers 8 through 1.

The top section, labeled "CPU VCORE DECOUPLING," shows a dense array of capacitors (C1600-C1679) connected to the PPVCORE_S0_CPU signal. It includes placement notes such as "Place close to U1000 on top side" and "Place near inductors on bottom side." A BOM table lists parts like C1655, C1656, C1657, C1658, C1659, C1660, C1661, C1662, C1663, C1664, C1665, and C1666.

The middle section, labeled "CPU VCCIO/VCCPQ DECOUPLING," illustrates capacitors (C1684-C1696) connected to the BP1V05_S0 signal. Placement notes include "Place on bottom side of U1000" and "Place near U1000 on bottom side."

The bottom section, labeled "CPU VCCPLL DECOUPLING," shows capacitors (C1697-C1699, C169A-C169D, C169E-C169F, C161A-C161C, C161D, C161E-C161H, C161I-C161L, C161M-C161N, C161O-C161P, C161Q-C161R, C161S-C161T, C161U-C161V, C161W-C161X, C161Y-C161Z) connected to the BP1V05_S0 signal. Placement notes include "Place near U1000 on top side" and "Place near U1000 on bottom side."

A detailed view of the CPU VCCPLL Low pass filter is shown, featuring a network of resistors (R1600, R1601), capacitors (C160X, C160Y, C160Z), and inductors (L1600, L1601).

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This figure displays three detailed circuit board layouts for CPU decoupling components, labeled A, B, and C, corresponding to different power planes.

Circuit Board A: CPU Vcore Decoupling

- Title:** CPU Vcore Decoupling
- Reference:** Intel recommendation (Section 6.2): 35x 2.2uF, 25x 22uF, 4x 470uF
- Components:** Capacitors C1600 through C1679 are shown, mostly rated at 2.2uF or 22uF, 4V, XSR, 482. Some are marked "NOSTUFF".
- Placement Notes:** Includes notes like "Place close to U1000 on top side." and "Place near inductors on bottom side."
- Table:** A table lists part numbers (e.g., 138S0691), quantities (16), descriptions (CAP, CER, XSR, 22UF, 20%, 6.3V, 0603, SAMSUNG), reference designations, criticality (CRITICAL), and BOM options.

Circuit Board B: CPU VCCIO/VCCPQ Decoupling

- Title:** CPU VCCIO/VCCPQ Decoupling
- Reference:** Intel recommendation (Section 6.5): 26x 1uF, 10x 10uF, 2x 330uF
- Components:** Capacitors C1684 through C1696 (1uF) and C1697 through C1699 (10uF) are shown. Larger capacitors C167D and C167E (330uF) are also present.
- Placement Notes:** Includes notes like "Place on bottom side of U1000" and "Place near U1000 on bottom side"
- Resistor:** R1601 (10mOhm resistor) is specified.

Circuit Board C: CPU VCCPLL Decoupling

- Title:** CPU VCCPLL Decoupling
- Reference:** Intel recommendation (section 6.4): 2x 1uF, 1x 330uF
- Components:** Capacitors C160X (1uF) and C160Y (10uF) are shown. A large capacitor C160Z (330uF) is also present.
- Placement Notes:** Includes notes like "Place near U1000 on top side" and "Place near U1000 AK65:2.54 mm:NO_VIA"
- Filter:** The section is titled "CPU VCCPLL Low pass filter".

This figure displays three detailed circuit board layouts for CPU decoupling components, labeled A, B, and C, corresponding to different power planes.

Circuit Board A: CPU Vcore Decoupling

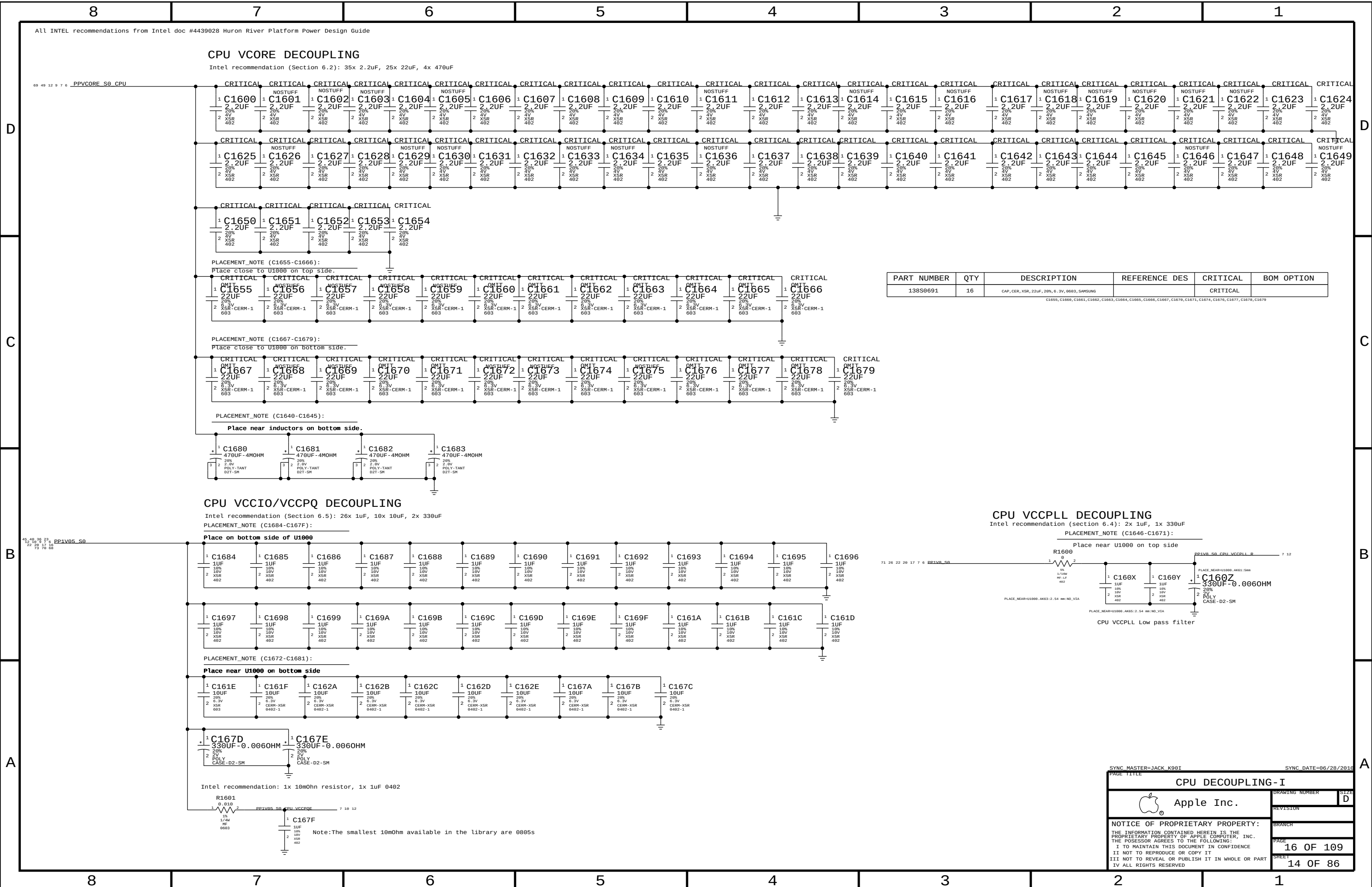
- Title:** CPU Vcore Decoupling
- Reference:** Intel recommendation (Section 6.2): 35x 2.2uF, 25x 22uF, 4x 470uF
- Components:** Capacitors C1600 through C1679 are shown, along with placement notes for U1000 on top and bottom sides, and near inductors on the bottom side.

Circuit Board B: CPU VCCIO/VCCPQ Decoupling

- Title:** CPU VCCIO/VCCPQ Decoupling
- Reference:** Intel recommendation (Section 6.5): 26x 1uF, 10x 10uF, 2x 330uF
- Components:** Capacitors C1684 through C1696 are shown, along with placement notes for U1000 on the bottom side and near U1000 on the bottom side.

Circuit Board C: CPU VCCPLL Decoupling

- Title:** CPU VCCPLL Decoupling
- Reference:** Intel recommendation (section 6.4): 2x 1uF, 1x 330uF
- Components:** Capacitors C160X, C160Y, and C160Z are shown, along with placement notes for U1000 on top side and near U1000 on top side.

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This figure displays three detailed circuit board layouts for CPU decoupling components, labeled A, B, and C, corresponding to different power planes.

Circuit Board A: CPU Vcore Decoupling

The top section shows the CPU Vcore decoupling layout. It includes a title "CPU Vcore DECOUPLING" and a reference to Intel recommendation (Section 6.2). The layout features a grid of capacitors (C1600 through C1679) connected to the PPVCore_S0_CPU signal. Each capacitor is specified with its value (e.g., 2.2uF, 4.7uF), tolerance (20%), and manufacturer (XSR, 482). A placement note indicates that capacitors C1655-C1666 should be placed close to U1000 on the top side, while C1667-C1679 should be placed close to U1000 on the bottom side. A table lists the part numbers, quantities, descriptions, and criticality of these capacitors.

The bottom section shows the CPU VCCIO/VCCPQ decoupling layout. It includes a title "CPU VCCIO/VCCPQ DECOUPLING" and a reference to Intel recommendation (Section 6.5). The layout features a grid of capacitors (C1684 through C1696) connected to the PPVBS_S0 signal. A placement note indicates that these capacitors should be placed on the bottom side of U1000. A table lists the part numbers, quantities, descriptions, and criticality of these capacitors.

Circuit Board B: CPU VCCPLL Decoupling

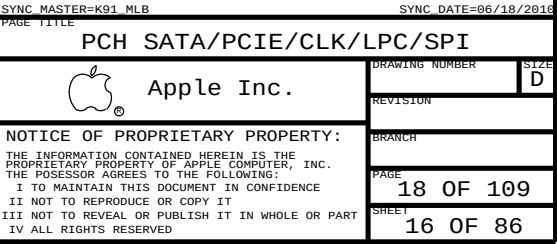
The top section shows the CPU VCCPLL decoupling layout. It includes a title "CPU VCCPLL DECOUPLING" and a reference to Intel recommendation (section 6.4). The layout features a grid of capacitors (C160X through C161D) connected to the PP1VB_S0_CPU_VCCPLL_R signal. A placement note indicates that these capacitors should be placed near U1000 on the top side. A table lists the part numbers, quantities, descriptions, and criticality of these capacitors.

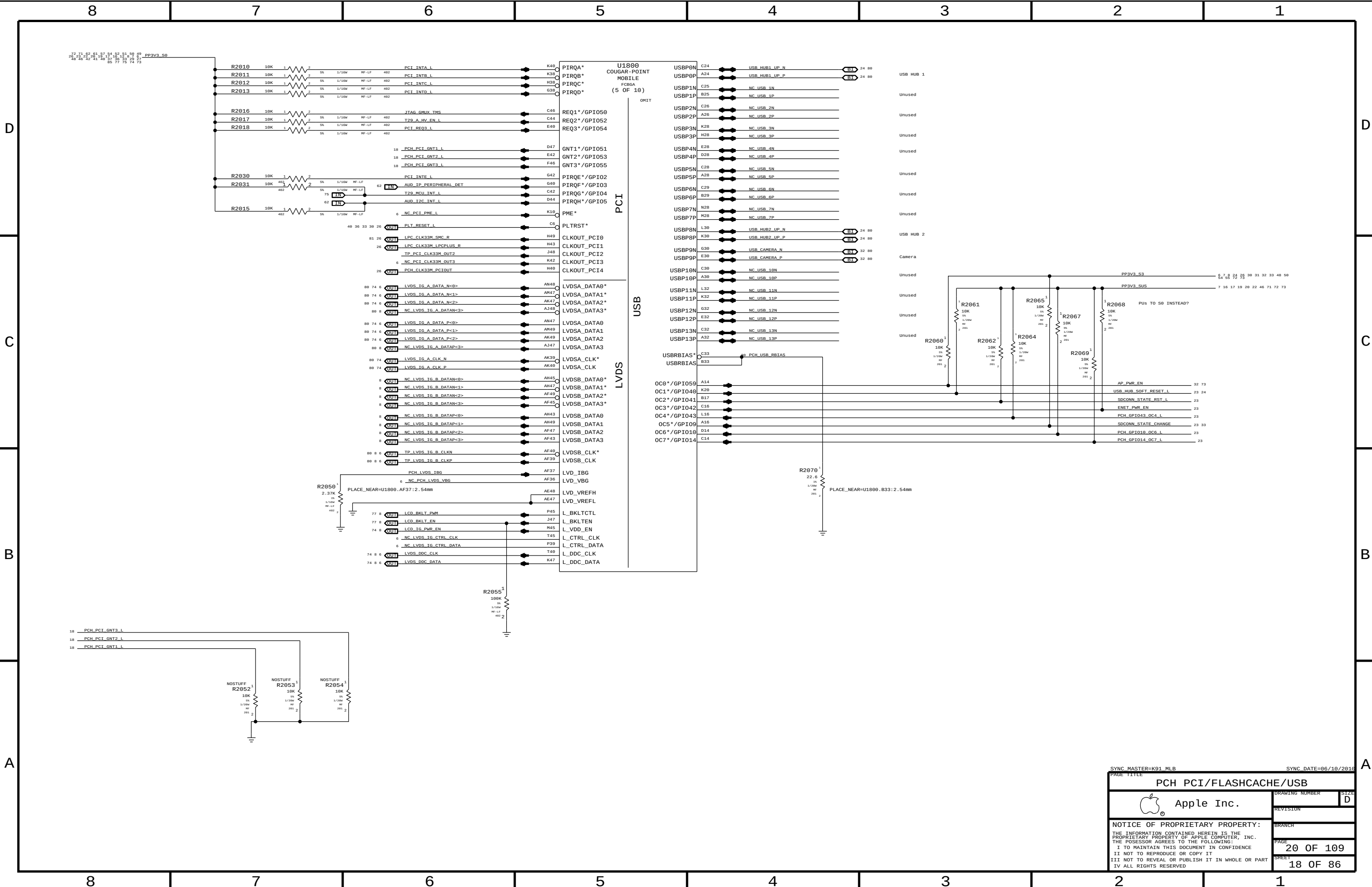
The bottom section shows the CPU VCCPLL Low pass filter layout. It includes a title "CPU VCCPLL Low pass filter" and a reference to Intel recommendation (Section 6.4). The layout features a low-pass filter network consisting of capacitors (C160X, C160Y, C160Z) and a resistor (R1600) connected to the PP1VB_S0_CPU_VCCPLL_R signal. A placement note indicates that these components should be placed near U1000 on the top side.

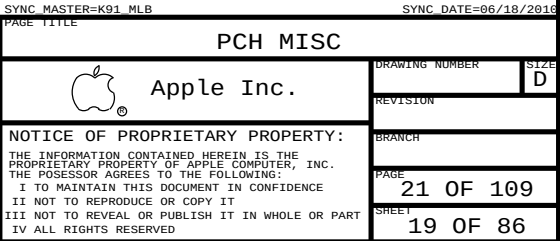
Circuit Board C: CPU Vcore Decoupling

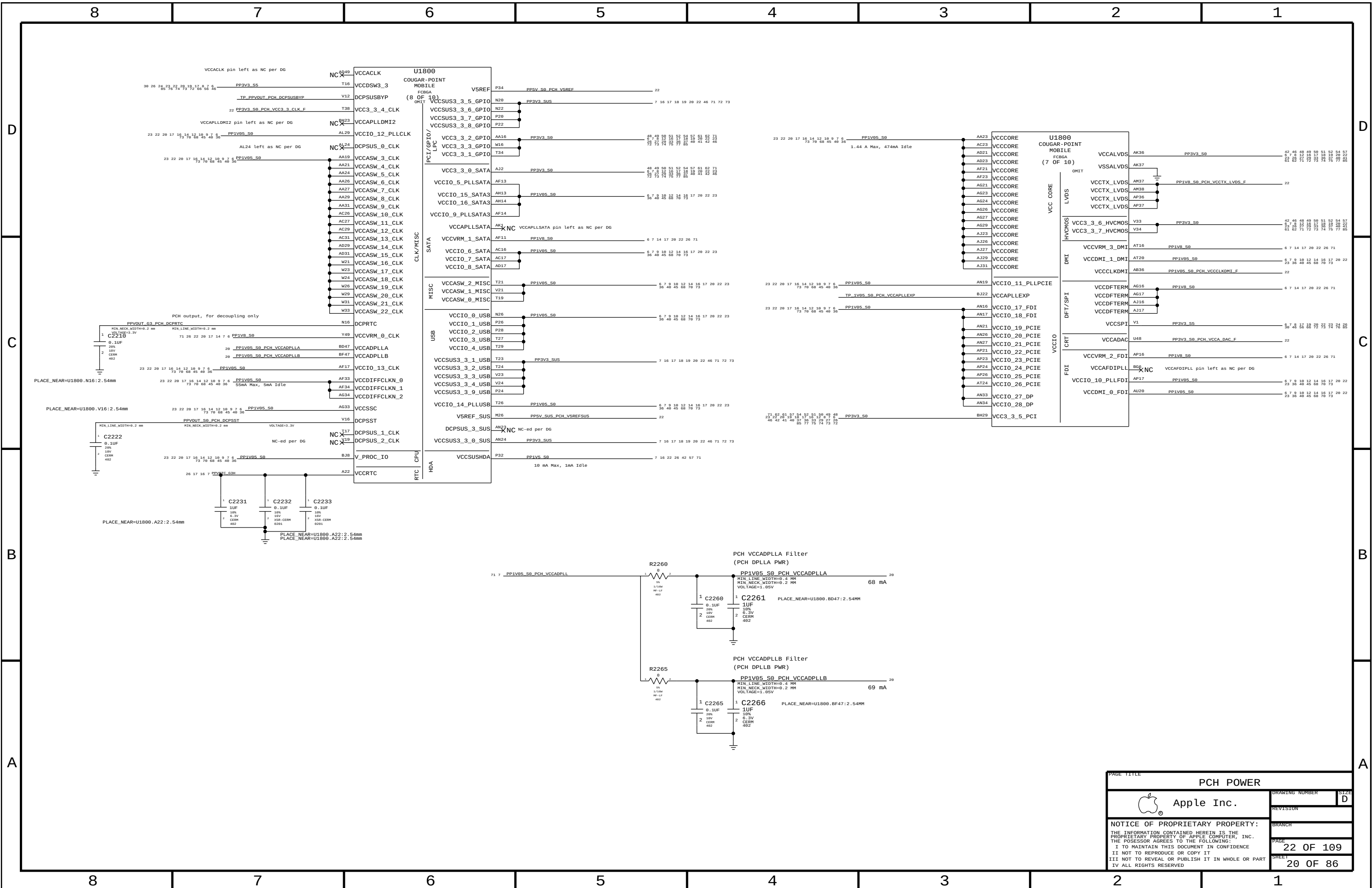
The top section shows the CPU Vcore decoupling layout. It includes a title "CPU Vcore DECOUPLING" and a reference to Intel recommendation (Section 6.2). The layout features a grid of capacitors (C1600 through C1679) connected to the PPVCore_S0_CPU signal. Each capacitor is specified with its value (e.g., 2.2uF, 4.7uF), tolerance (20%), and manufacturer (XSR, 482). A placement note indicates that capacitors C1655-C1666 should be placed close to U1000 on the top side, while C1667-C1679 should be placed close to U1000 on the bottom side. A table lists the part numbers, quantities, descriptions, and criticality of these capacitors.

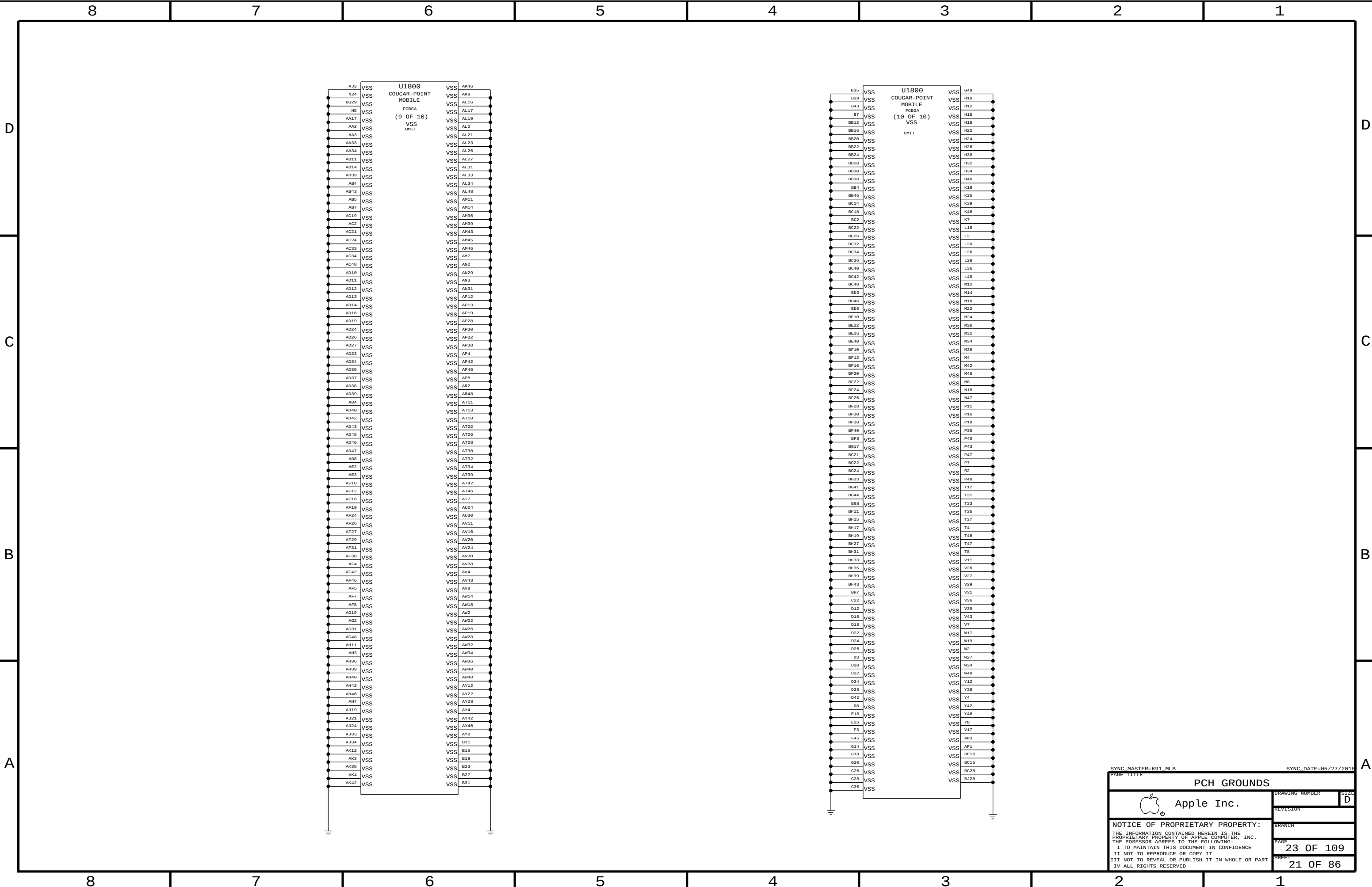
The bottom section shows the CPU VCCIO/VCCPQ decoupling layout. It includes a title "CPU VCCIO/VCCPQ DECOUPLING" and a reference to Intel recommendation (Section 6.5). The layout features a grid of capacitors (C1684 through C1696) connected to the PPVBS_S0 signal. A placement note indicates that these capacitors should be placed on the bottom side of U1000. A table lists the part numbers, quantities, descriptions, and criticality of these capacitors.







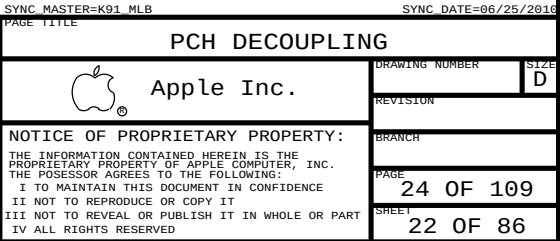


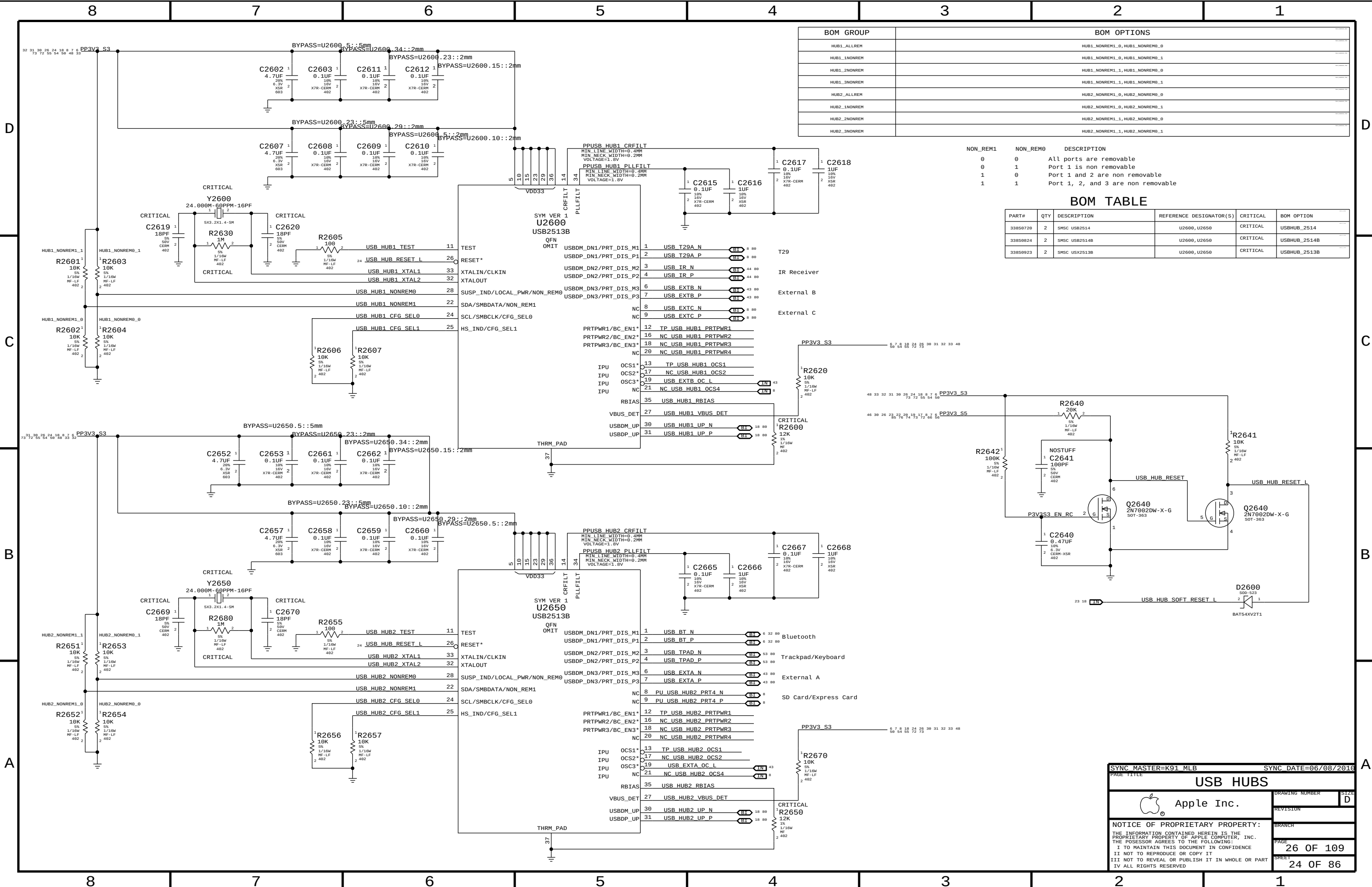


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BOM GROUP		BOM OPTIONS	
HUB1_ALLREM		HUB1_NONREM1_0, HUB1_NONREM0_0	
HUB1_1NONREM		HUB1_NONREM1_0, HUB1_NONREM0_1	
HUB1_2NONREM		HUB1_NONREM1_1, HUB1_NONREM0_0	
HUB1_3NONREM		HUB1_NONREM1_1, HUB1_NONREM0_1	
HUB2_ALLREM		HUB2_NONREM1_0, HUB2_NONREM0_0	
HUB2_1NONREM		HUB2_NONREM1_0, HUB2_NONREM0_1	
HUB2_2NONREM		HUB2_NONREM1_1, HUB2_NONREM0_0	
HUB2_3NONREM		HUB2_NONREM1_1, HUB2_NONREM0_1	

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33850728	2	SMSC USB2514	U2600, U2650	CRITICAL	USBHUB_2514
33850824	2	SMSC USB2514B	U2600, U2650	CRITICAL	USBHUB_2514B
33850923	2	SMSC USX2513B	U2600, U2650	CRITICAL	USBHUB_2513B

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SYNC DATE=06/08/2016

USB HUBS

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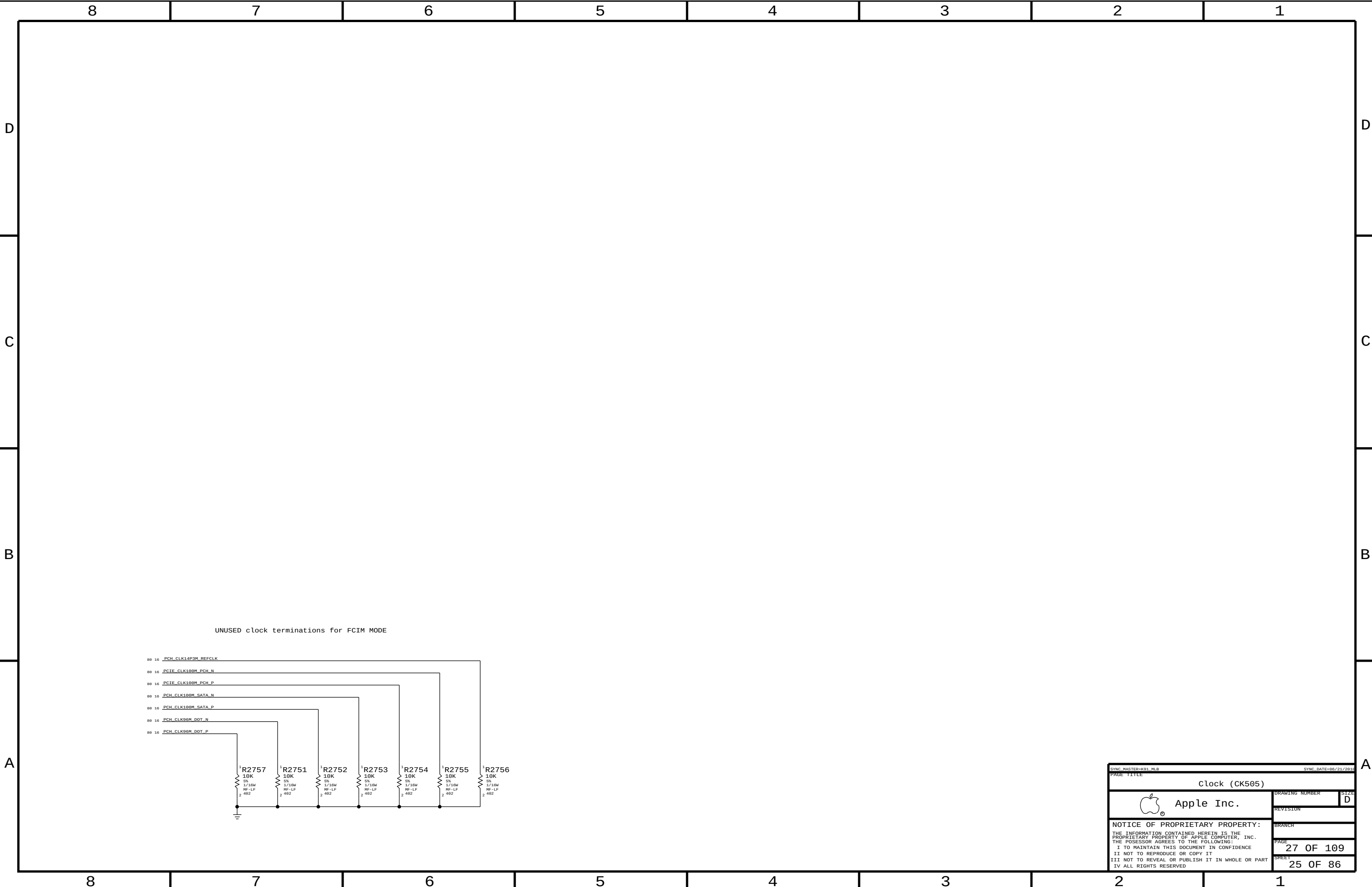
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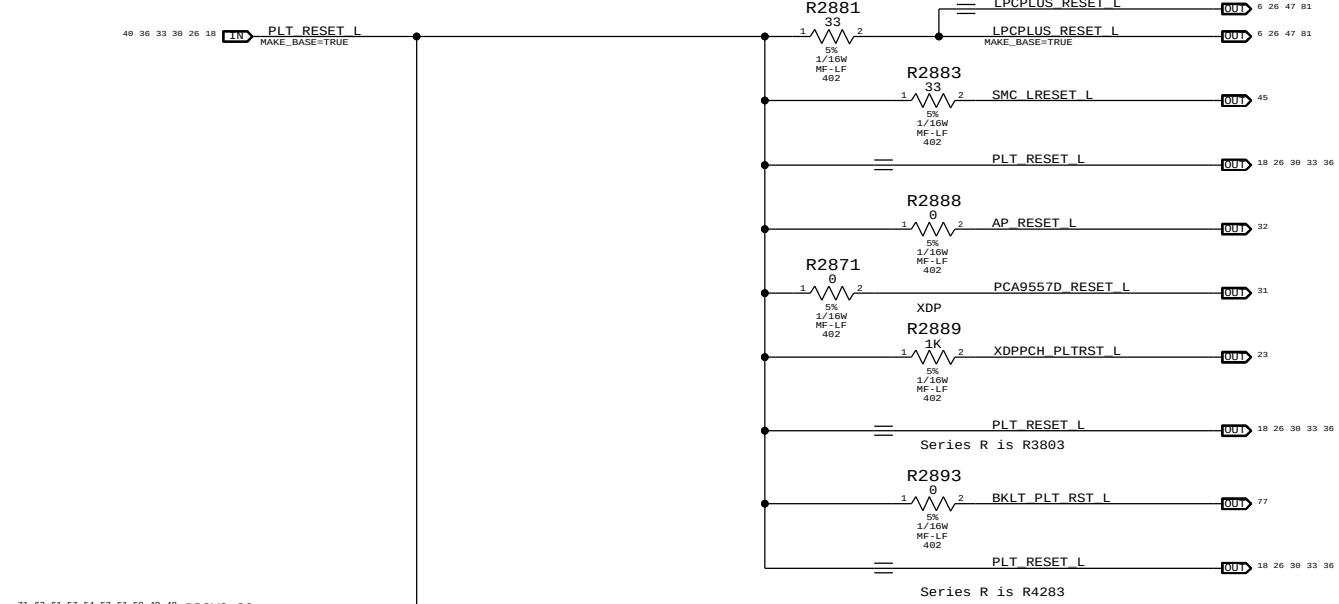
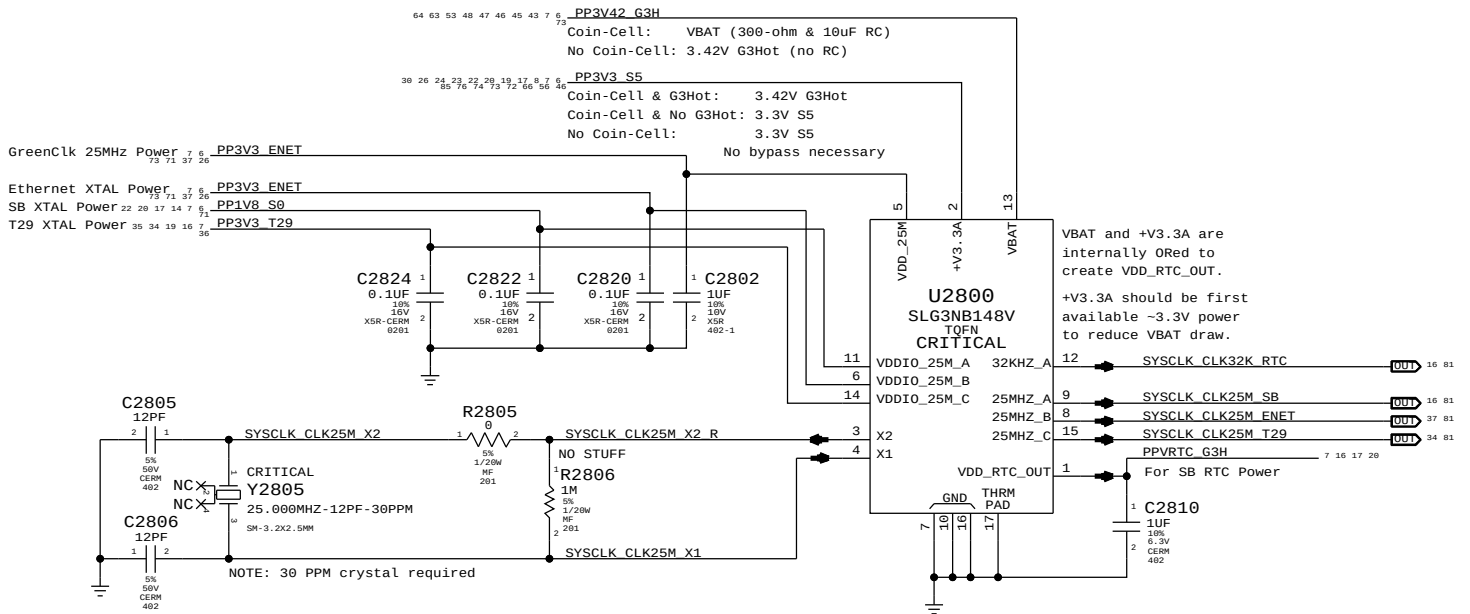
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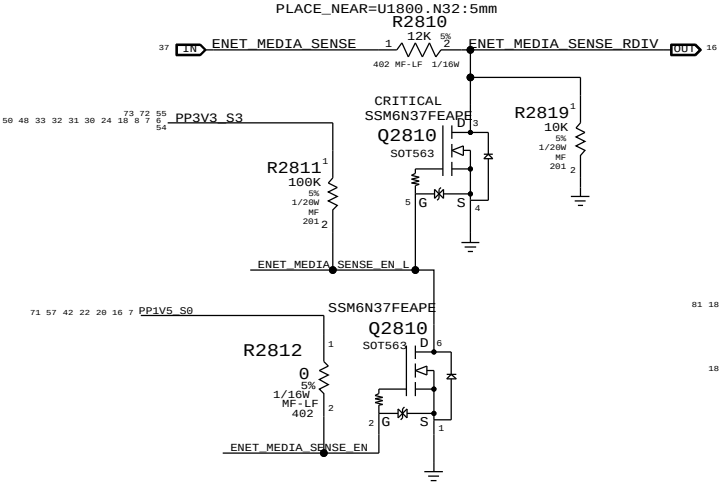
System RTC Power Source & 32kHz / 25MHz Clock Generator

Platform Reset Connections

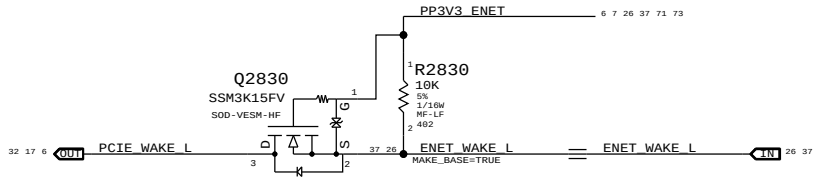
Unbuffered



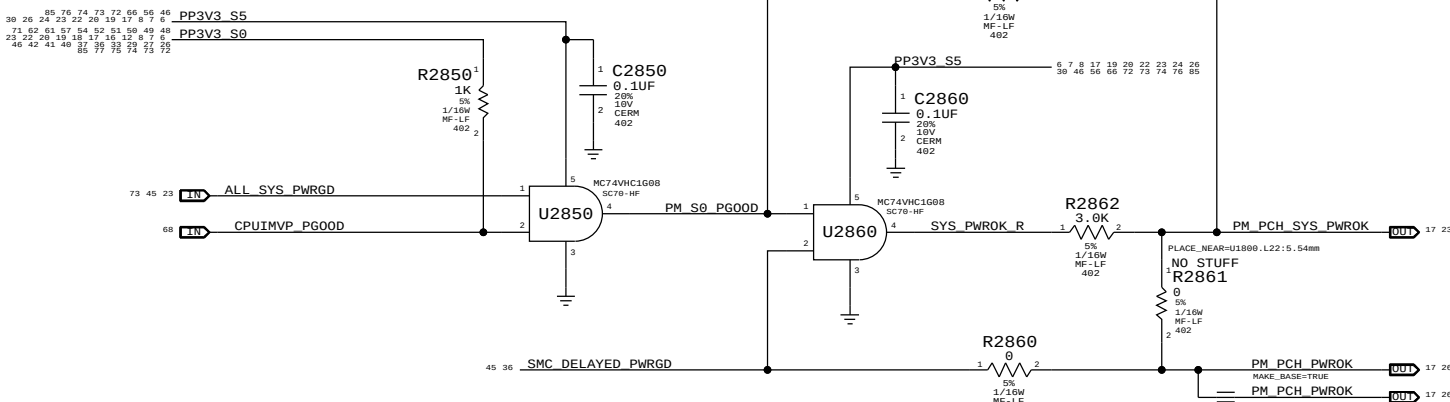
ENET_MEDIA_SENSE ISOLATION CIRCUIT



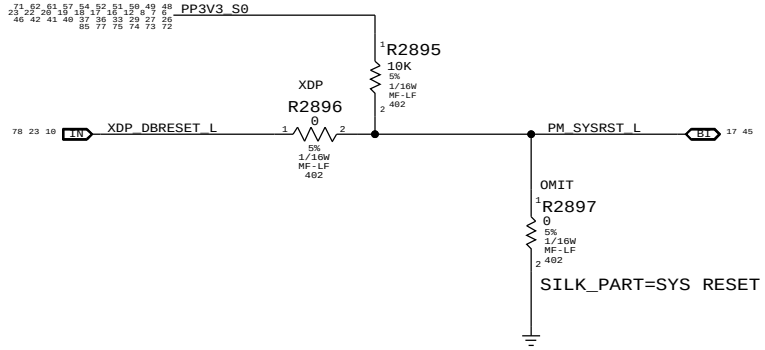
Ethernet WAKE# Isolation



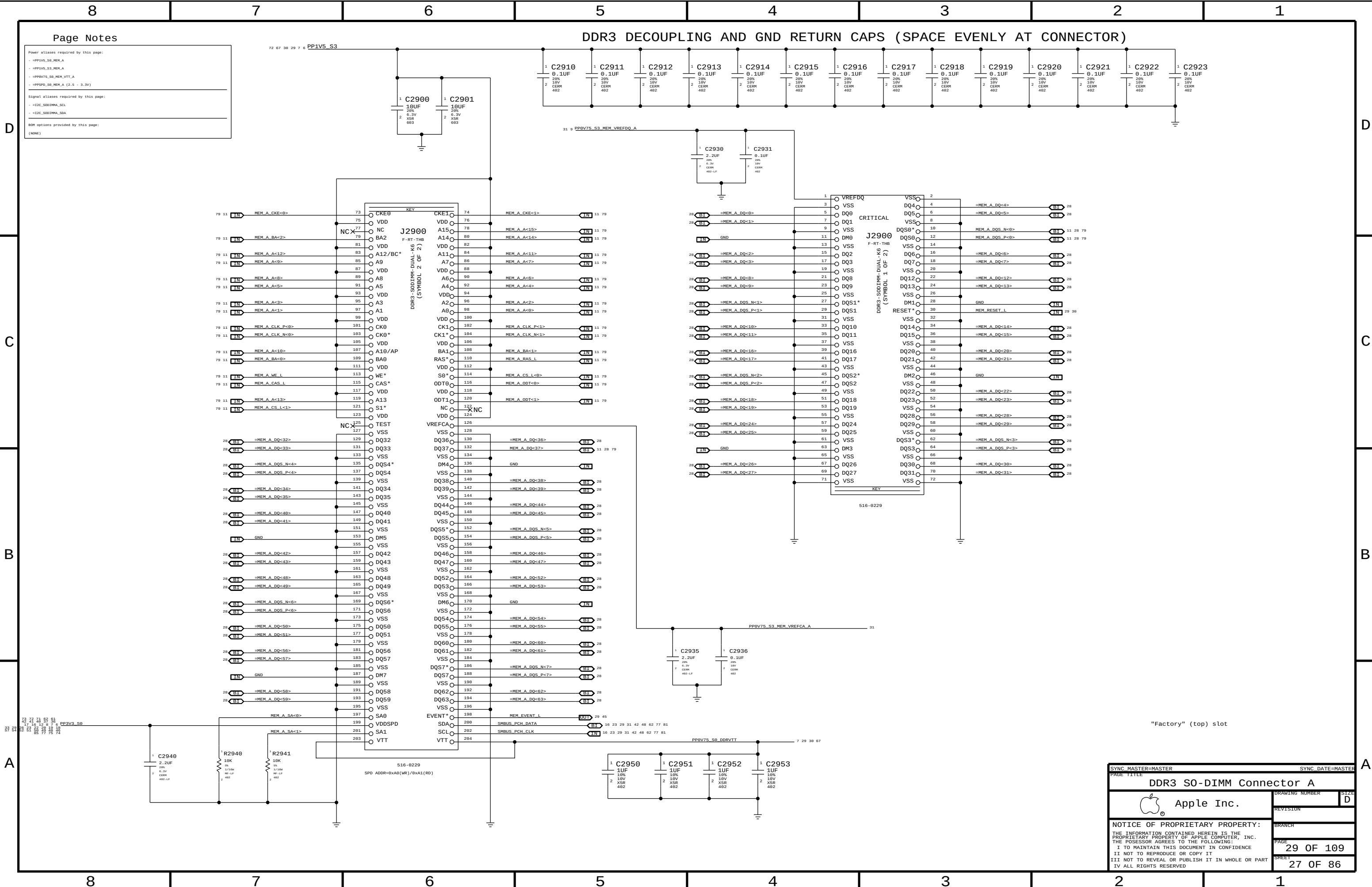
PCH S0 PWRGD



PCH Reset Button



SYNC MASTER=LINDA K901		SYNC DATE=07/08/2010	
PAGE TITLE		Chipset Support	
Apple Inc.		DRAWING NUMBER	SIZE D
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Page Notes

Power aliases required by this page:

- PP1V5_S3_MEM_A
- PP1V5_S3_MEM_A
- PP0V75_S3_MEM_VTT_A
- PPSPD_S3_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- I2C_S0DIMM_SCL
- I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE			
DDR3 S0-DIMM Connector A			
Apple Inc.		DRAWING NUMBER	SIZE
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8	7	6	5	4	3	2	1
D	CPU CHANNEL A DQS 0 -> DIMM A DQS 0		CPU CHANNEL B DQS 0 -> DIMM B DQS 0				
	79 28 27 11	MEM A DQS N<0>	11 27 28 79 79 29 28 11	MEM B DQS N<0>	11 28 29 79		
	79 28 27 11	MEM A DQS P<0>	11 27 28 79 79 29 28 11	MEM B DQS P<0>	11 28 29 79		
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
C	CPU CHANNEL A DQS 1 -> DIMM A DQS 1		CPU CHANNEL B DQS 1 -> DIMM B DQS 1				
	79 11	MEM A DQS N<1>	79 11	MEM B DQS N<1>	79 11		
	79 11	MEM A DQS P<1>	79 11	MEM B DQS P<1>	79 11		
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
B	CPU CHANNEL A DQS 2 -> DIMM A DQS 2		CPU CHANNEL B DQS 2 -> DIMM B DQS 2				
	79 11	MEM A DQS N<2>	79 11	MEM B DQS N<2>	79 11		
	79 11	MEM A DQS P<2>	79 11	MEM B DQS P<2>	79 11		
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
A	CPU CHANNEL A DQS 3 -> DIMM A DQS 3		CPU CHANNEL B DQS 3 -> DIMM B DQS 3				
	79 11	MEM A DQS N<3>	79 11	MEM B DQS N<3>	79 11		
	79 11	MEM A DQS P<3>	79 11	MEM B DQS P<3>	79 11		
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
		MAKE_BASE=TRUE		MAKE_BASE=TRUE			
NOTE: Sandybridge does not use DM signals per doc 438297 Huron River SFF DG rev1.0 Section 2.6.13							
8	7	6	5	4	3	2	1

DDR3 Byte/Bit Swaps

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SYNC MASTER=ANNE_K901

SYNC DATE=06/22/2010

DDR3 Byte/Bit Swaps

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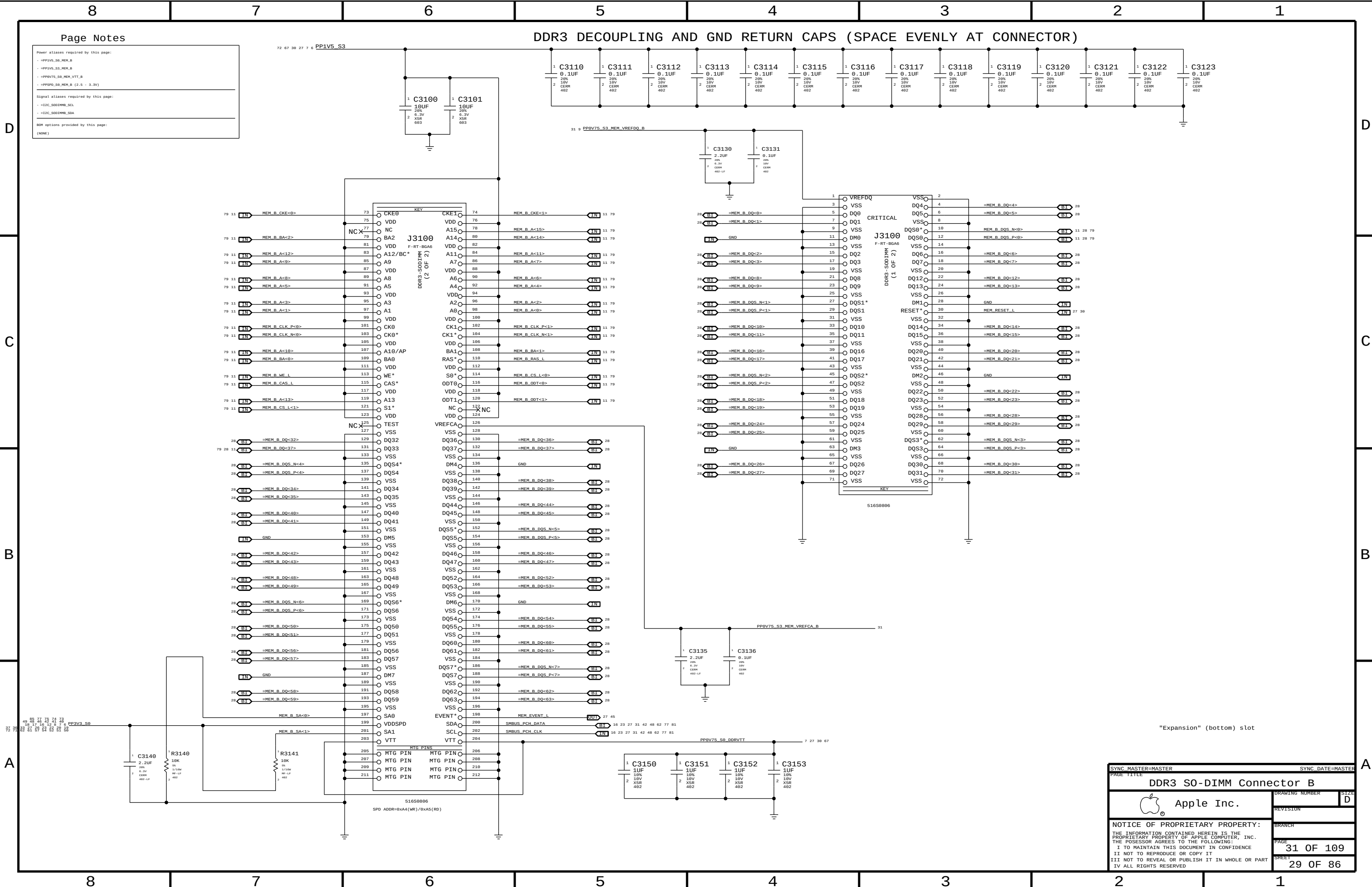
PAGE

SHEET

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SIZE
D



Page Notes

Power aliases required by this page:

- PP1V5_S3_MEM_B
- PP1V5_S3_MEM_B
- PP0V75_S0_MEM_VTT_B
- PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- I2C_S0D1MMB_SCL
- I2C_S0D1MMB_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

"Expansion" (bottom) slot

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE			
DDR3 S0-DIMM Connector B			
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
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The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

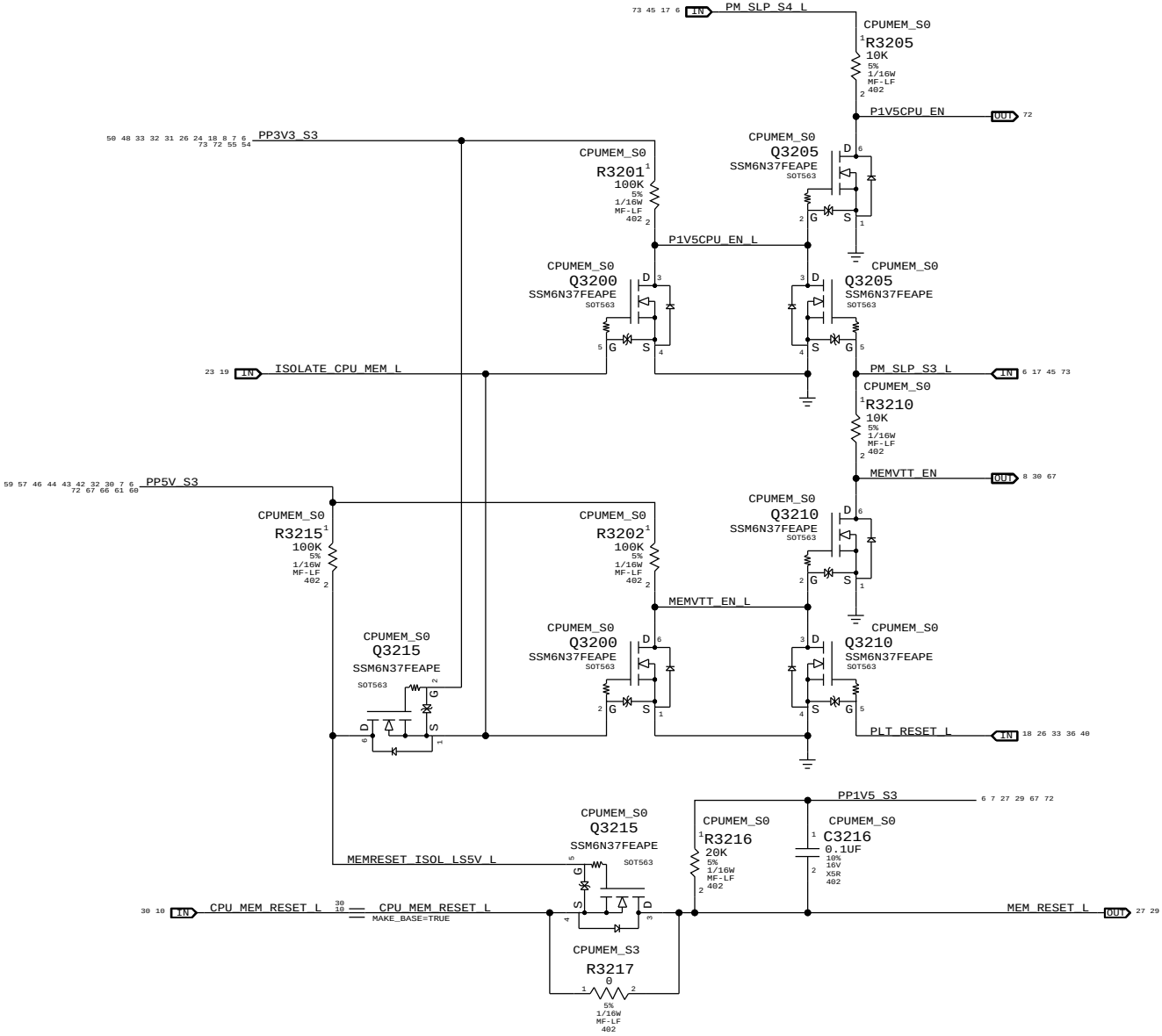
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

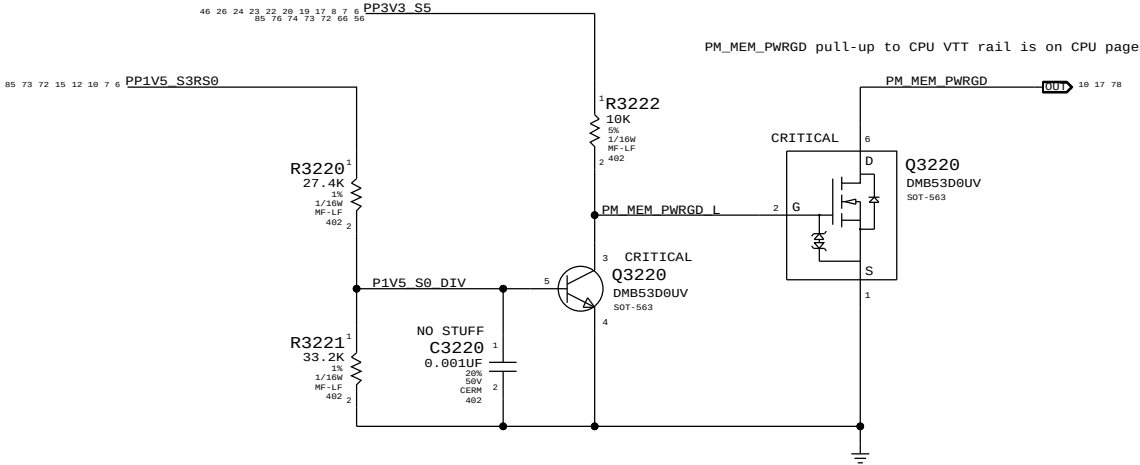
$P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L$

$MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L$

$MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L$

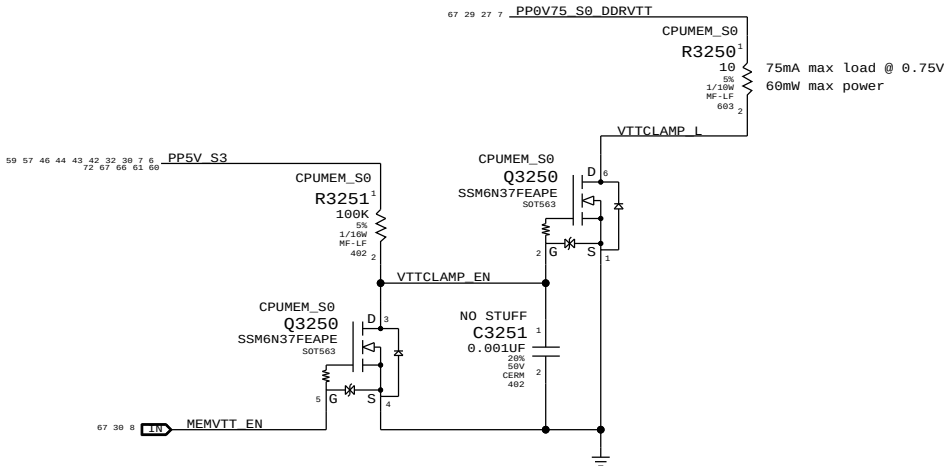


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

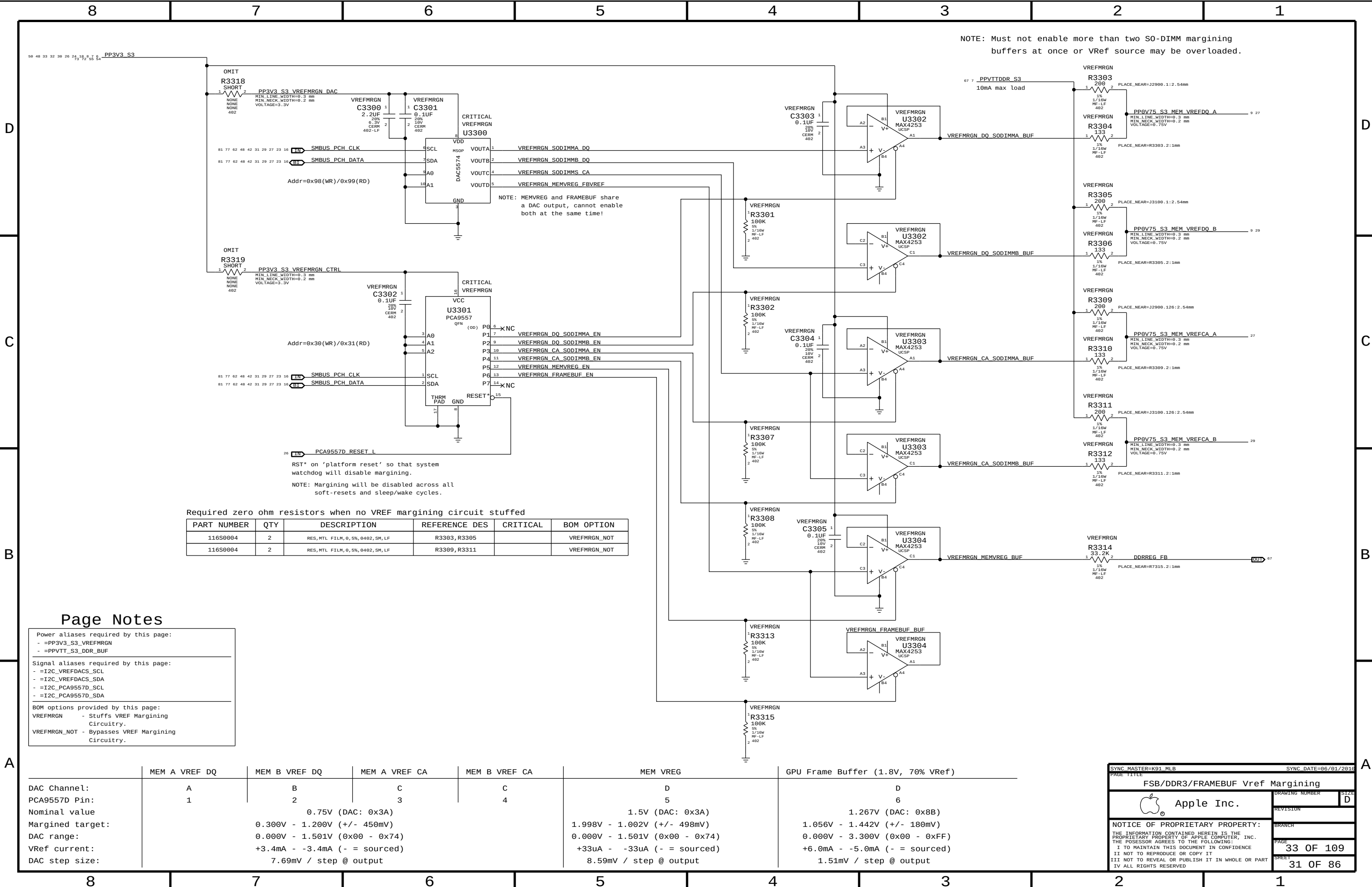


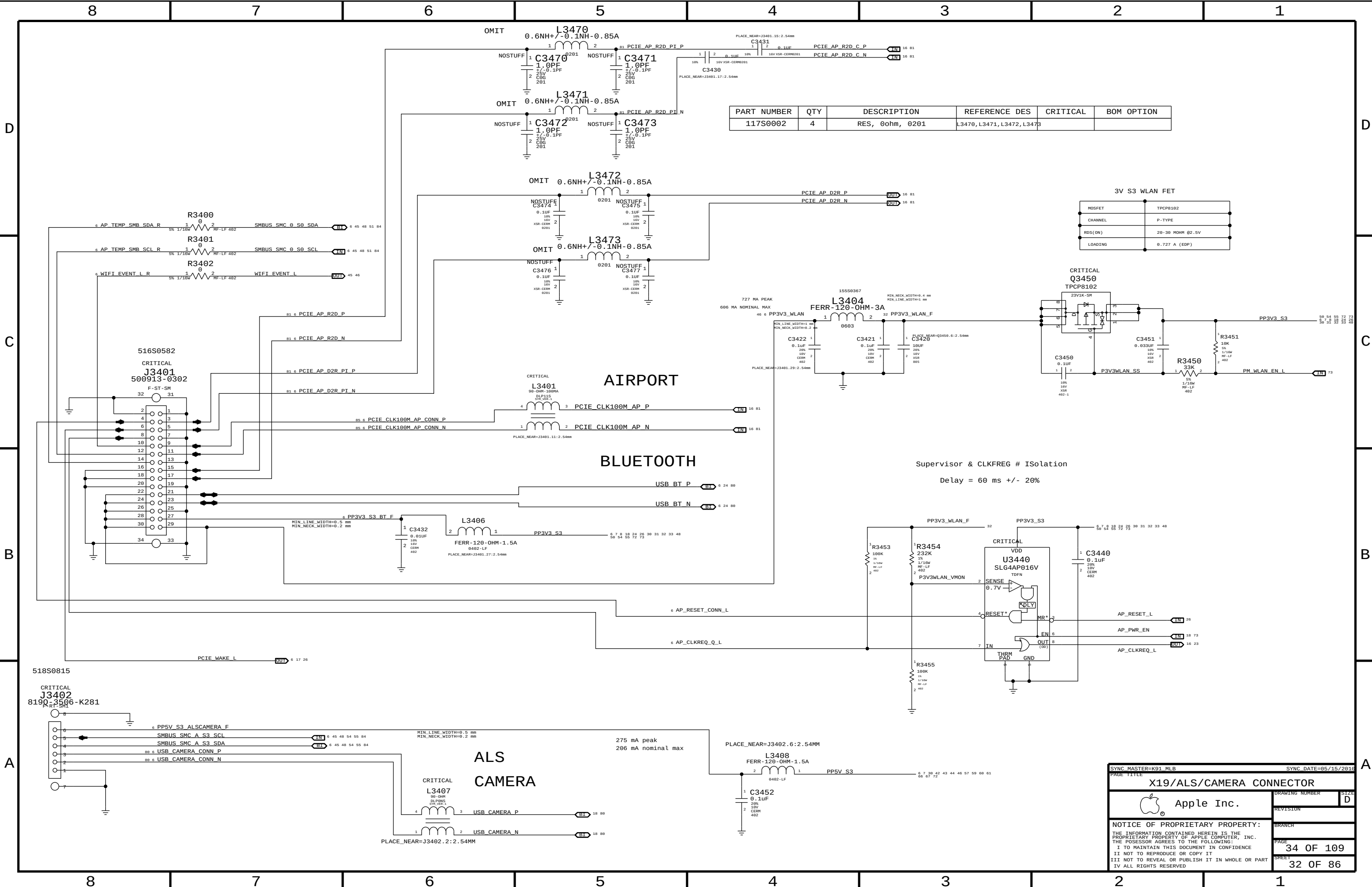
Step	SOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

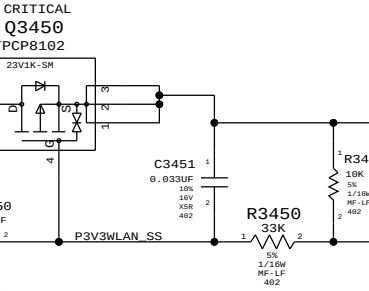
SYNC MASTER=ANNE_K901		SYNC DATE=06/22/2010	
PAGE TITLE			
CPU Memory S3 Support			
Apple Inc.		DRAWING NUMBER	SIZE D
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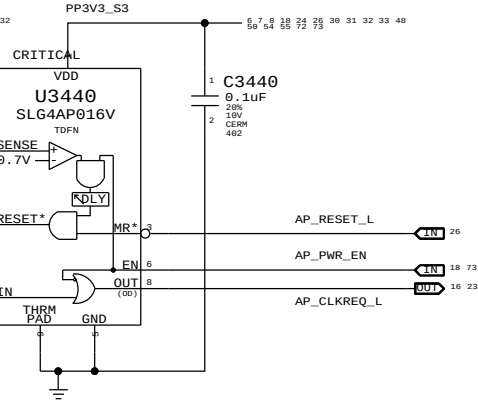


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0002	4	RES, 90hm, 0201	L3470, L3471, L3472, L3473		

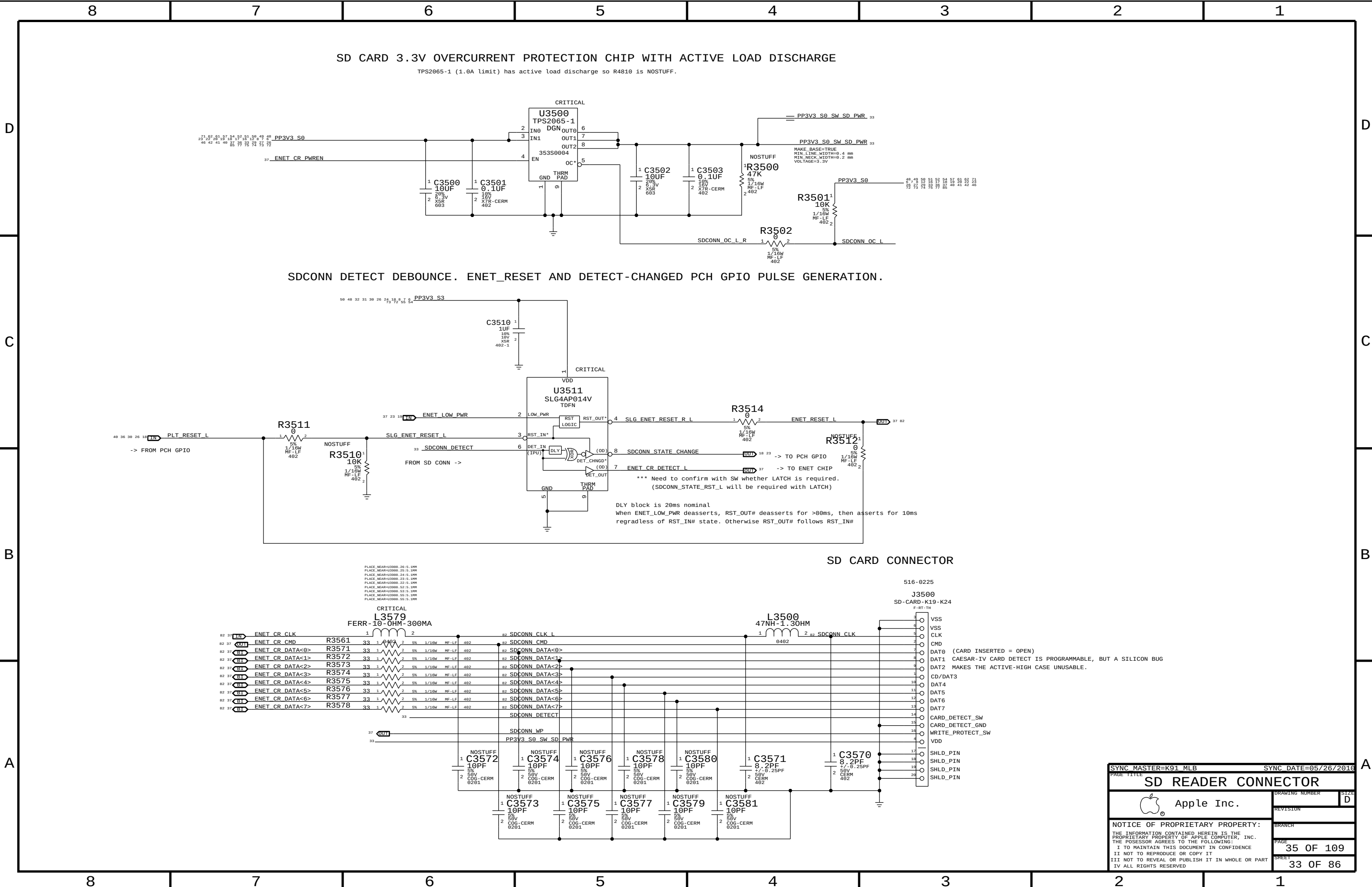
3V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	26-30 MOHM @2.5V
LOADING	0.727 A (EDP)

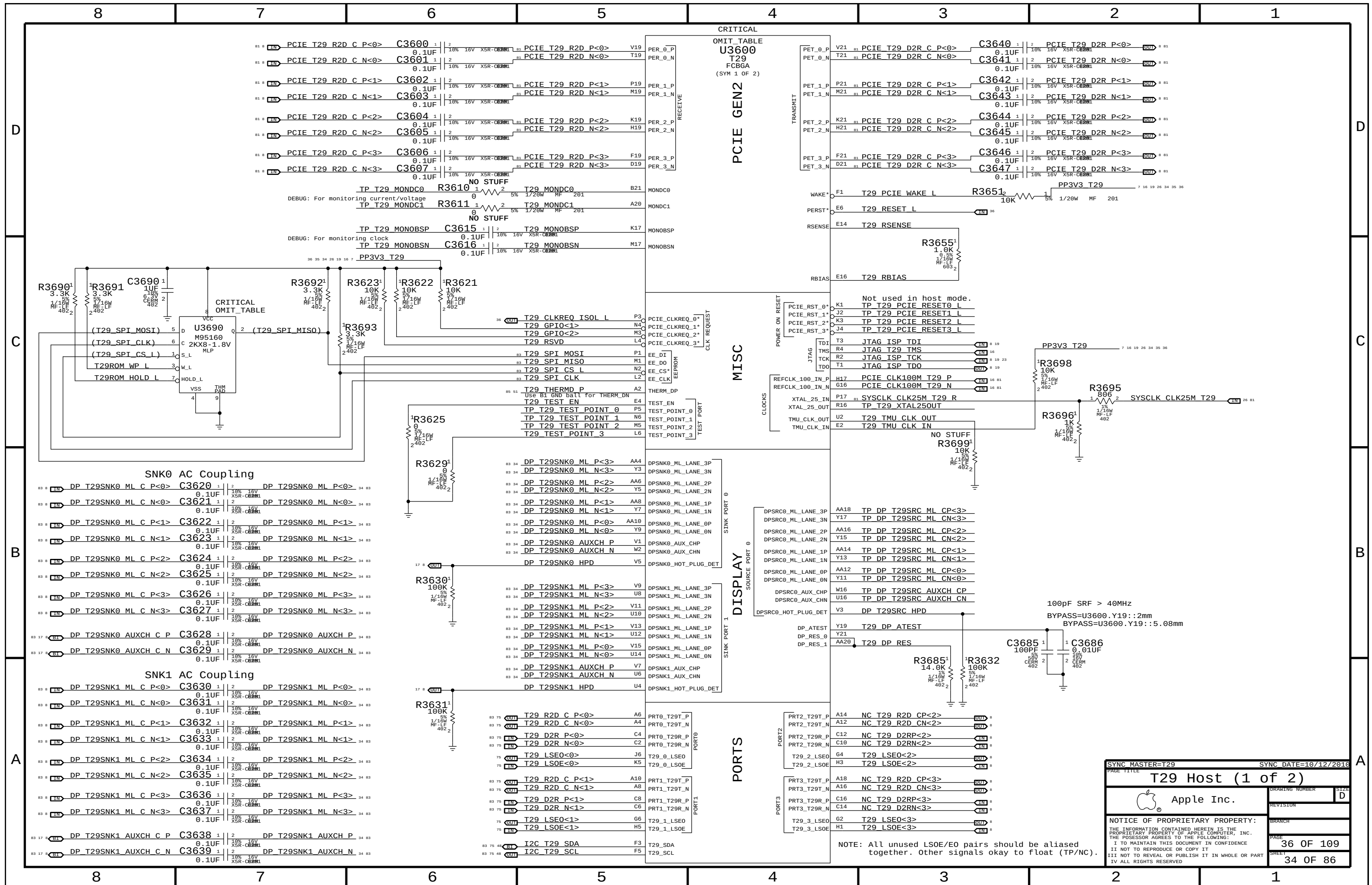


Supervisor & CLKFREQ # Isolation
Delay = 60 ms +/- 20%



SYNC MASTER=K91 MLB		SYNC DATE=05/15/2010	
PAGE TITLE		DRAWING NUMBER	SIZE
X19/ALS/CAMERA CONNECTOR			D
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D

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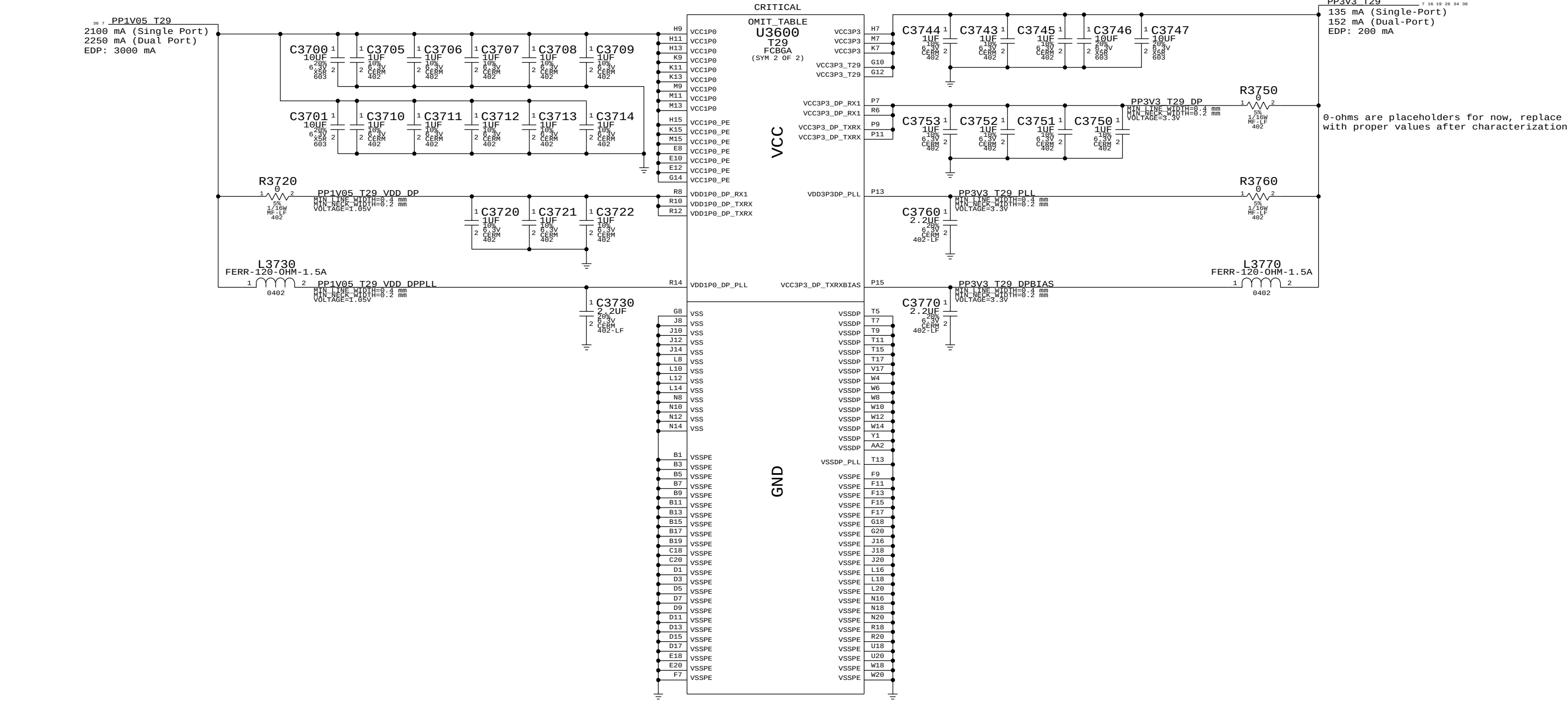
A

8 7 6 5 4 3 2 1

36 7 PP1V05 T29
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA

PP3V3 T29
135 mA (Single-Port)
152 mA (Dual-Port)
EDP: 200 mA

0-ohms are placeholders for now, replace with proper values after characterization.



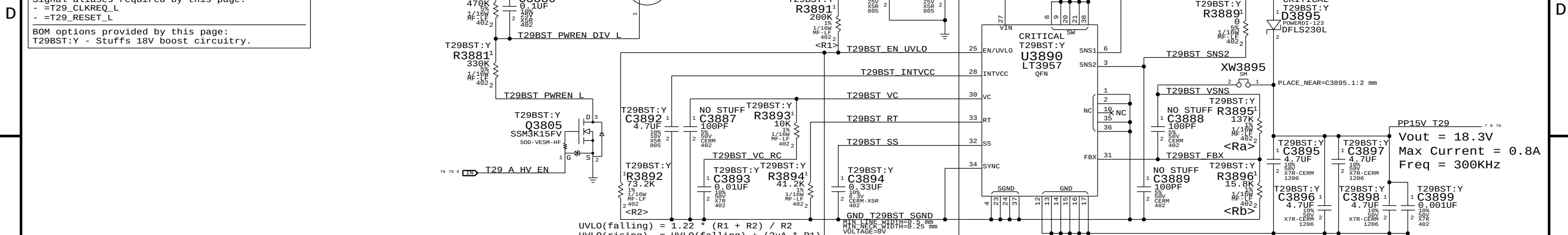
Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

8 7 6 5 4 3 2 1

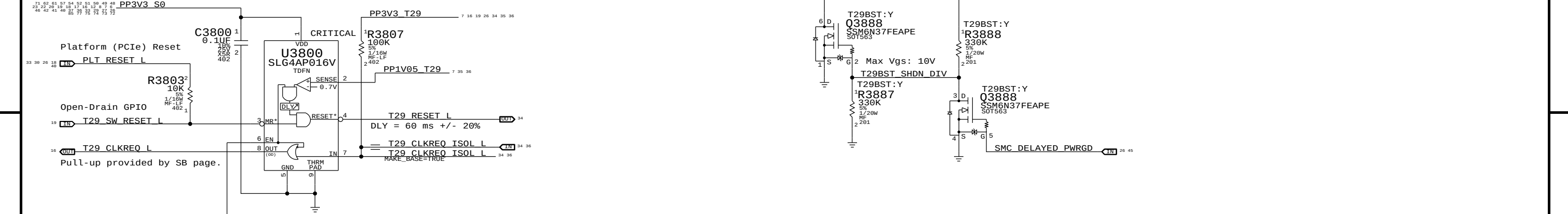
SYNC MASTER=T29		SYNC DATE=10/12/2010	
PAGE TITLE			
T29 Host (2 of 2)			
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8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

T29BST:Y - Stuffs 18V boost circuitry.



C	Supervisor & CLKREQ# Isolation	$UVLO(\text{rising}) = UVLO(\text{falling}) + (20\mu A \times R1)$ $UVLO = 4.55V (\text{falling}), 4.95 (\text{rising})$	SGND shorted to GND inside package, no Xw necessary.	$V_{out} = 1.6V \times (1 + R_a / R_b)$	C
---	--------------------------------	--	--	---	---



B	3.3V T29 Switch	B
---	-----------------	---



	1.05V T29 Switch	
	U3815	



							IV ALL RIGHTS RESERVED	30 OF 80
8	7	6	5	4	3	2	1	

Page Notes

Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

ENET MDI P<0>

ENET MDI N<0>

ENET MDI P<1>

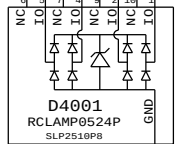
ENET MDI N<1>

ENET MDI P<3>

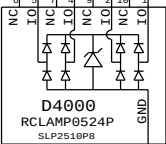
ENET MDI N<3>

ENET MDI N<2>

ENET MDI P<2>



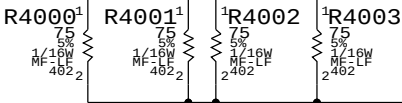
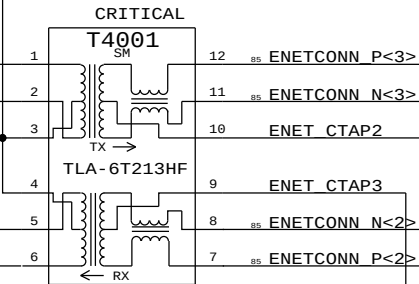
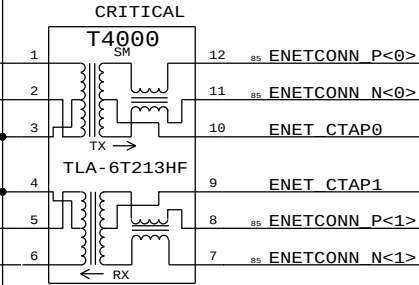
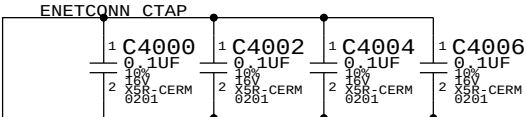
CRITICAL
NOSTUFF



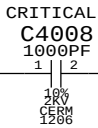
CRITICAL
NOSTUFF

Transformers should be mirrored on opposite sides of the board

Place one of 0.1uF cap close to each centertap pin of transformer



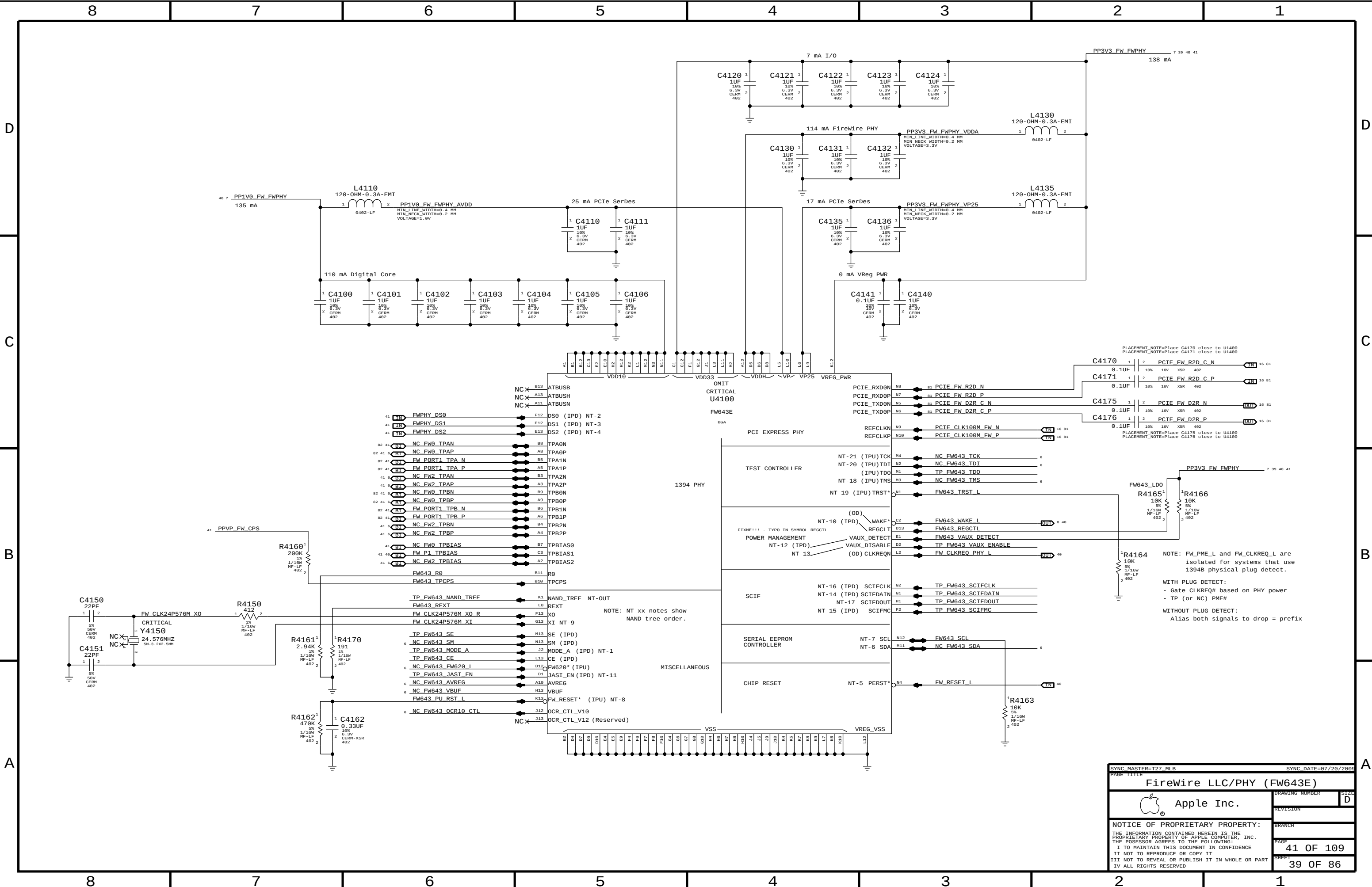
ENET BOB SMITH CAP
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.25 mm



CRITICAL
J4000
RJ45-M97-3
F-RT-TH

514-0636

SYNC_MASTER=K91_MLB		SYNC_DATE=05/26/2010	
PAGE TITLE		Ethernet Connector	
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
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SYNC MASTER=T27_MLB

SYNC DATE=07/20/2005

PAGE TITLE

FireWire LLC/PHY (FW643E)

 Apple Inc.

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SIZE
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Page Notes

Power aliases required by this page:

- =PPBUS_S5_FWPWRSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_P3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPWRCTL
- =PP1V05_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_FW_P1V0FWFET (1.0V FET Input)
- =PP1V0_FW_FET_R (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

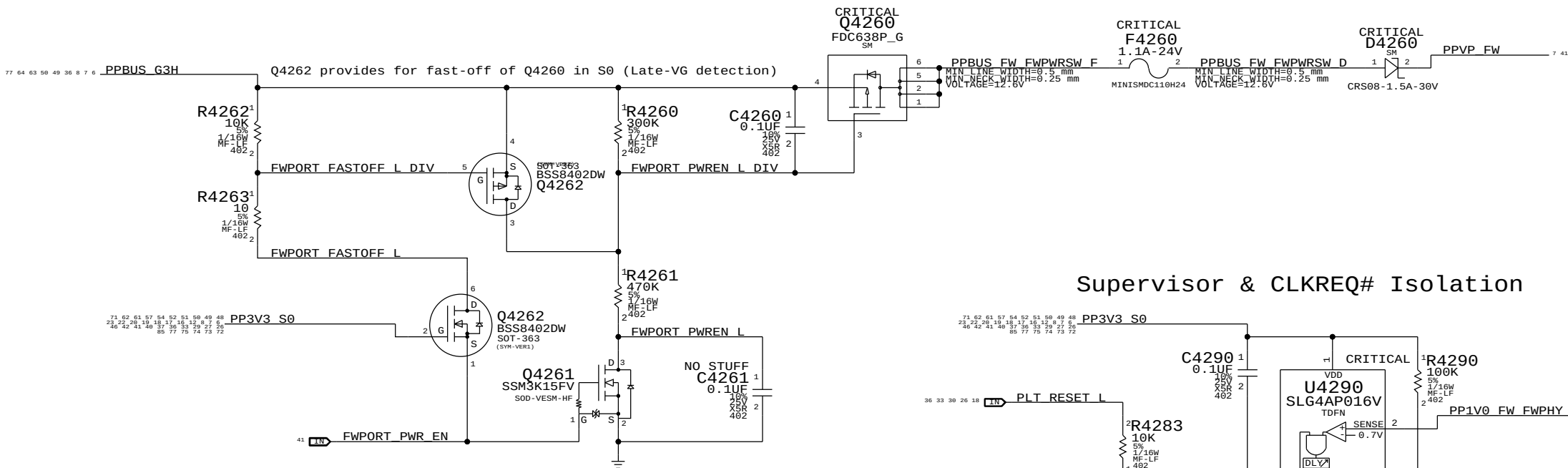
Signal aliases required by this page:

- =FW_CLKREQ_L
- =FW_PME_L

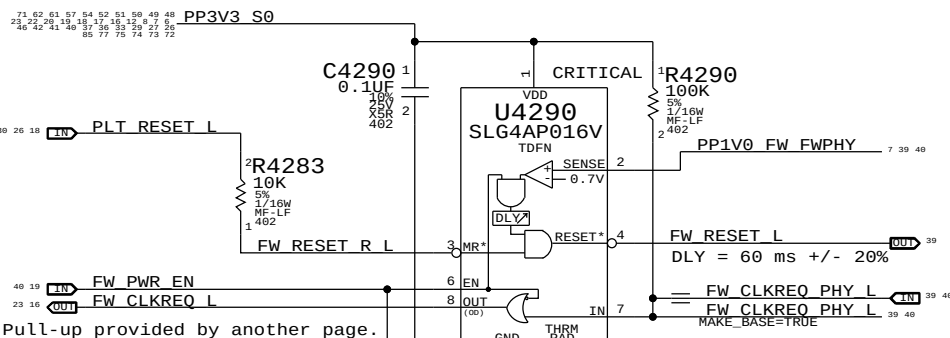
BOM options provided by this page:

(NONE)

FireWire Port Power Switch

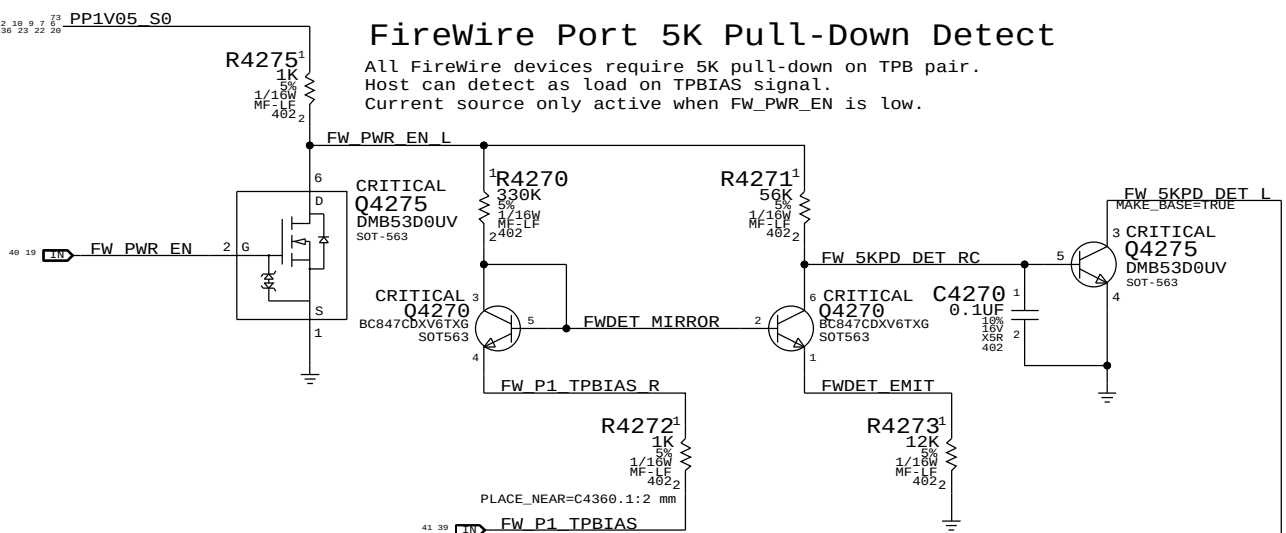


Supervisor & CLKREQ# Isolation



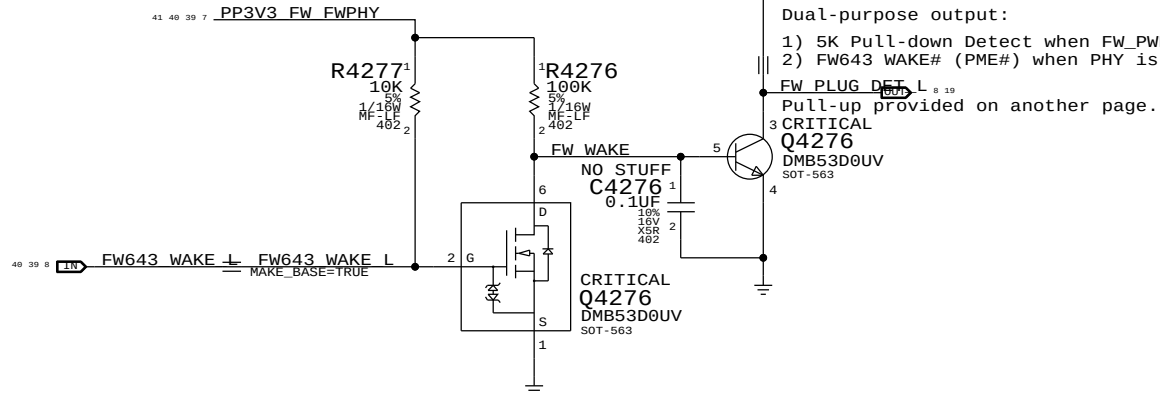
FireWire Port 5K Pull-Down Detect

All FireWire devices require 5K pull-down on TPB pair.
Host can detect as load on TPBIAS signal.
Current source only active when FW_PWR_EN is low.



FireWire PHY WAKE# Support

When PHY is powered, FW_5KPD_DET_L acts as legacy PME# signal.

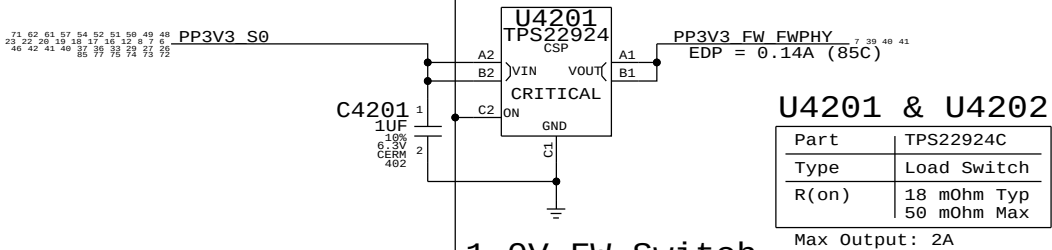


Dual-purpose output:

- 1) 5K Pull-down Detect when FW_PWR_EN is low.
- 2) FW643 WAKE# (PME#) when PHY is powered.

Pull-up provided on another page.

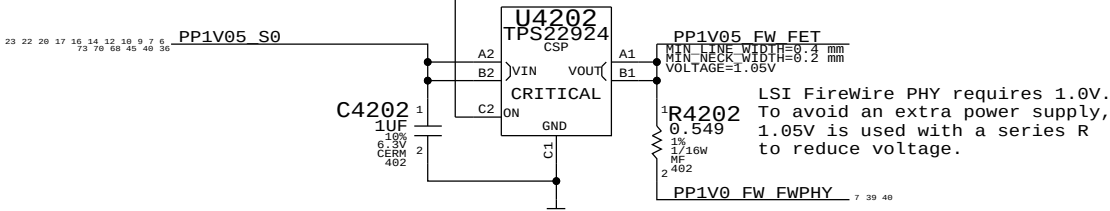
3.3V FW Switch



Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

1.0V FW Switch



TEXT NOTE FOR 3.3V RAIL CURRENT CHANGED TO EDP NUMBER.

SYNC MASTER=T27_MLB		SYNC DATE=12/15/2009	
PAGE 12/12		DRAWING NUMBER	
FireWire Port & PHY Power		SIZE D	
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Page Notes

Power aliases required by this page:

- PPVP_FW_PORT1
- PPVP_FW_PHY_CPS_FET (From Port)
- PPVP_FW_PHY_CPS (To PHY)
- PP3V3_FW_FWPHY
- PP3V3_S0_FWLATEVG

Signal aliases required by this page:

- FW_PHY_DS0
- FW_PHY_DS1
- FW_PHY_DS2

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

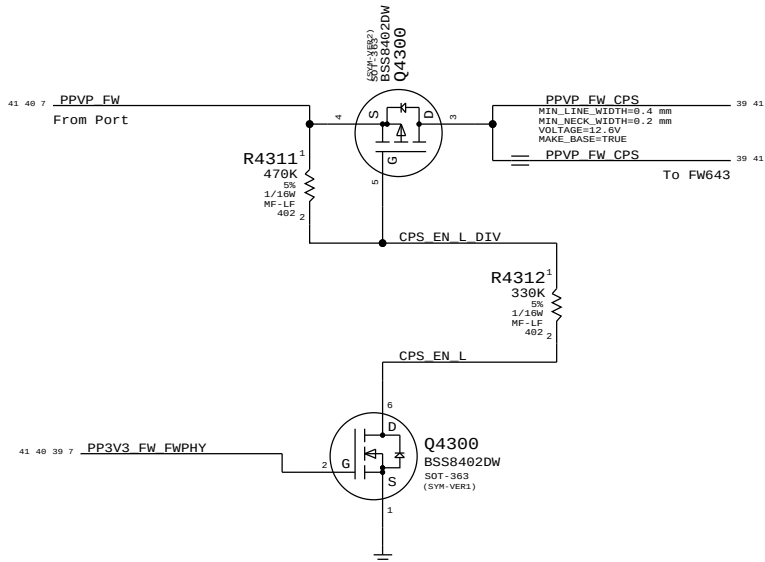
BOM options provided by this page:

(NONE)

1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)

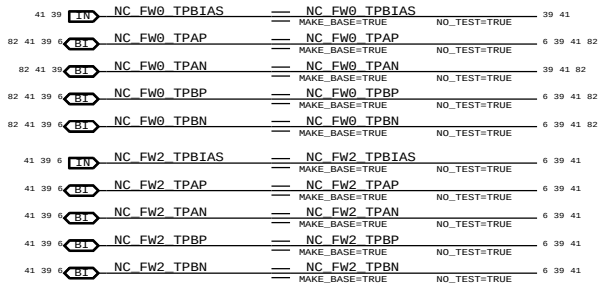
FW643 TPCPS Leakage Protection

FW643 has internal leakage path from TPCPS pin to VDD33.
FET blocks current to TPCPS until VDD33 is powered.



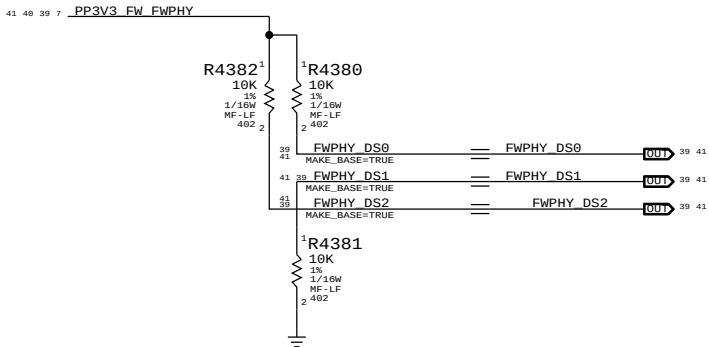
Unused FireWire Ports

Disabled per LSI instructions
(All unused port signals TP/NC)



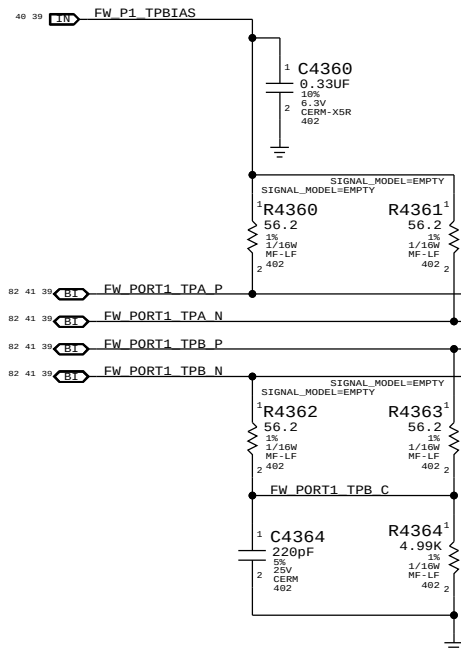
FireWire PHY Config Straps

Configures PHY for:
- Port "1" Bilingual (1394B)



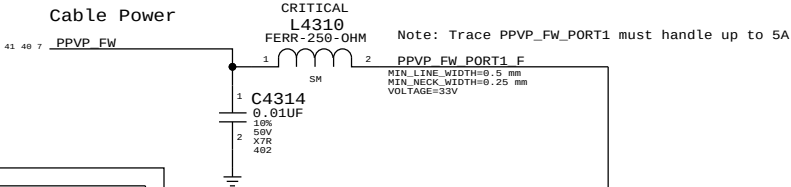
Termination

Place close to FireWire PHY

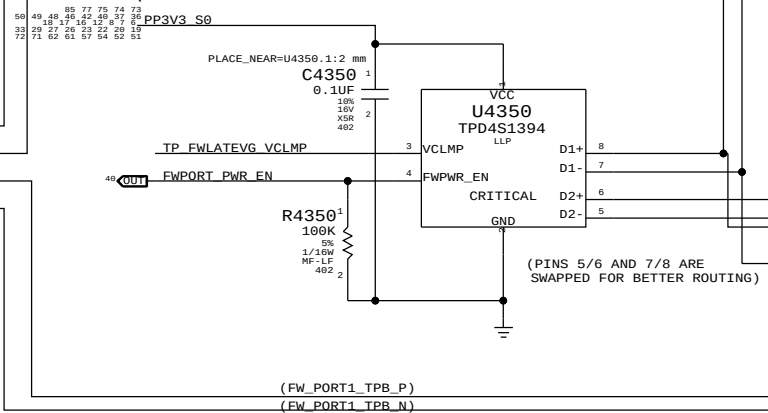


Cable Power

Note: Trace PPVP_FW_PORT1 must handle up to 5A



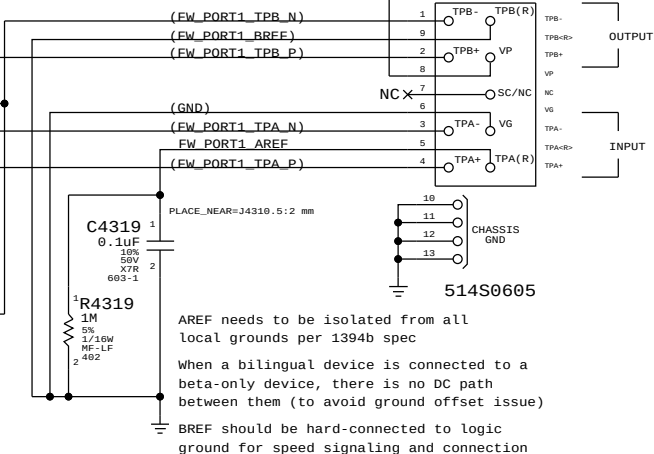
"Snapback" & "Late VG" Protection



PORT 1

BILINGUAL

CRITICAL
J4310
1394B-M97
F-RT-TH



AREF needs to be isolated from all local grounds per 1394b spec
When a bilingual device is connected to a beta-only device, there is no DC path between them (to avoid ground offset issue)
BREF should be hard-connected to logic ground for speed signaling and connection

CANNOT SYNC THIS PAGE FROM T27, TPA AND TPB FOR U4350 IS SWAPPED

FireWire Connector



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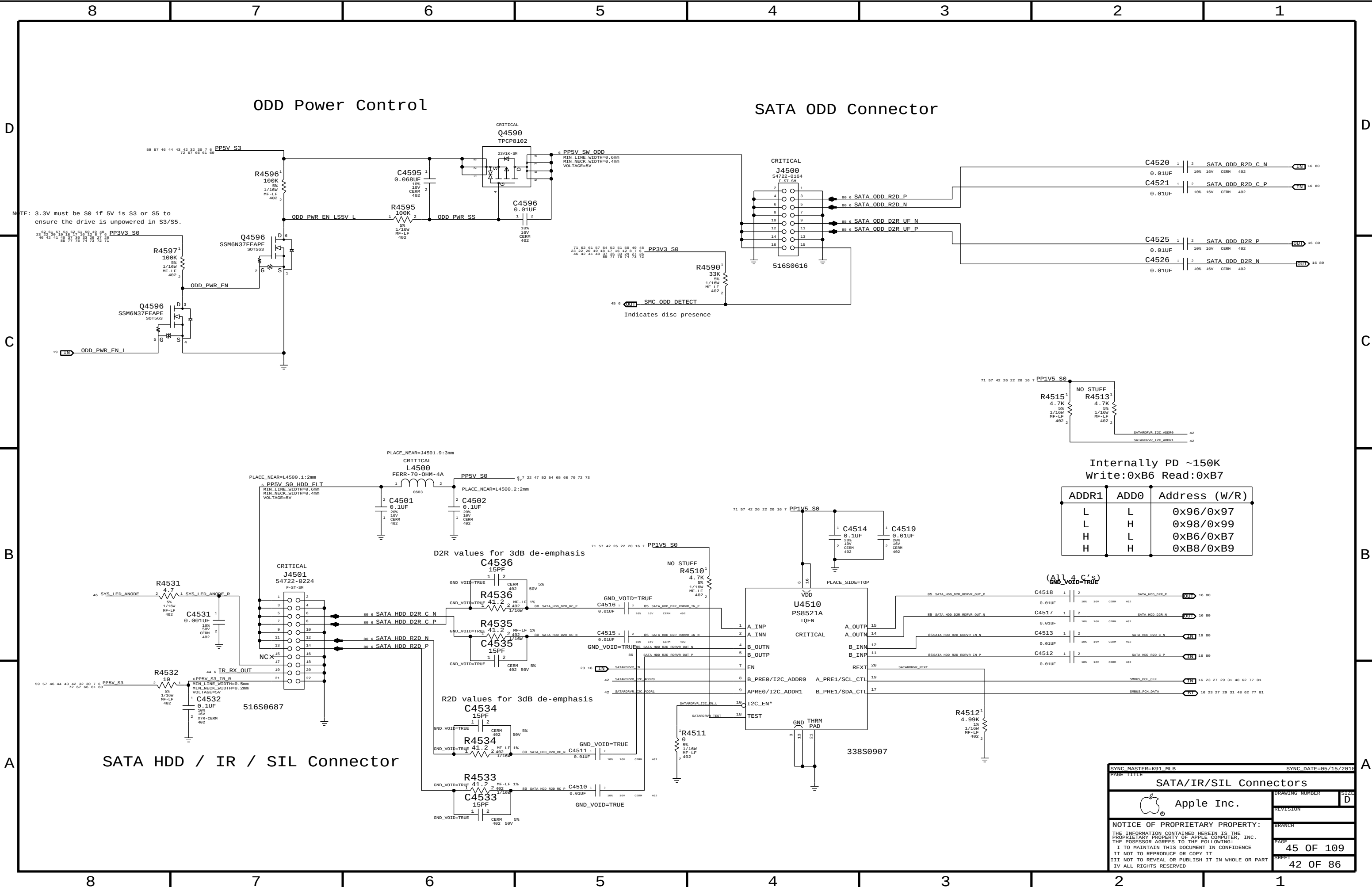
DRAWING NUMBER

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Internally PD ~150K
Write:0xB6 Read:0xB7

ADDR1	ADD0	Address (W/R)
L	L	0x96/0x97
L	H	0x98/0x99
H	L	0xB6/0xB7
H	H	0xB8/0xB9

SYNC MASTER=K91 MLB

SYNC DATE=85/15/2810

PAGE TITLE

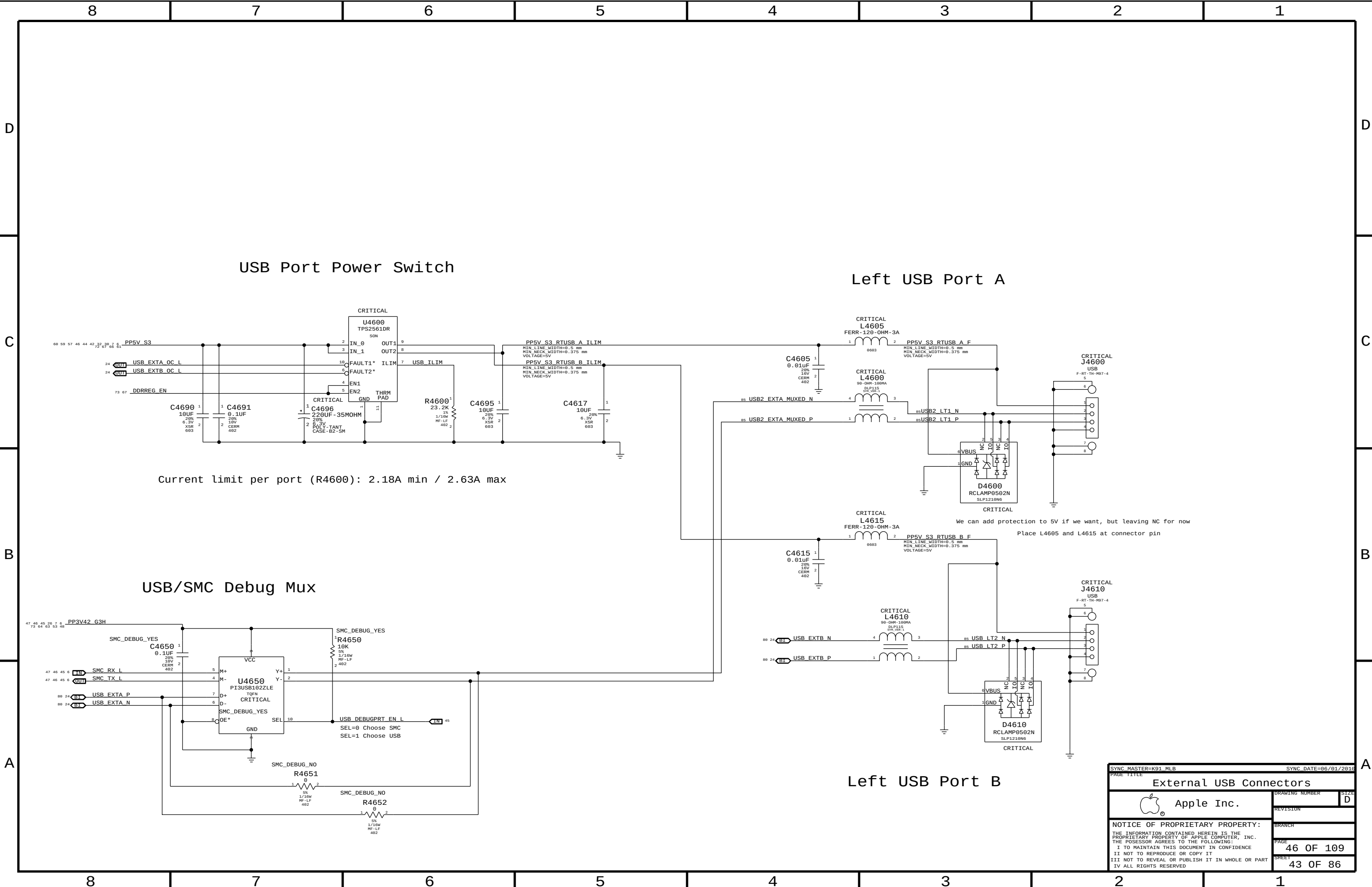
SATA/IR/SIL Connectors

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Current limit per port (R4600): 2.18A min / 2.63A max

We can add protection to 5V if we want, but leaving NC for now
Place L4605 and L4615 at connector pin

8 7 6 5 4 3 2 1

D

C

B

A

8 7 6 5 4 3 2 1

IR SUPPORT

PP5V_S3

C4801
0.1UF
10%
16V
X7R-CERM
402

USB_IR_P
DIFFERENTIAL_PAIR=USB2_IR

USB_IR_N
DIFFERENTIAL_PAIR=USB2_IR

IR_VREF_FILTER

C4803
1UF
10%
16V
X5R
402-1

U4800
CY7C63803-LQXC
QFN

VCC

P1.0/D+ P0.0
P1.1/D- P0.1
P1.2/VREG INT0/P0.2
P1.3/SSEL INT1/P0.3
P1.4/SCLK INT2/P0.4
P1.5/SMOSI TI00/P0.5
P1.6/SMISO TI01/P0.6

CRITICAL
OMIT

P/N 33850633

NC

THRML
PAD

VSS

IR_RX_OUT_RC

C4804
0.001UF
10%
50V
CERM
402

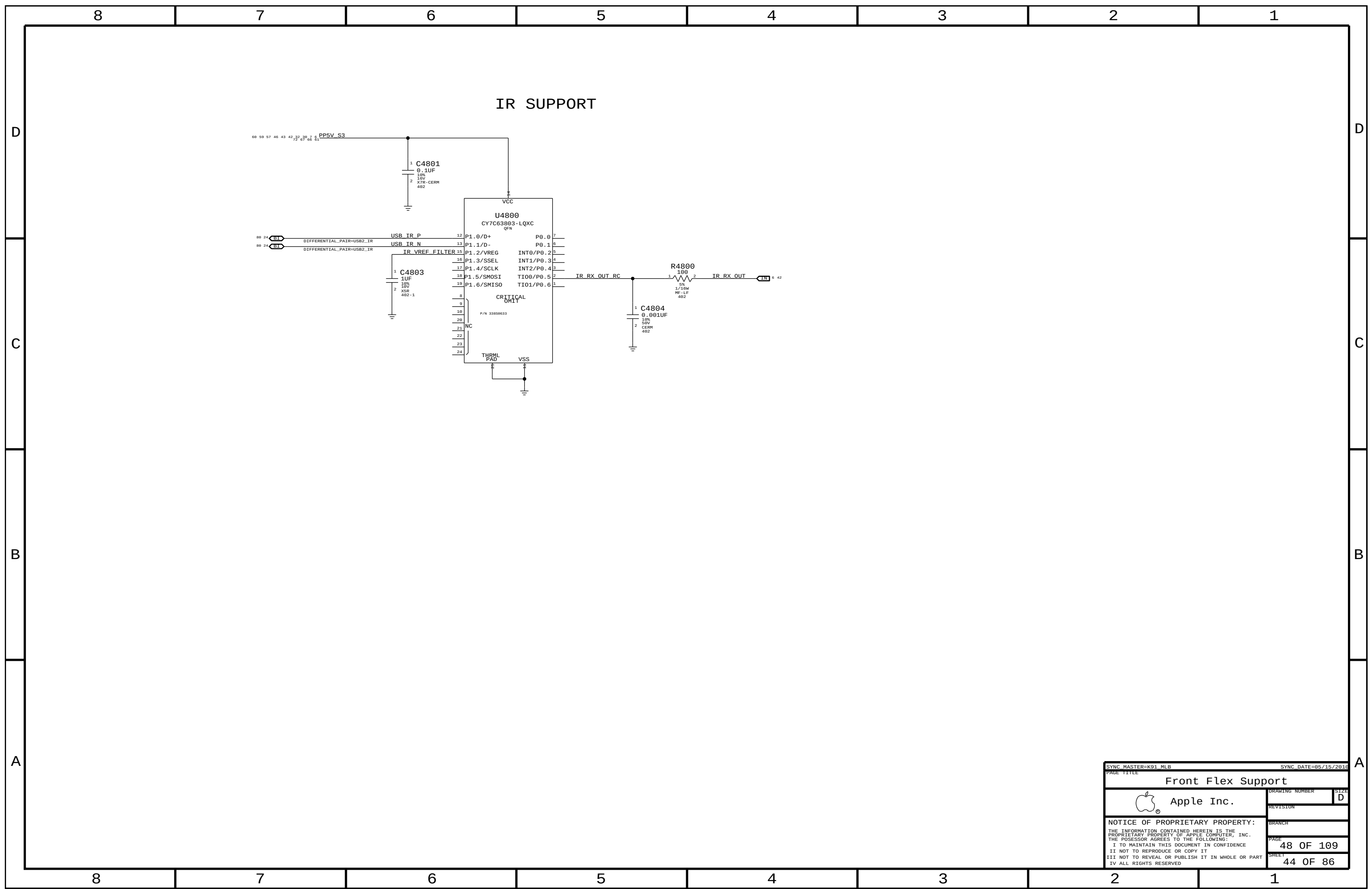
R4800
100
5%
1/16W
MF-LF
402

IR_RX_OUT

10

42

SYNC MASTER=K91 MLB		SYNC DATE=05/15/2016	
PAGE TITLE			
Front Flex Support		DRAWING NUMBER	
Apple Inc.		SIZE	
		D	
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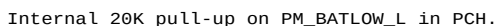
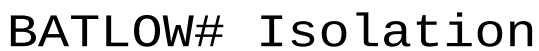


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SMC			
 Apple Inc.		DRAWING NUMBER	SHEET
			D
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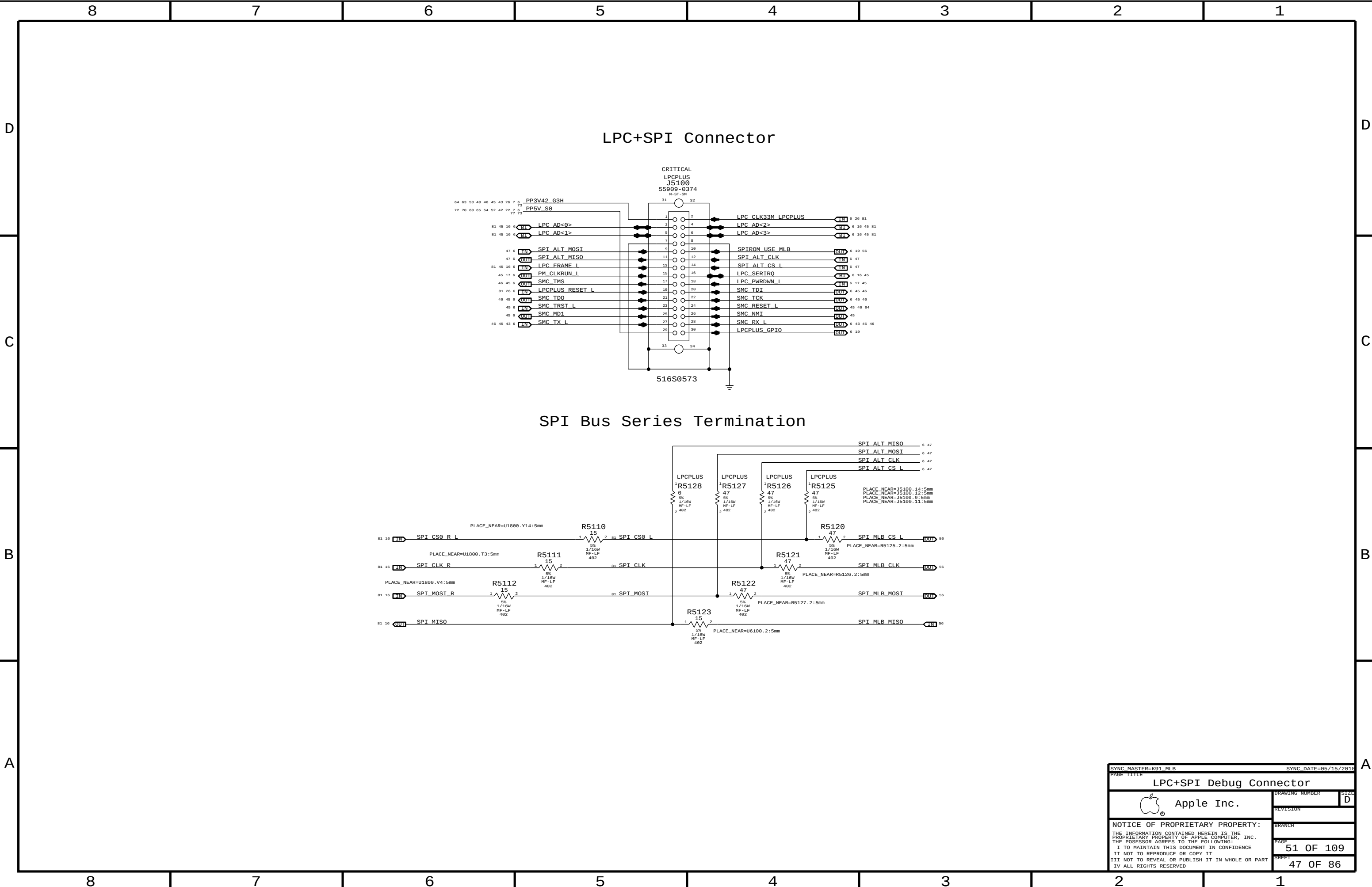
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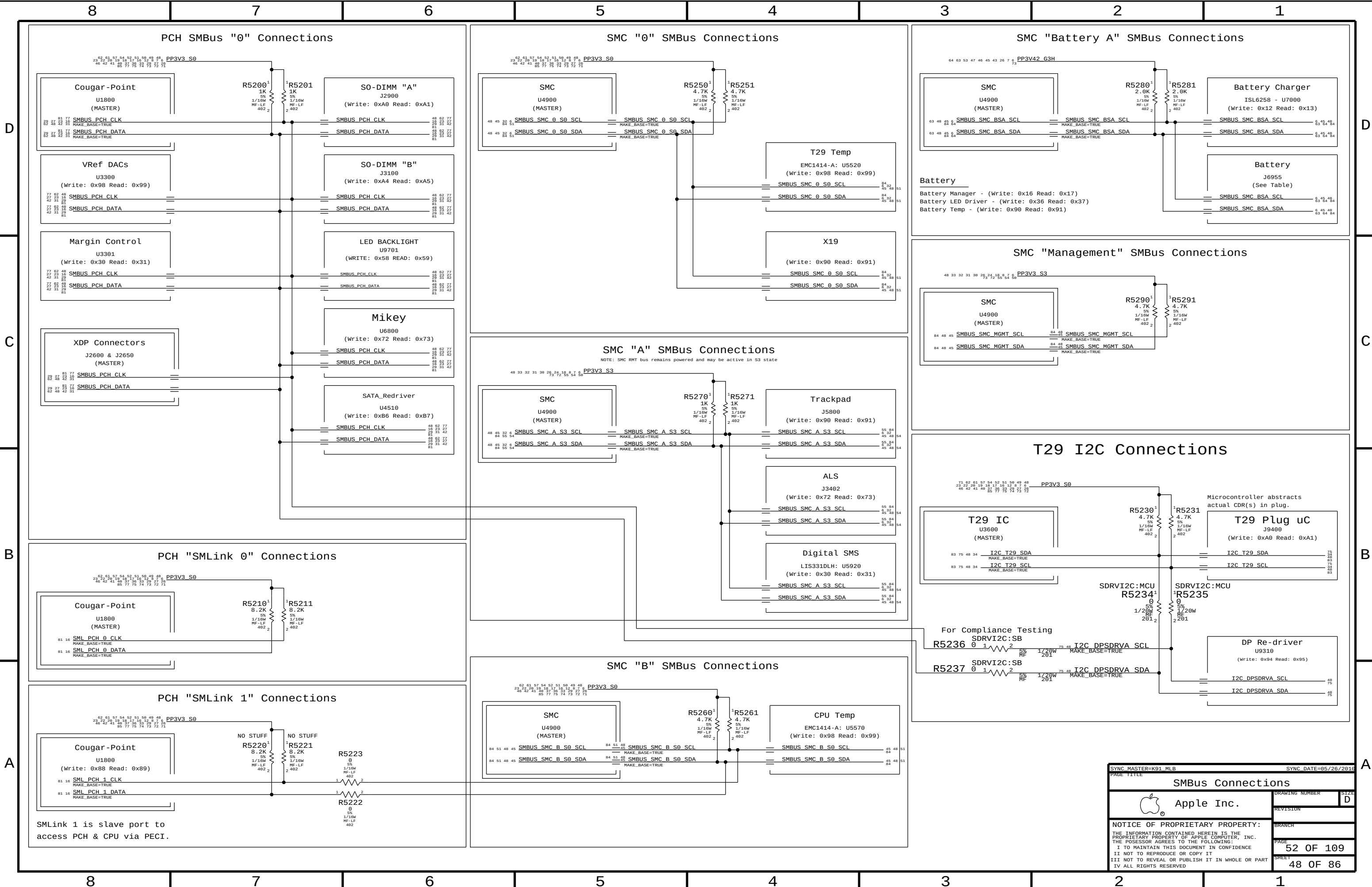


SMC GFX OVERTEMP_L R5095 10K 1 2 5% 1/16W MF-LF 402

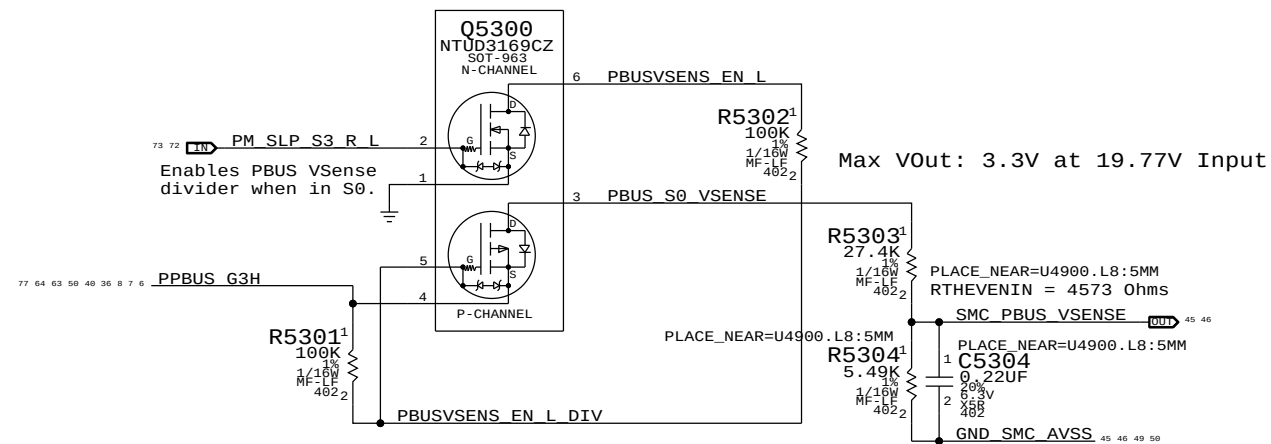
SMC GFX_OVERTEMP_L R5095 10K 1  2 5% 1/16W MF-LF 402

SYNC MASTER=LINDA K90I		SYNC DATE=07/08/20	
PAGE TITLE			
 Apple Inc.		DRAWING NUMBER	
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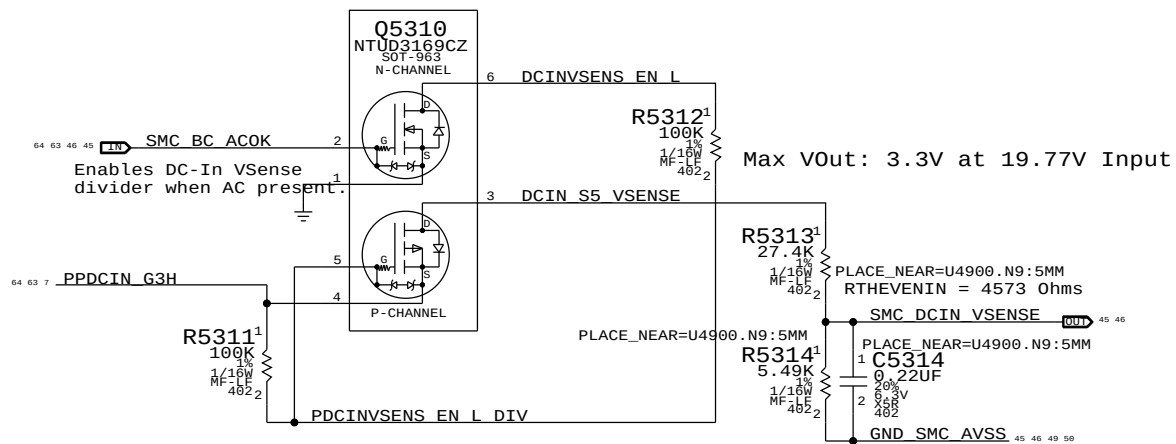




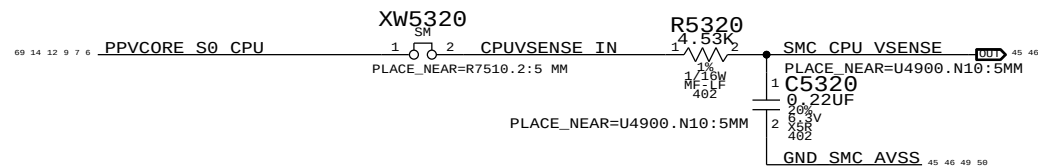
PBUS Voltage Sense Enable & Filter



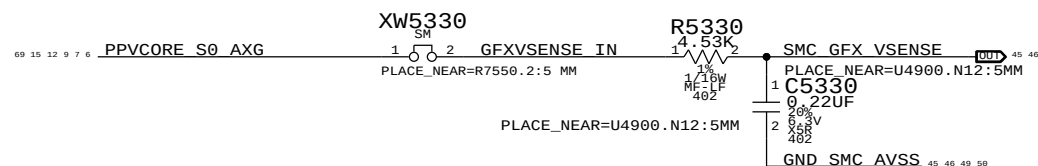
DC-In Voltage Sense Enable & Filter



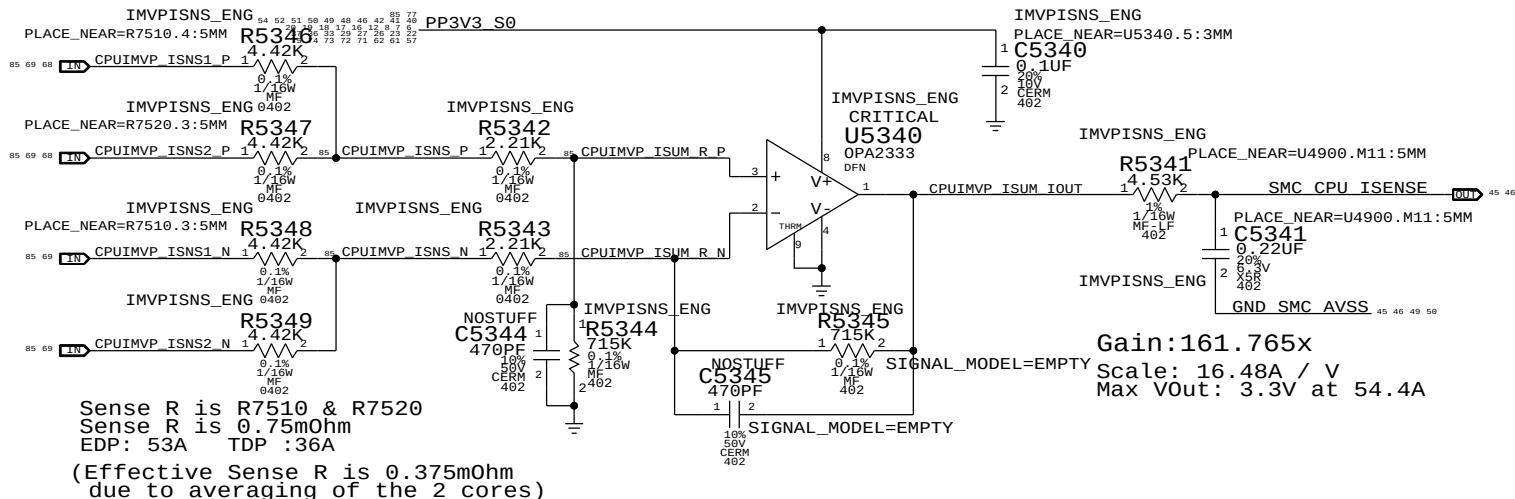
CPU Vcore Voltage Sense / Filter



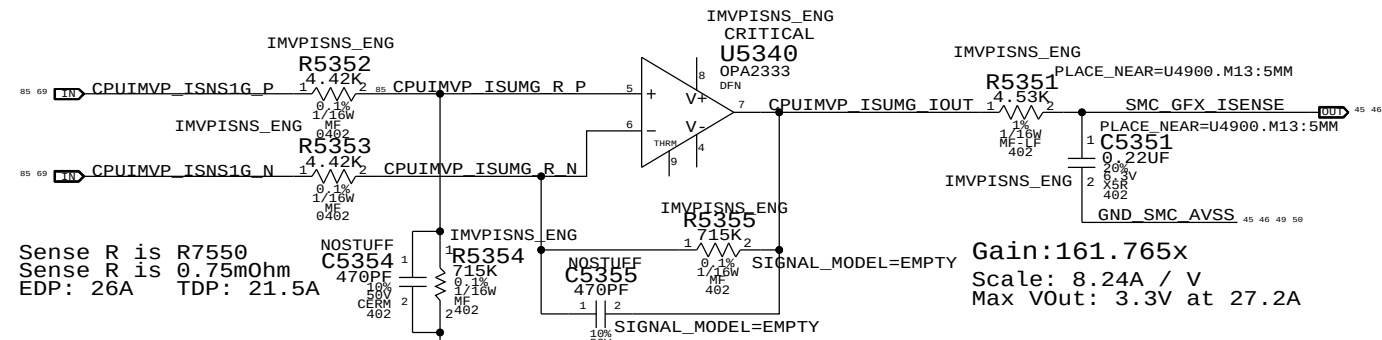
GFX/IG Vcore Voltage Sense / Filter



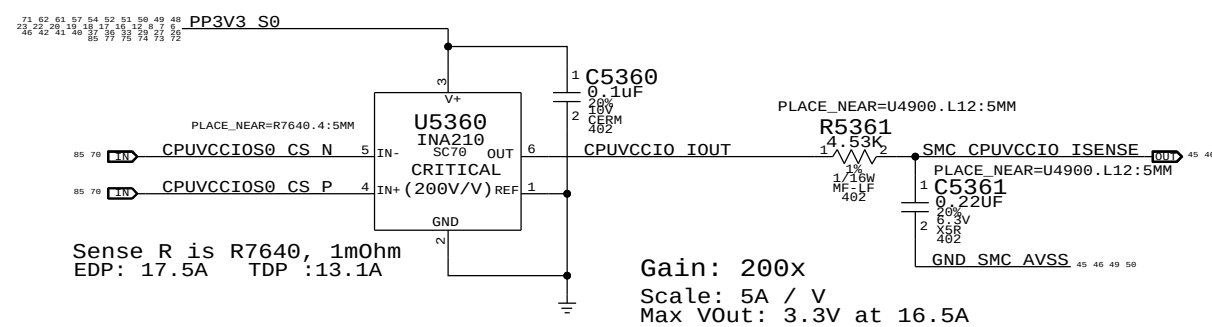
CPU VCore Load Side Current Sense / Filter



GFX/IG VCore Load Side Current Sense / Filter



CPU 1.05V VCCIO Current Sense / Filter



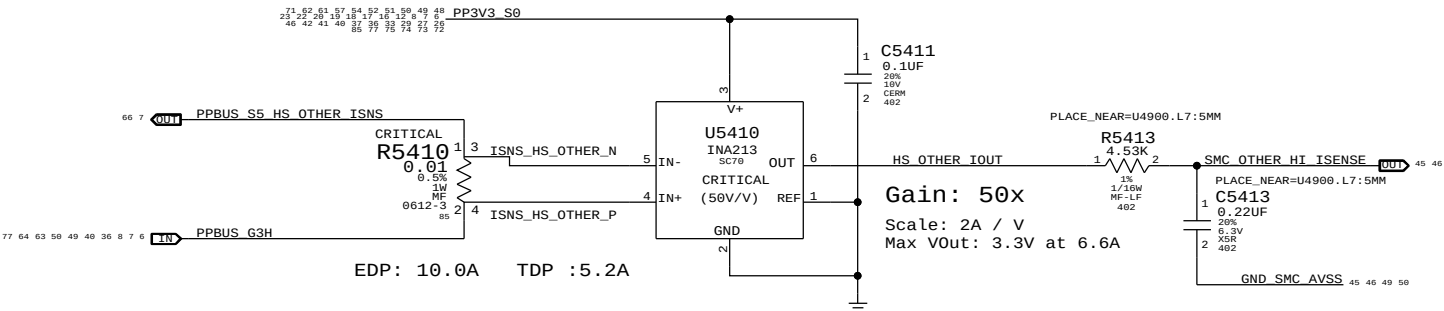
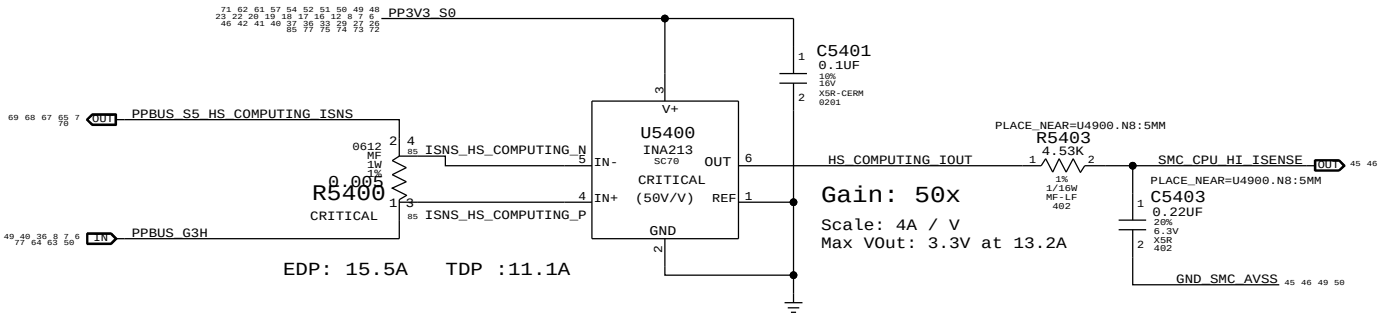
SYNC MASTER=LINDA K90I		SYNC DATE=10/22/2016	
PAGE TITLE			
Voltage & Load Side Current Sensing			
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D

D

COMPUTING High Side Current Sense / Filter

OTHER High Side Current Sense / Filter

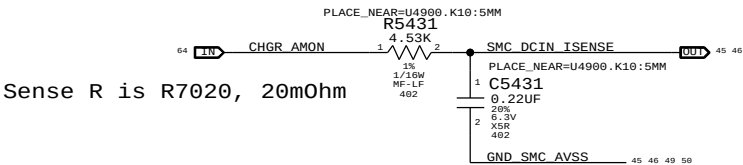
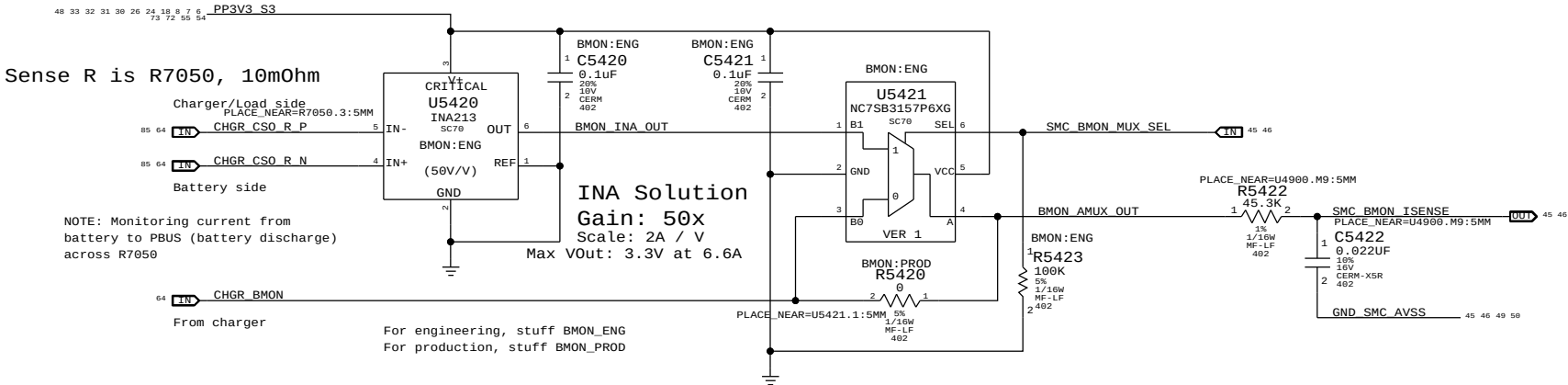


C

C

CHARGER BMON High Side (BATTERY DISCHARGE) Current Sense, MUX & Filter

DC-IN (AMON) Current Sense Filter



INA (Engineering) Solution
Gain: 50x
Scale: 2A / V
Max VOut: 3.3V at 6.6A

Charger BMON (Production) Solution
ISL6259 Gain: 36x
Scale: 2.78A / V
Max VOut: 3.3V at 9.167A

A

A

SYNC MASTER=LINDA K901		SYNC DATE=10/22/2010	
PAGE TITLE		High Side Current Sensing	
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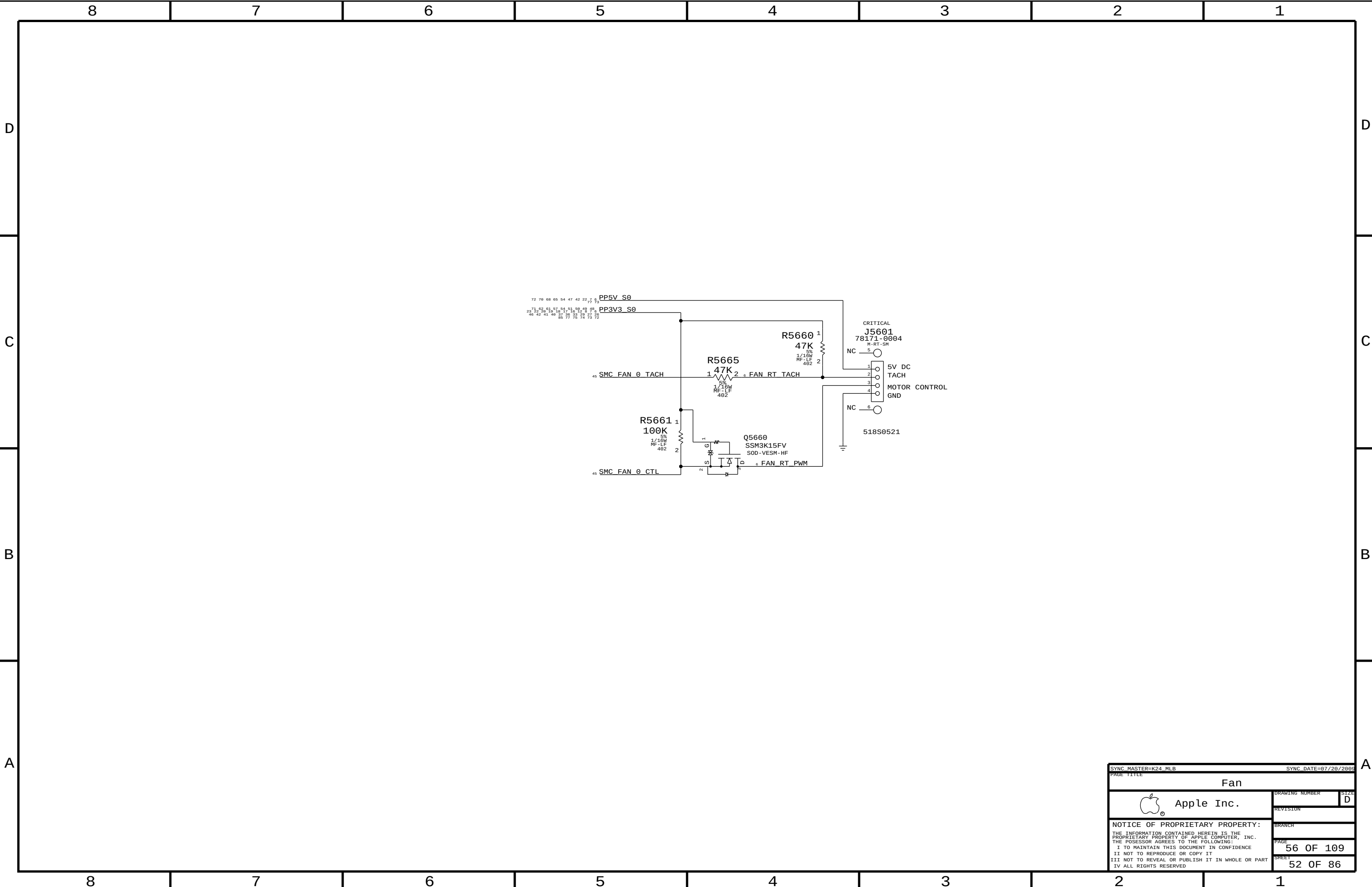


B



D





PSOC USB CONTROLLER

- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER

IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA 80UA	2.55 KOHM	0.0255 V 0.204 V	0.255E-6 W 16.32E-6 W
3V3 LDO	VDD VOUT	60MA (MAX) 60MA (MAX)	10 OHM 0.2 OHM	0.6 V 0.012 V	36E-3 W 0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V 0.021 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

Keyboard Connector

SMC Manual Reset & Isolation

Left shift, option & control keys combined with power button cause SMC RESET# assertion.
Keys ANDed with PSOC power to isolate when PSOC is not powered.

TPAD Buttons Disable

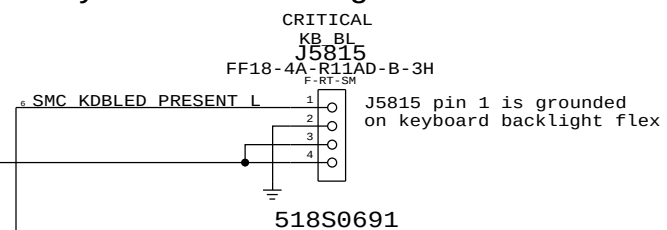
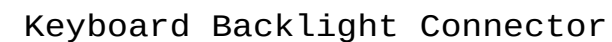
PLACE THESE COMPONENTS CLOSE TO J5800
THIS ASSUMES THERE'S A PP3V42_G3H PULL UP ON MLB

THE TPAD BUTTONS WILL BE DISABLE
WHEN THE LID IS CLOSED
LID OPEN => SMC_LID_LC ~ 3.42V
LID CLOSE => SMC_LID_LC < 0.50V

PAGE TITLE		SYNC DATE=07/12/2016	
WELLSPRING 1		SIZE D	
Apple Inc.		DRAWING NUMBER	
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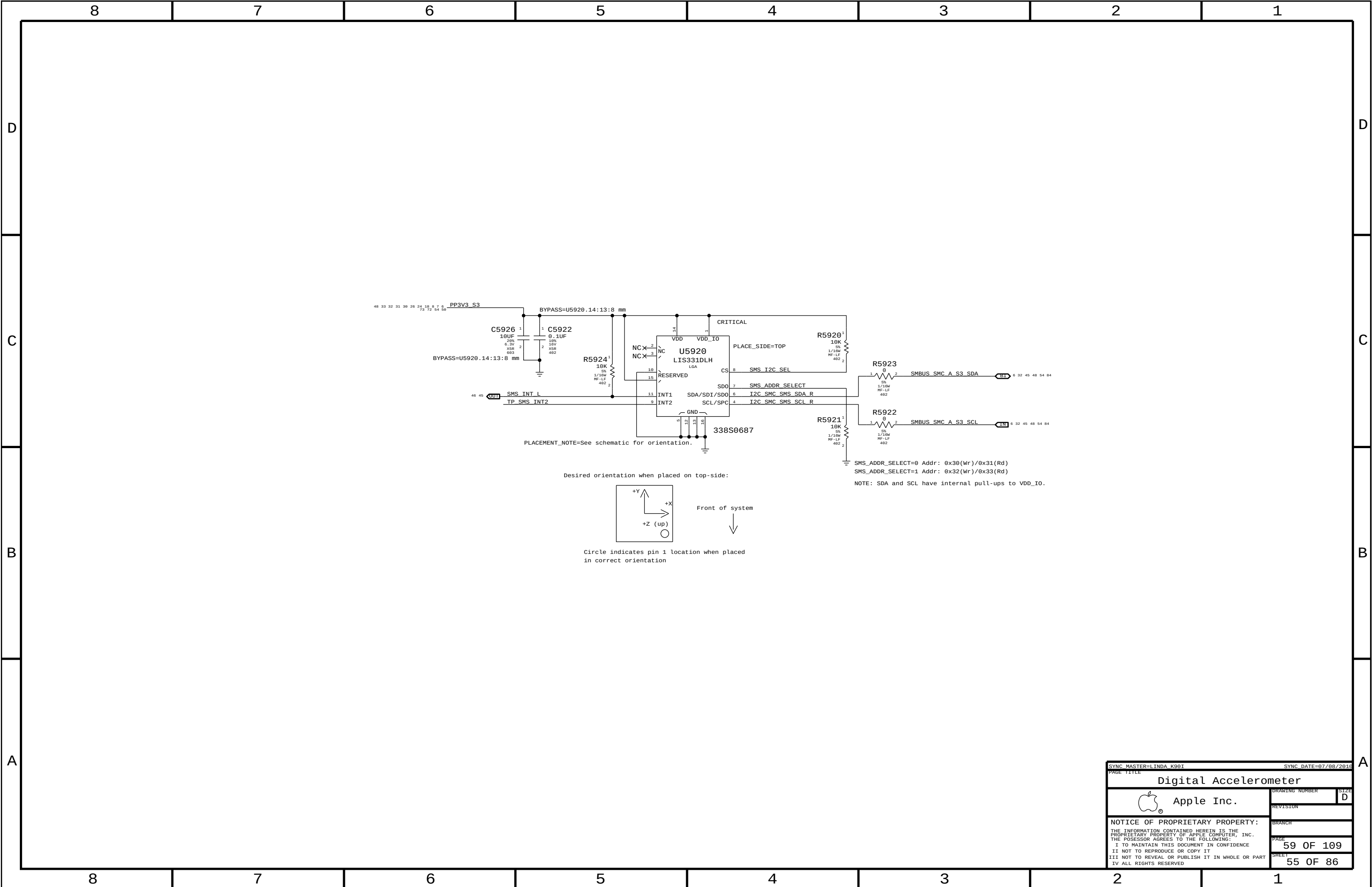
BOOSTER DESIGN CONSIDERATION:

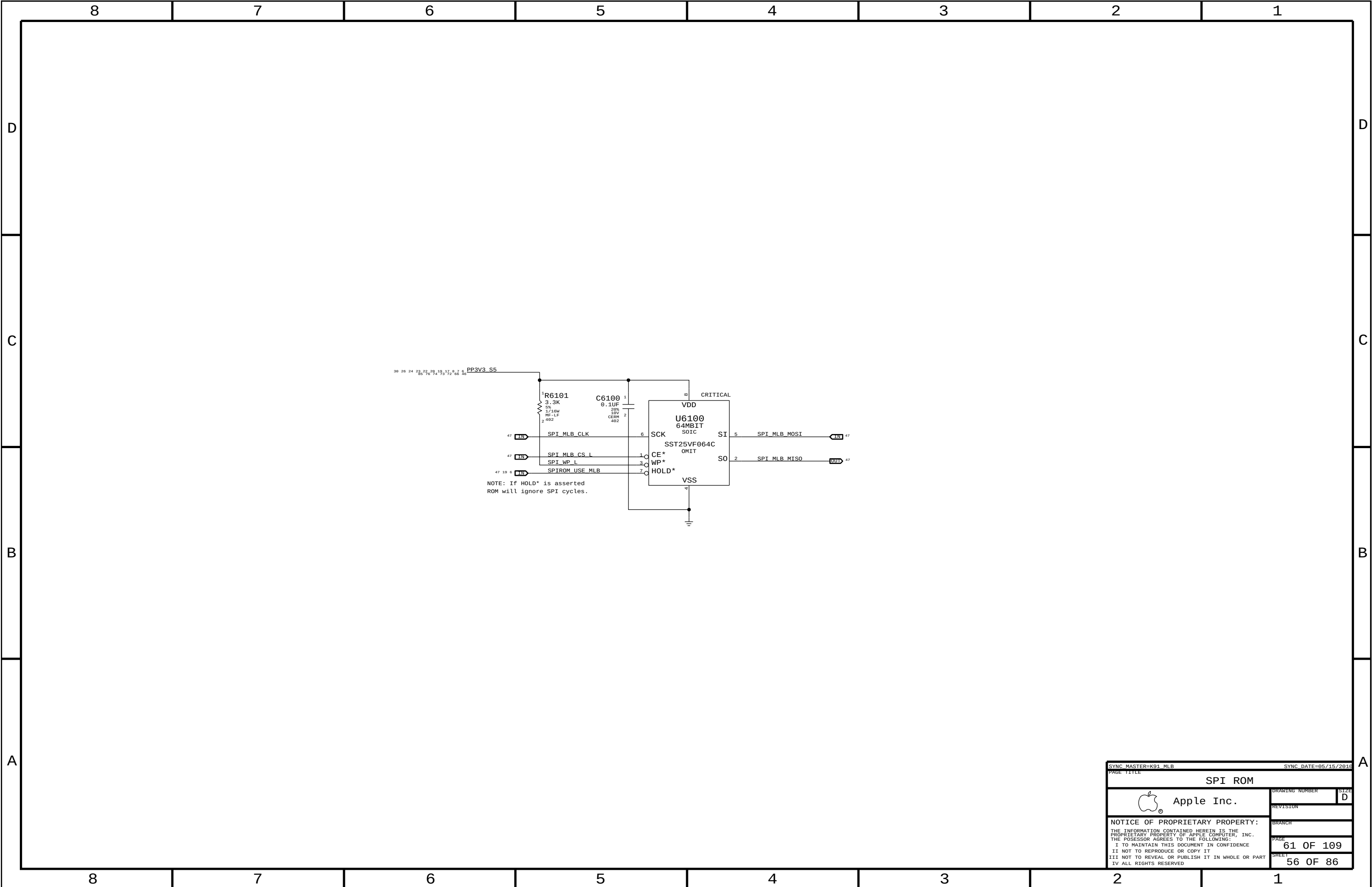
- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812,R5813,C5818 MODIFIED

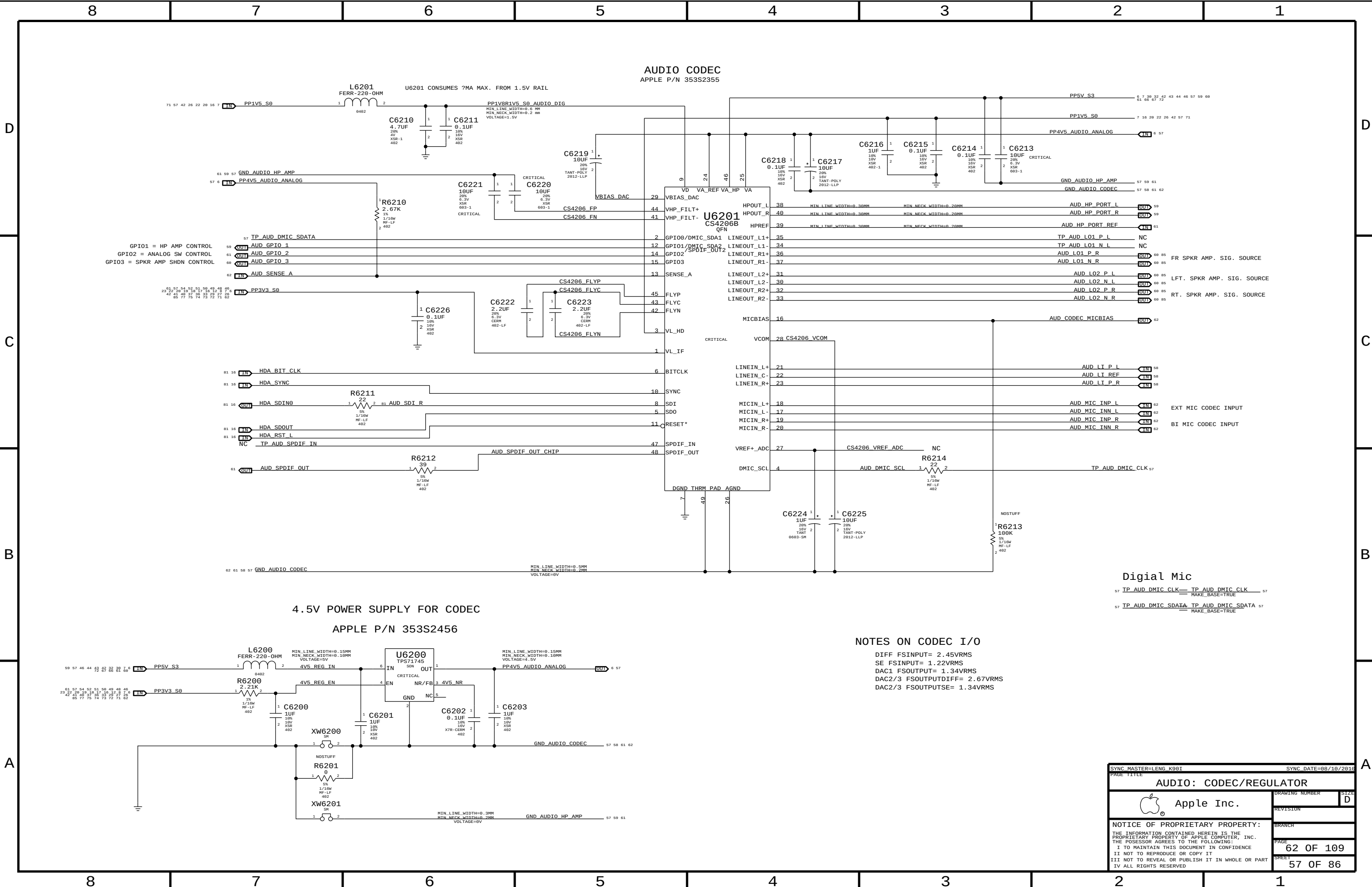


K6 NOTES : C5850 HAS BYPASS PROPERTY, SHOULD BE ADDED INCASE THIS PAGE IS SYNC'ED FROM T27

SYNC MASTER=LINDA K90I		SYNC DATE=07/12/2016	
PAGE TITLE		A	
WELLSPRING 2			
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ALL RIGHTS RESERVED			







NOTES ON CODEC I/O

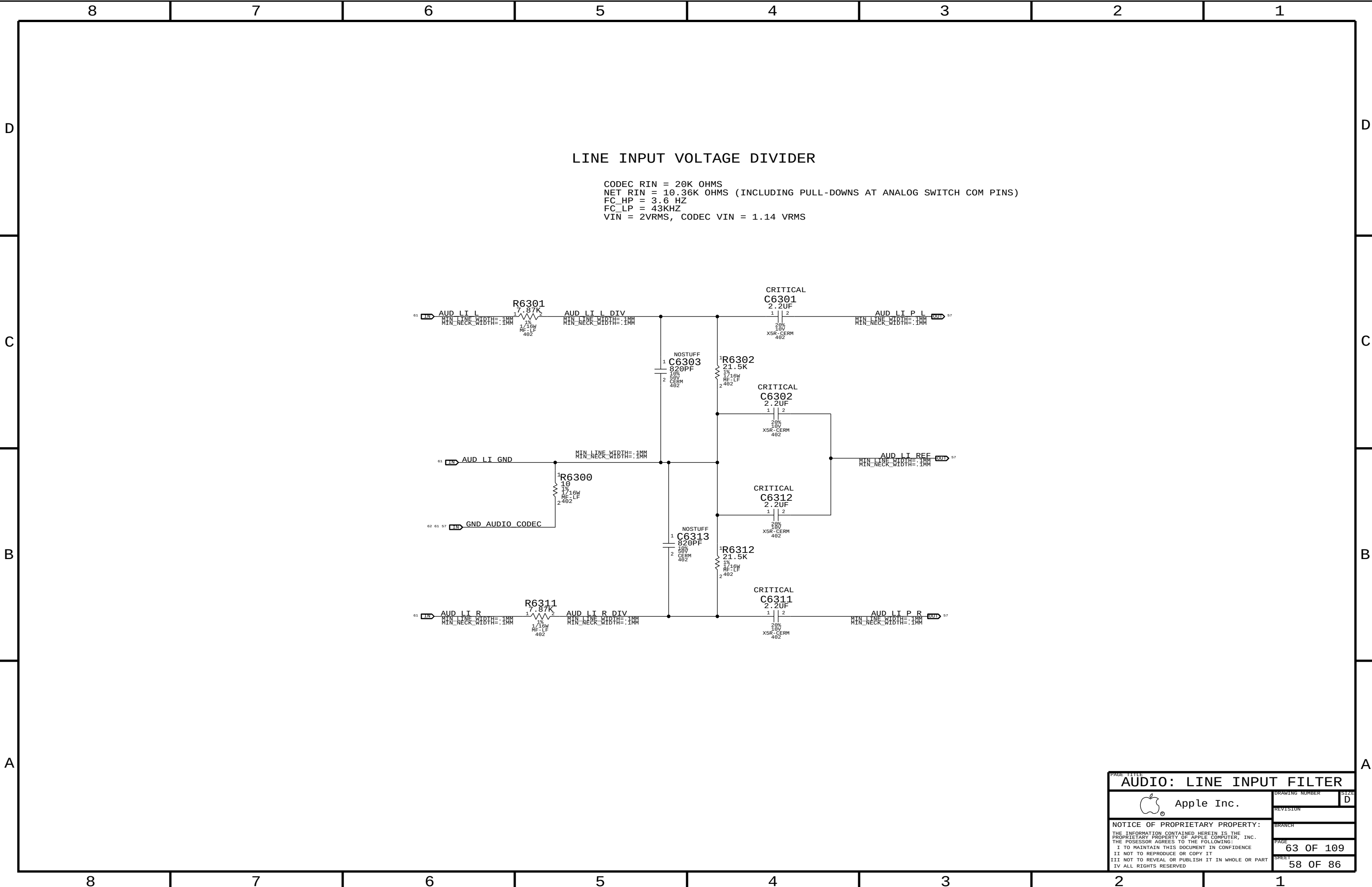
DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

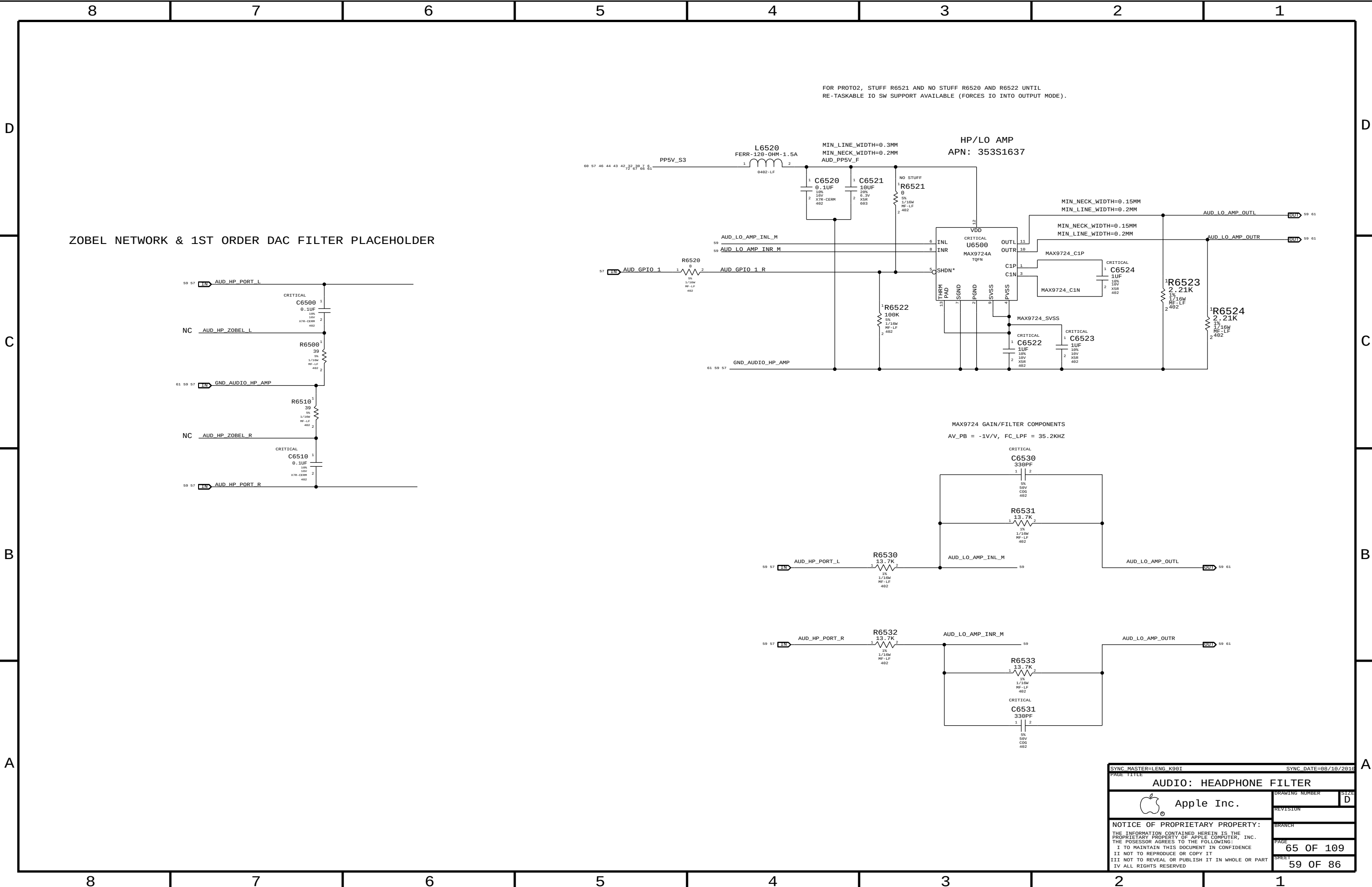
Digital Mic

57 TP_AUD_DMIC_CLK TP_AUD_DMIC_CLK 57
MAKE_BASE=TRUE

57 TP_AUD_DMIC_SDATA TP_AUD_DMIC_SDATA 57
MAKE_BASE=TRUE

SYNC MASTER=LENG K901		SYNC DATE=08/10/2010	
PAGE TITLE			
AUDIO: CODEC/REGULATOR			
Apple Inc.		DRAWING NUMBER	SIZE
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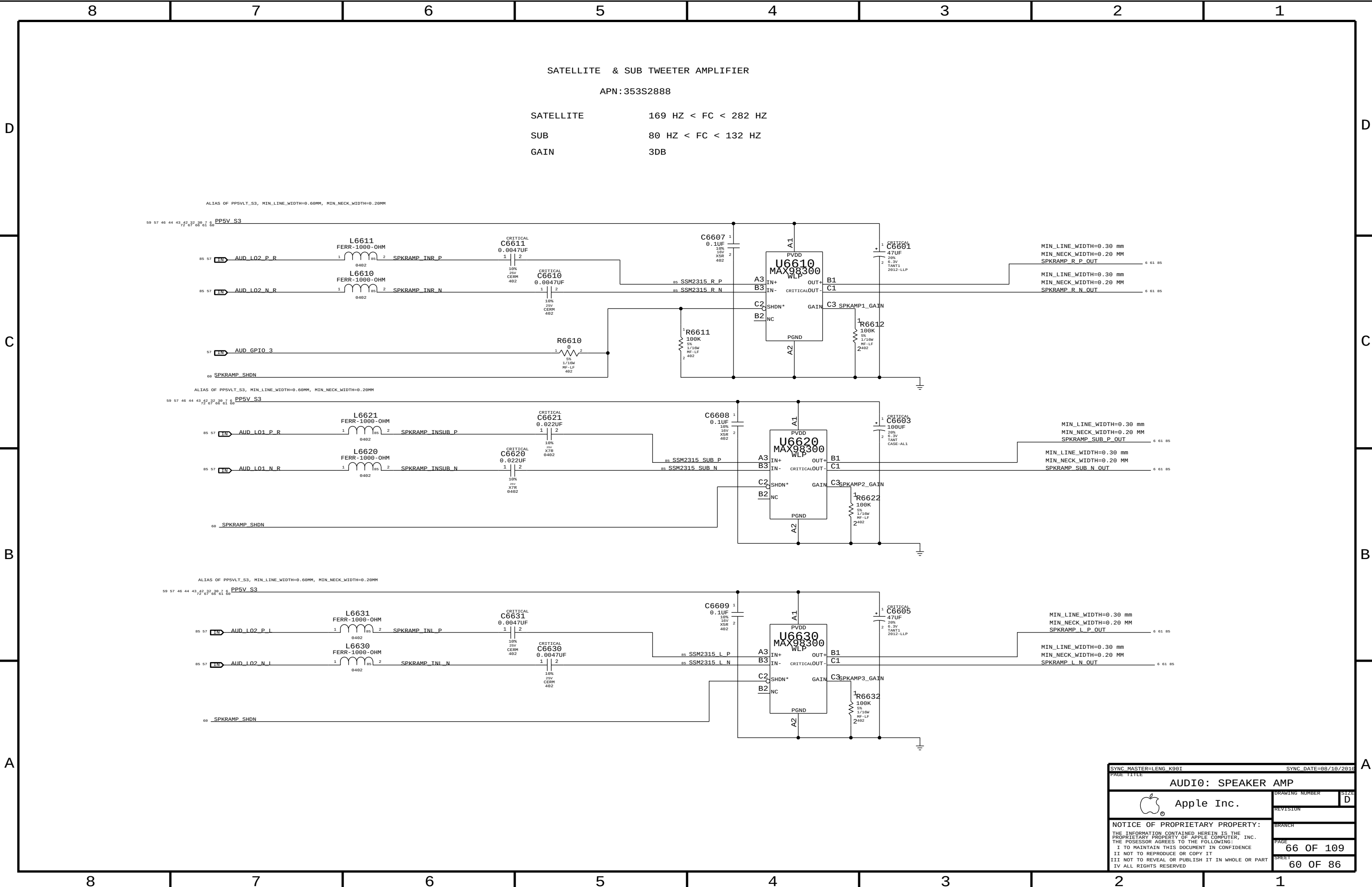


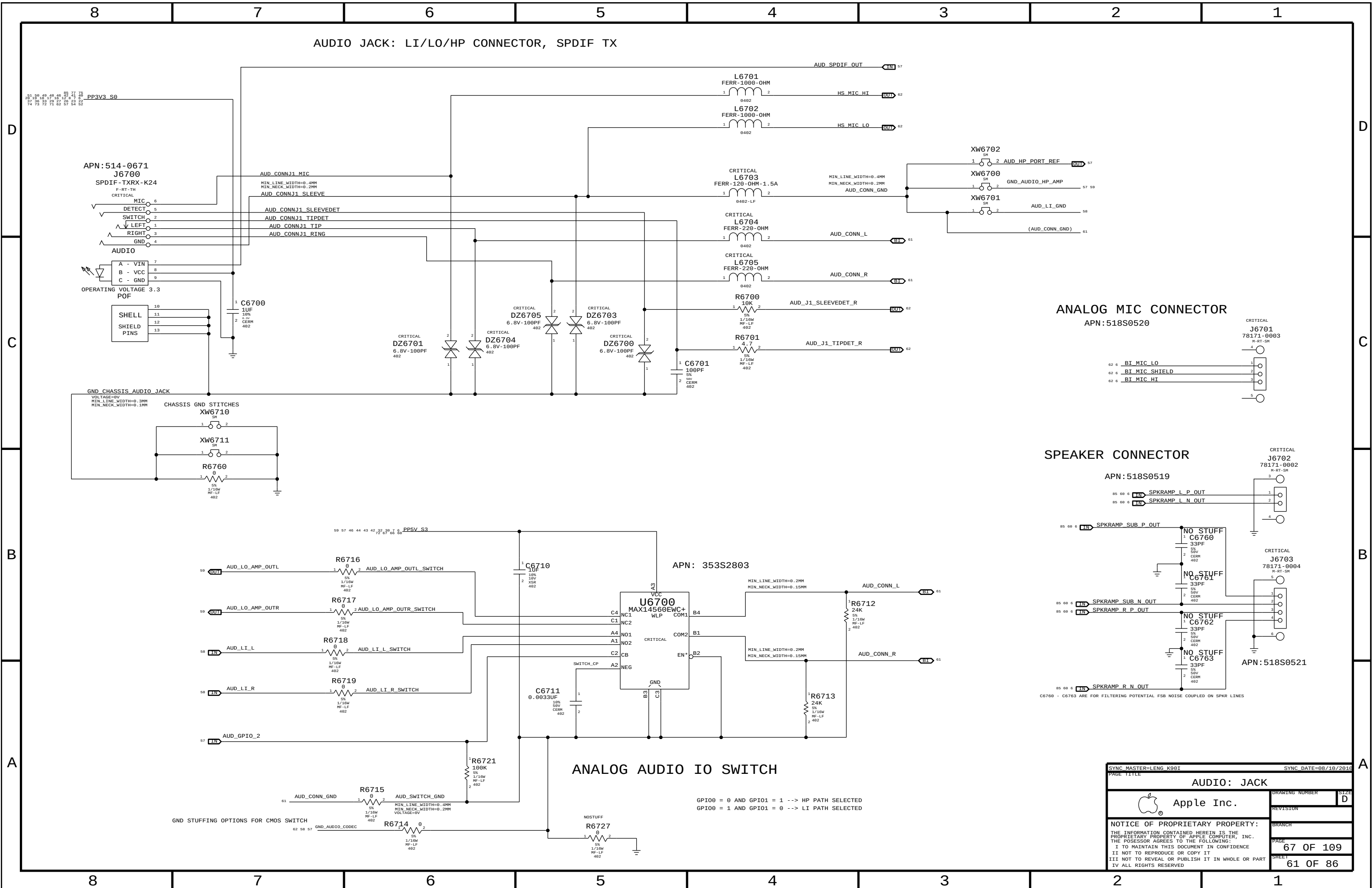
FOR PROT02, STUFF R6521 AND NO STUFF R6520 AND R6522 UNTIL
RE-TASKABLE IO SW SUPPORT AVAILABLE (FORCES IO INTO OUTPUT MODE).

HP/LO AMP
APN: 353S1637

MAX9724 GAIN/FILTER COMPONENTS
AV_PB = -1V/V, FC_LPF = 35.2KHZ

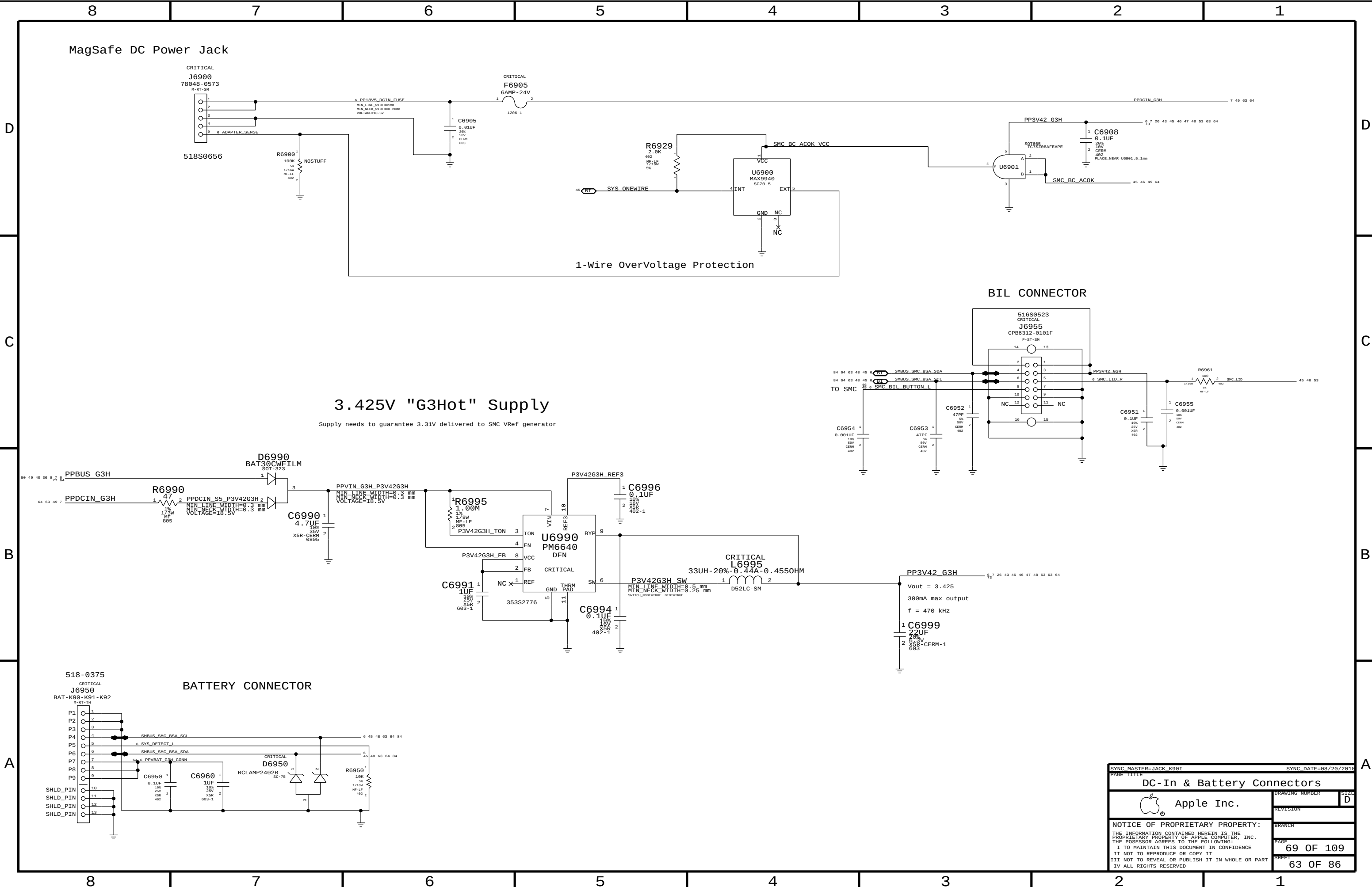
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PAGE TITLE			
AUDIO: HEADPHONE FILTER			
 Apple Inc.		DRAWING NUMBER	SIZE
			D
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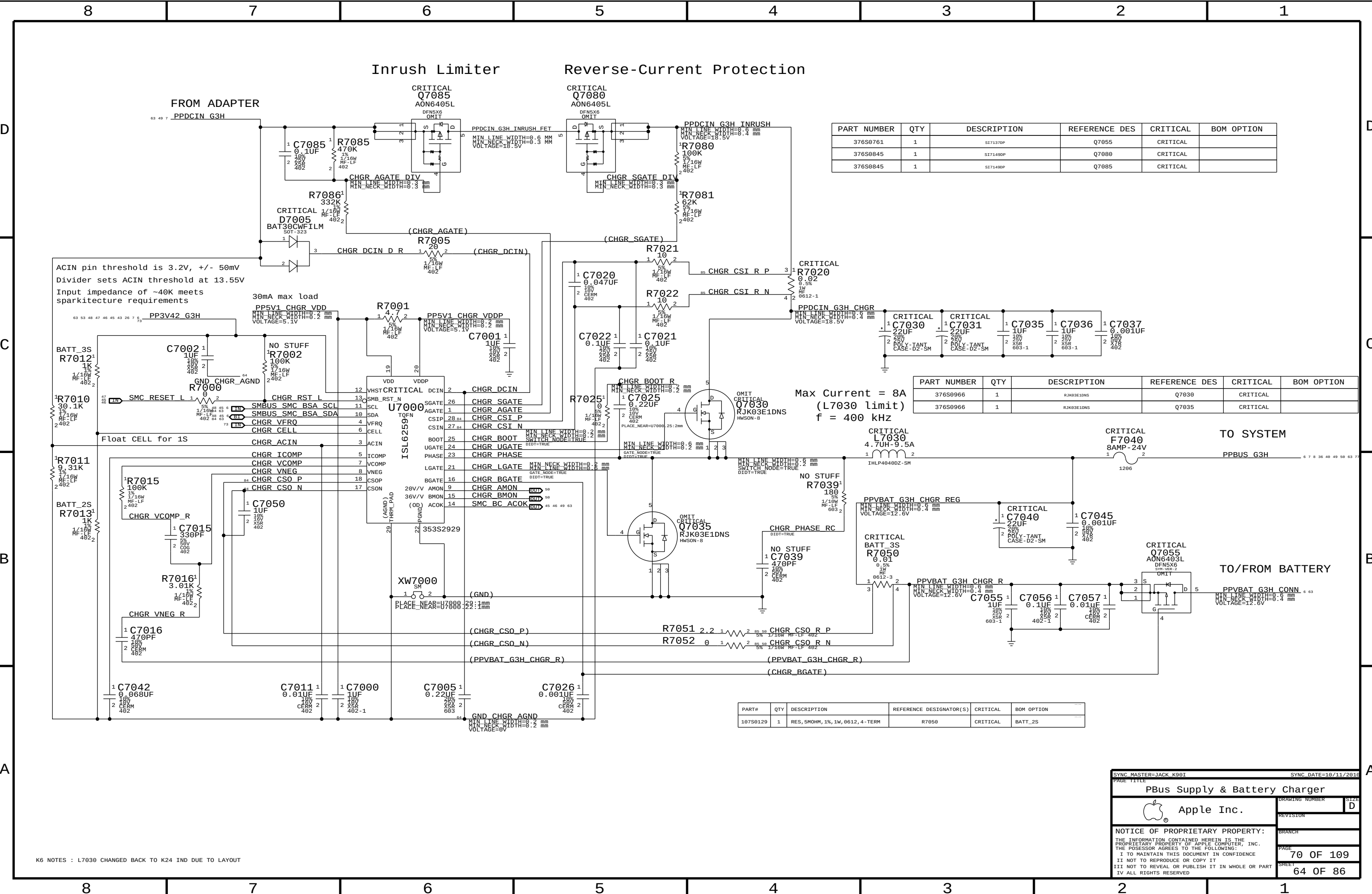




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Apple Inc.		DRAWING NUMBER	SIZE D
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SYNCS MASTER-LENG K90I		SYNC DATE=08/10/2016	
PAGE TITLE			
AUDIO: JACK TRANSLATORS			
 Apple Inc.		DRAWING NUMBER	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0761	1	SL7137DP	Q7055	CRITICAL	
376S0845	1	SL7149DP	Q7080	CRITICAL	
376S0845	1	SL7149DP	Q7085	CRITICAL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0966	1	R3K03E1DNS	Q7030	CRITICAL	
376S0966	1	R3K03E1DNS	Q7035	CRITICAL	

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
107S0129	1	RES, 5MOHM, 1%, 1W, 0612, 4-TERM	R7050	CRITICAL	BATT_2S

SYNC MASTER=JACK K901

SYNC DATE=10/11/2010

PAGE TITLE

PBus Supply & Battery Charger

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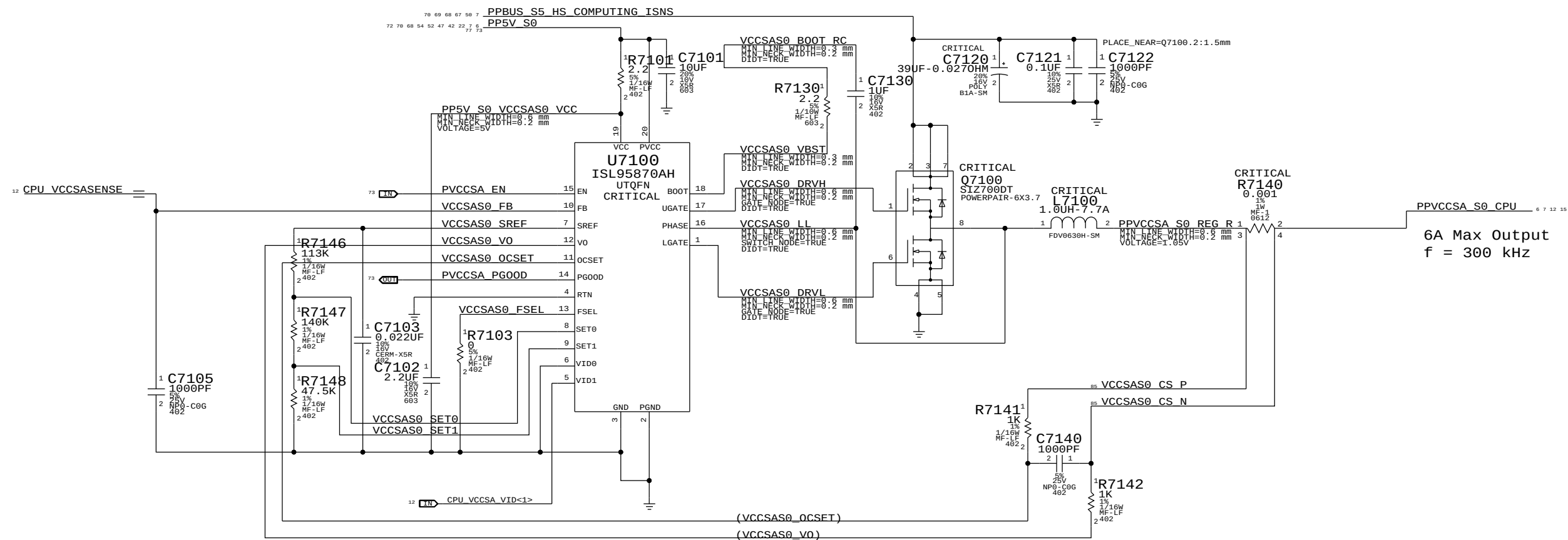
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K6 NOTES : L7030 CHANGED BACK TO K24 IND DUE TO LAYOUT

System Agent Power Supply



5V_S3/3.3V_S5 POWER SUPPLY

VOUT = (2 * RA / RB) + 2 VOUT = (2 * RC / RD) + 2

MAX CURRENT = 12.947A
PWM FREQ. = 300 KHZ

PWM FREQ. = 375 KHZ
MAX CURRENT = 7.45A

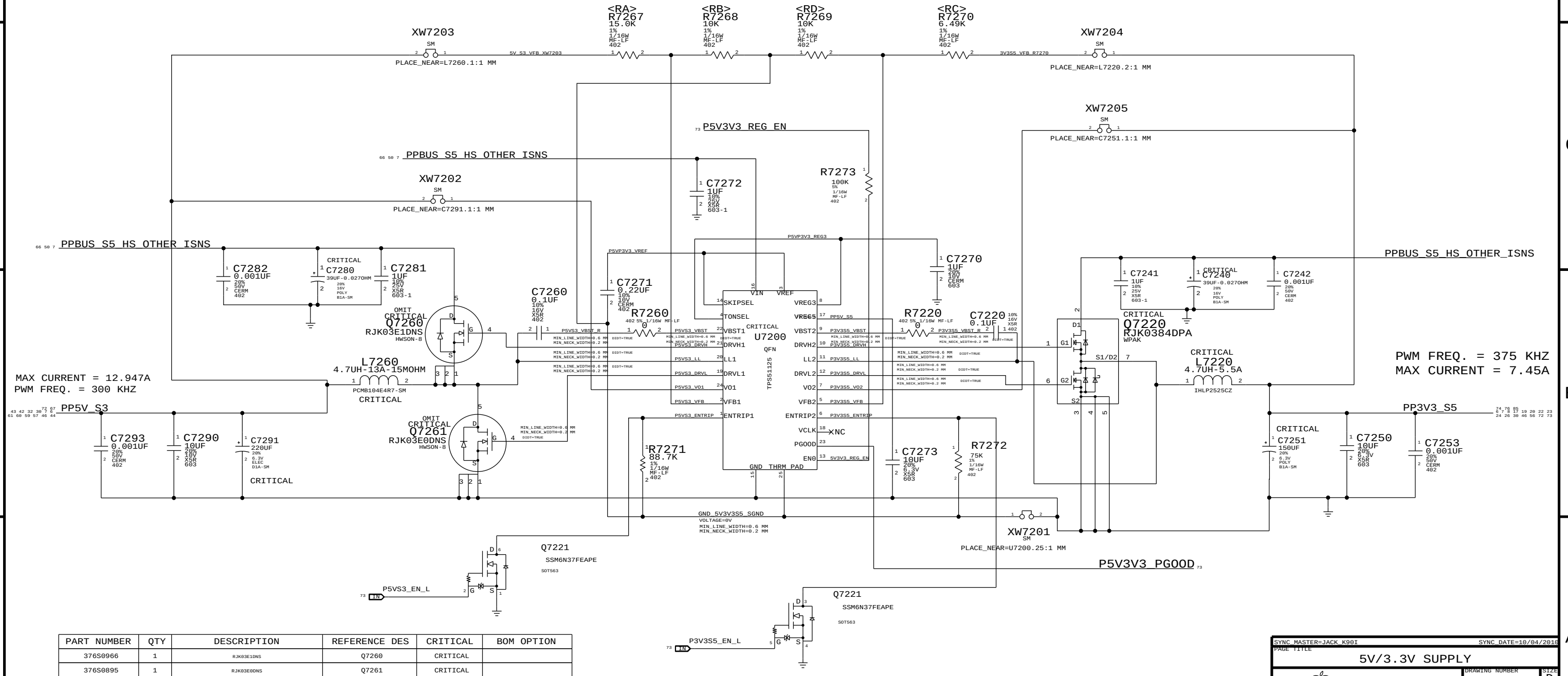
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S9966	1	RJK03E1DNS	Q7260	CRITICAL	
376S9895	1	RJK03E0DNS	Q7261	CRITICAL	

SEPERATED MASTER PG00D FOR BOTH 5V AND 3V3.

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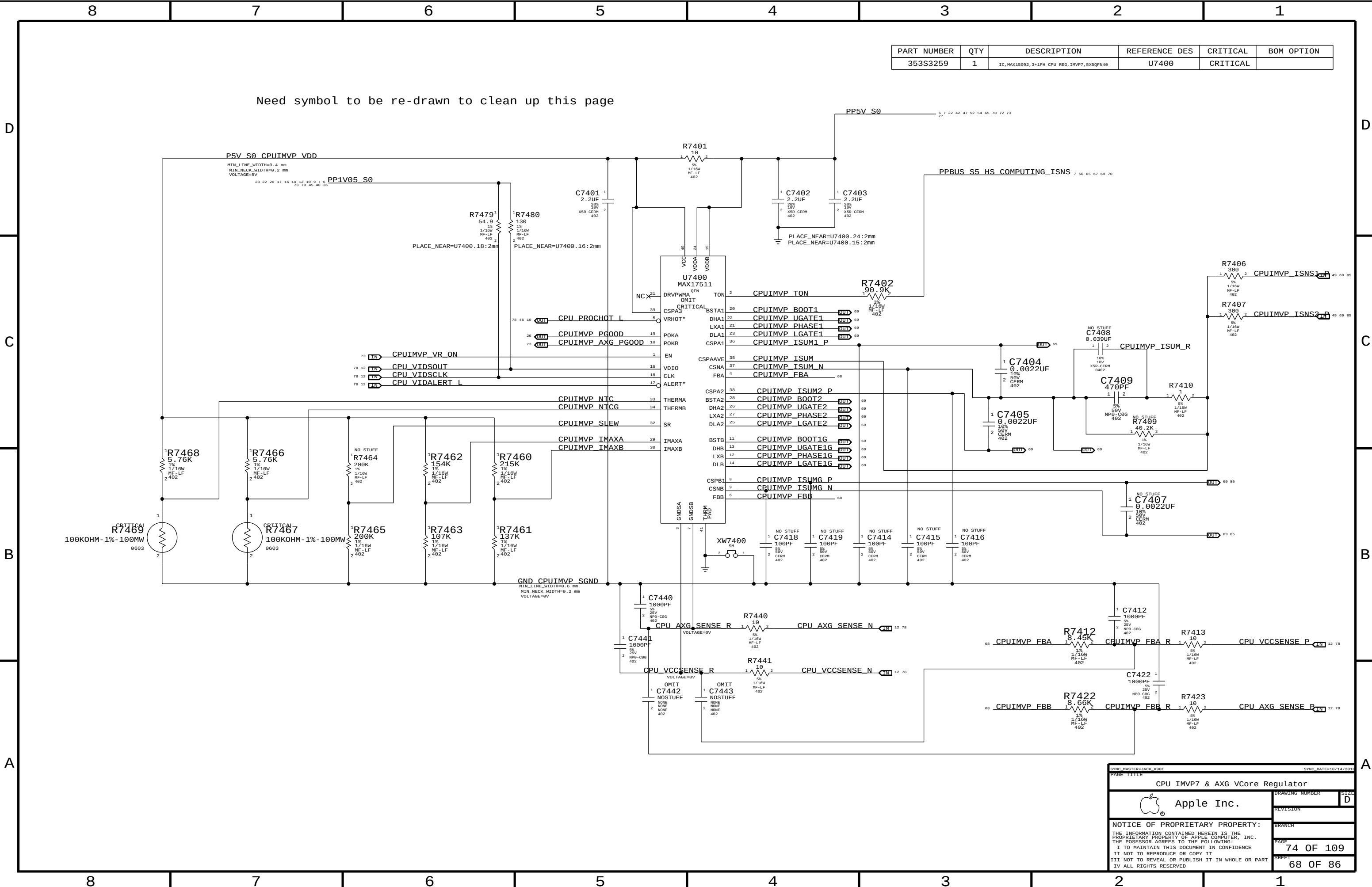
$$V_{OUT} = (2 * R_C / R_D) + 2$$

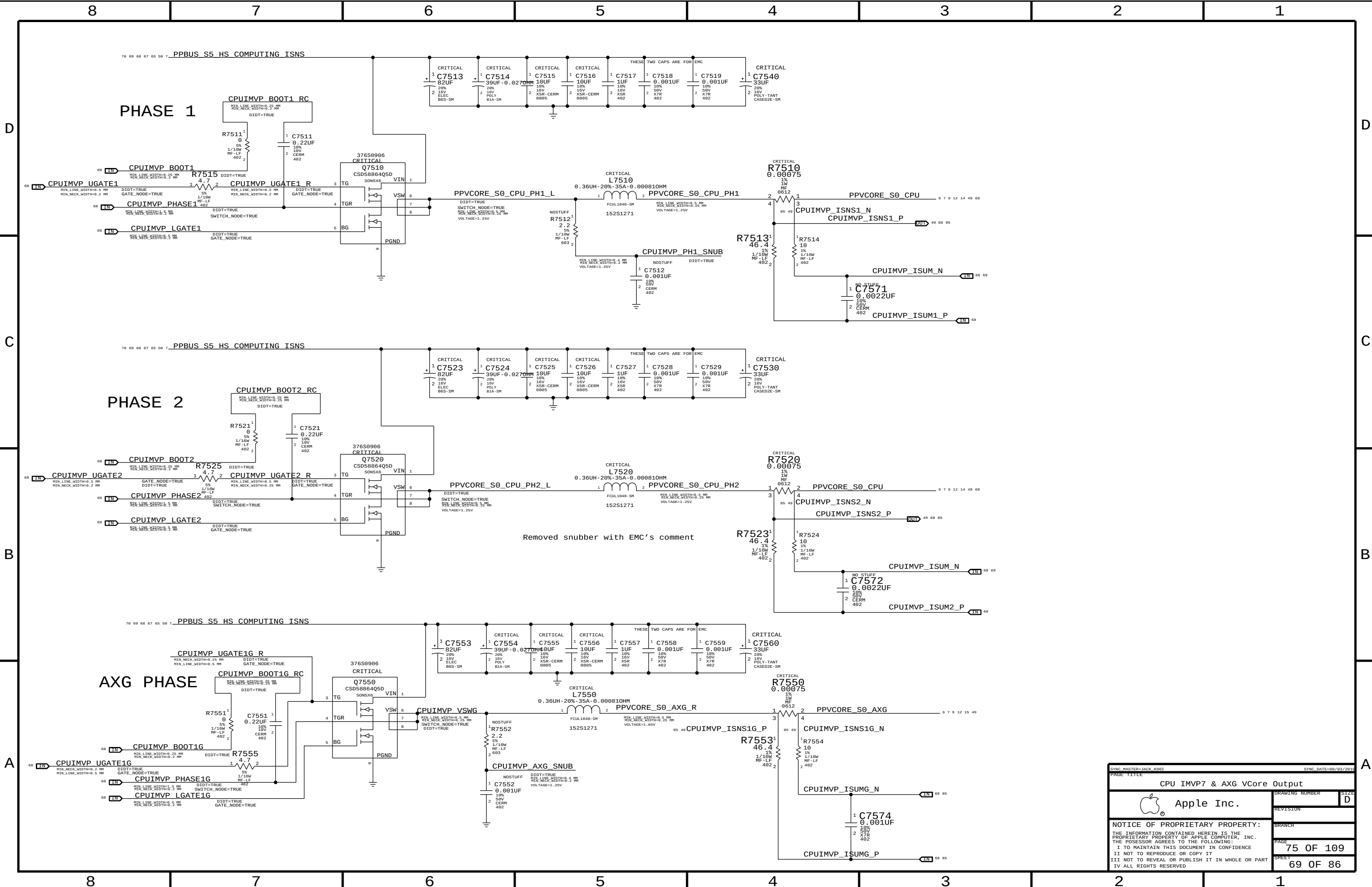


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0966	1	RJK03E1D05	Q7260	CRITICAL	
376S0895	1	RJK03E0D05	Q7261	CRITICAL	

SEPERATED MASTER PG00D FOR BOTH 5V AND 3V3.

SYNC MASTER=JACK K901		SYNC DATE=10/04/2010	
PAGE TITLE			
5V/3.3V SUPPLY			
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CPU IMVP7 & AXG VCore Output	
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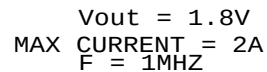
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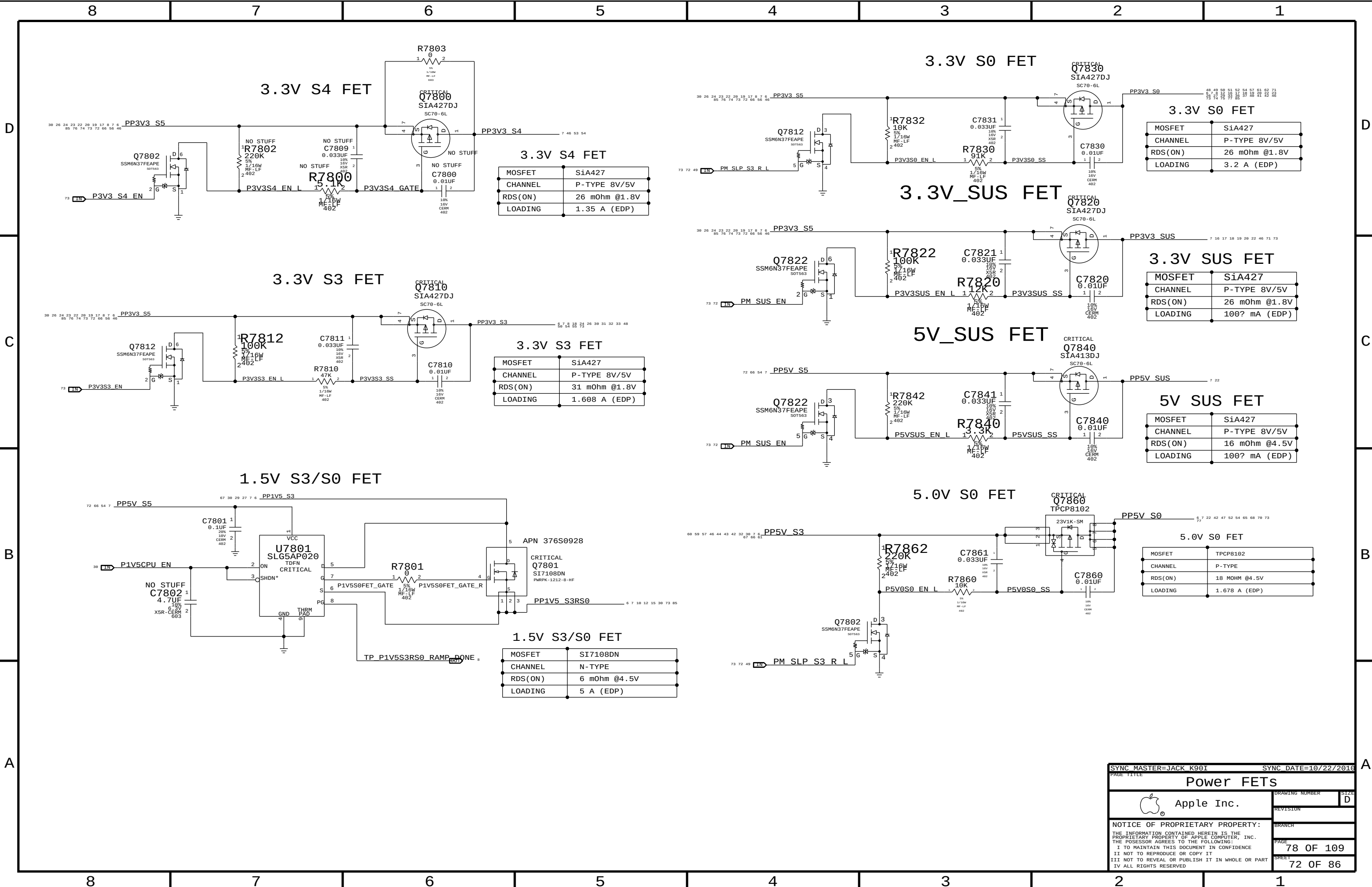
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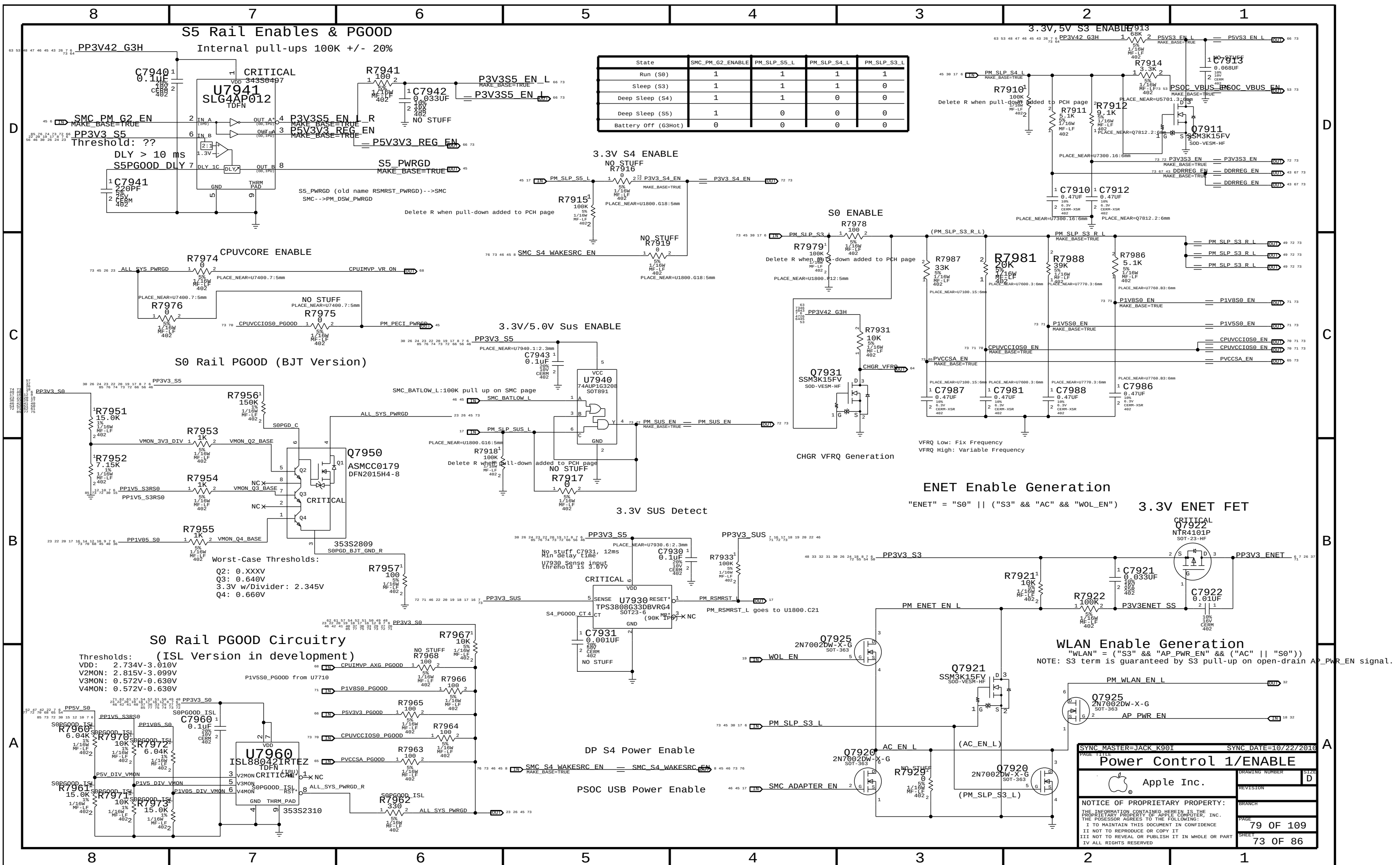


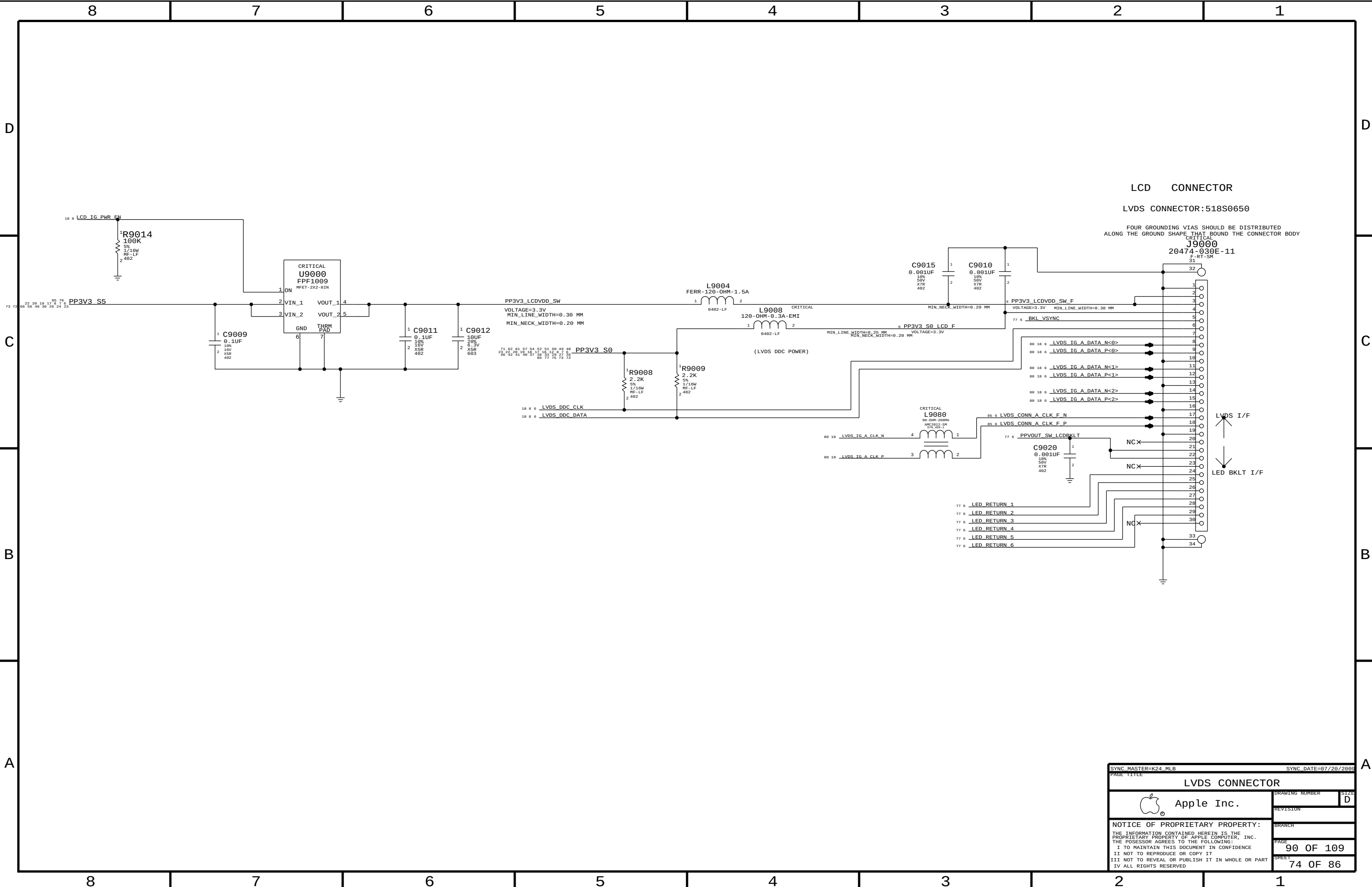
CRITICAL
YDB, BCU

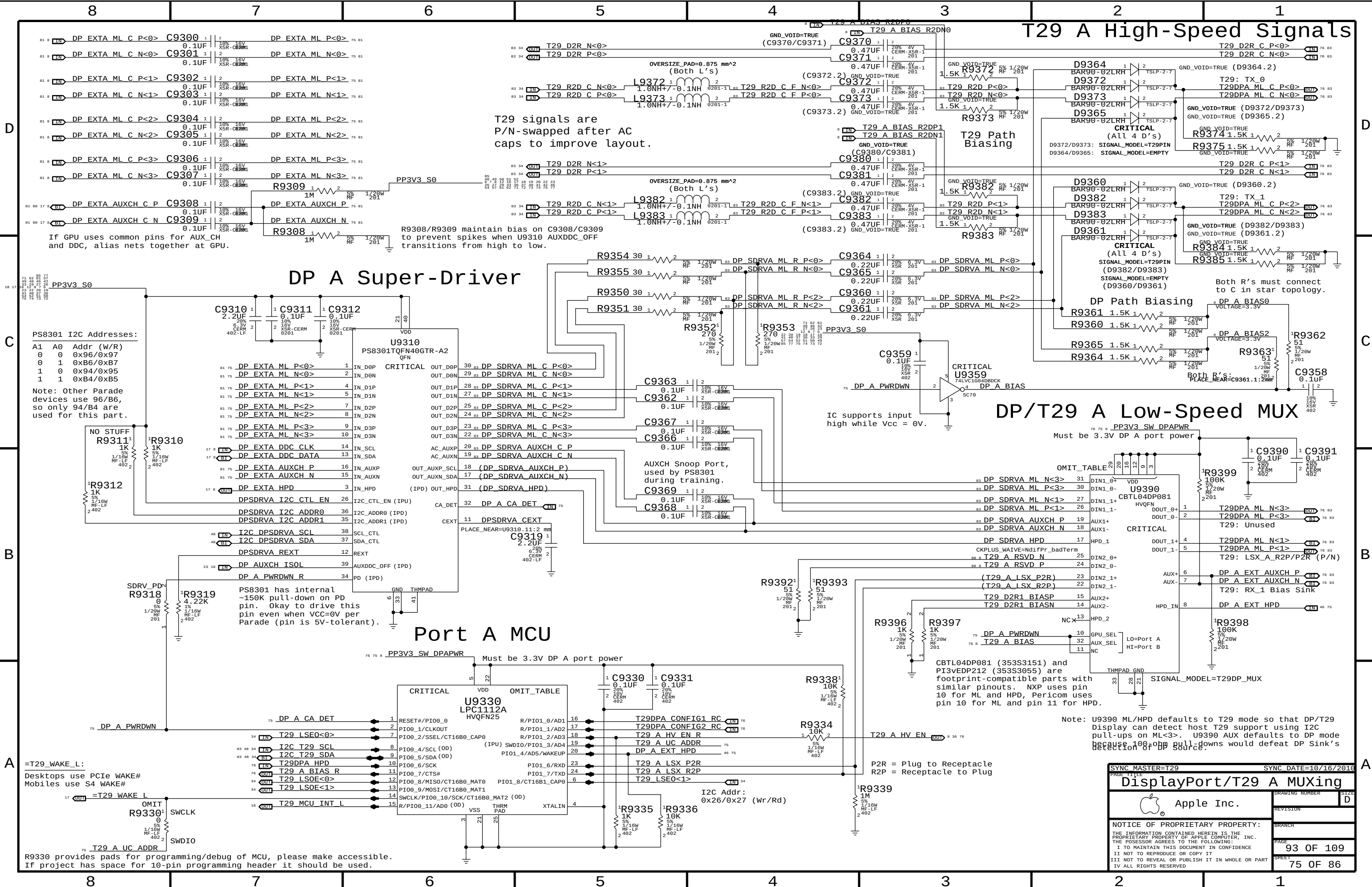


SYNC MASTER=JACK K90I		SYNC DATE=08/19/2010	
PAGE TITLE			
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		REVISION 	
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T29 signals are P/N-swapped after AC caps to improve layout.

R9308/R9309 maintain bias on C9308/C9309 to prevent spikes when U9310 AUXDDC_OFF transitions from high to low.

PS8301 I2C Addresses:
A1 A0 Addr (W/R)
0 0 0x96/0x97
0 1 0xB6/0xB7
1 0 0x94/0x95
1 1 0xB4/0xB5
Note: Other Parade devices use 96/B6, so only 94/B4 are used for this part.

PS8301 has internal ~150K pull-down on PD pin. Okay to drive this pin even when VCC=0V per Parade (pin is 5V-tolerant).

=T29_WAKE_L:
Desktops use PCIe WAKE#
Mobiles use S4 WAKE#

R9330 provides pads for programming/debug of MCU, please make accessible.
If project has space for 10-pin programming header it should be used.

T29 A High-Speed Signals

DP A Super-Driver

DP/T29 A Low-Speed MUX

Port A MCU

SYNC MASTER=T29

SYNC DATE=10/16/2016

DisplayPort/T29 A MUXing

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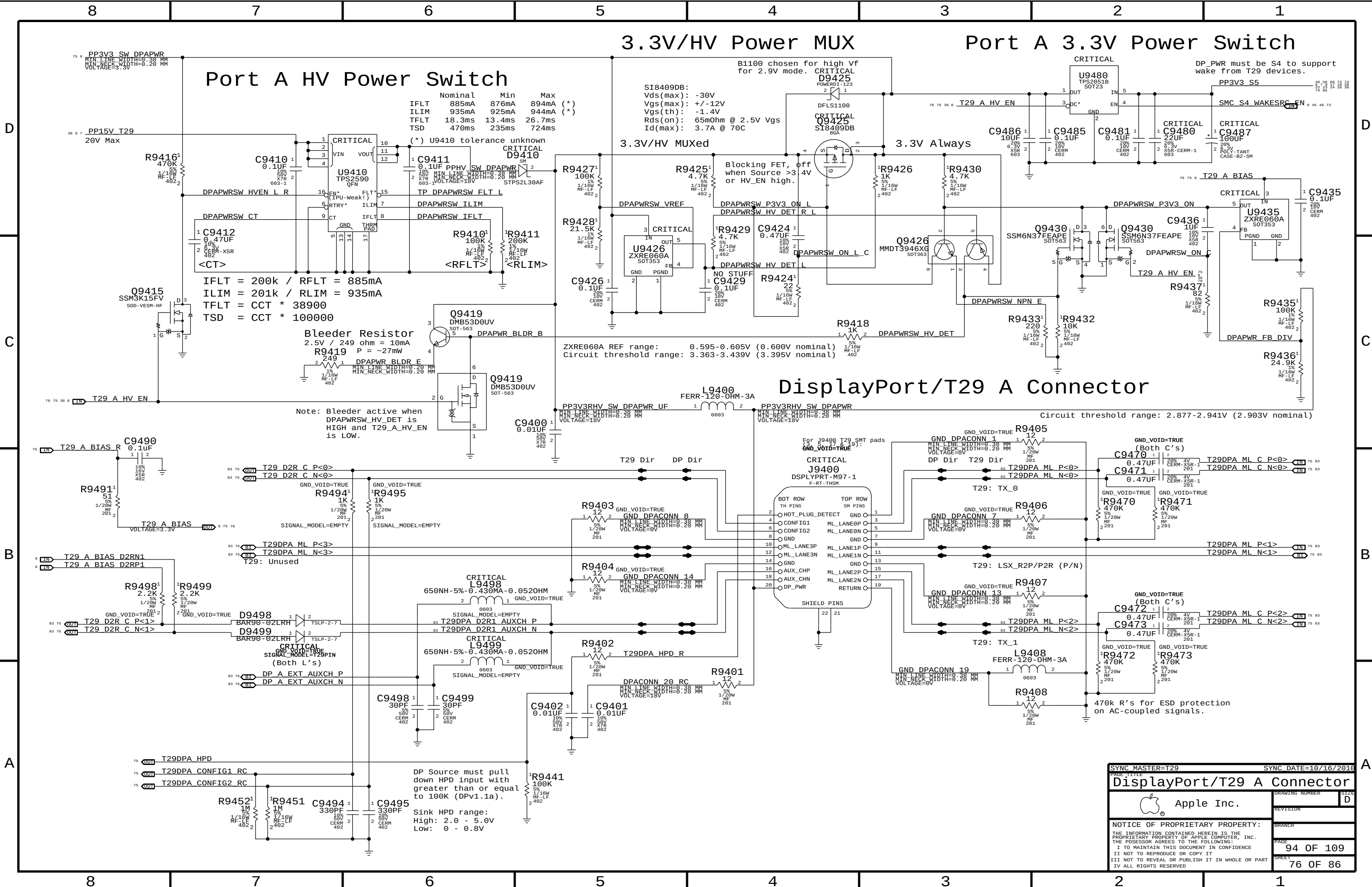
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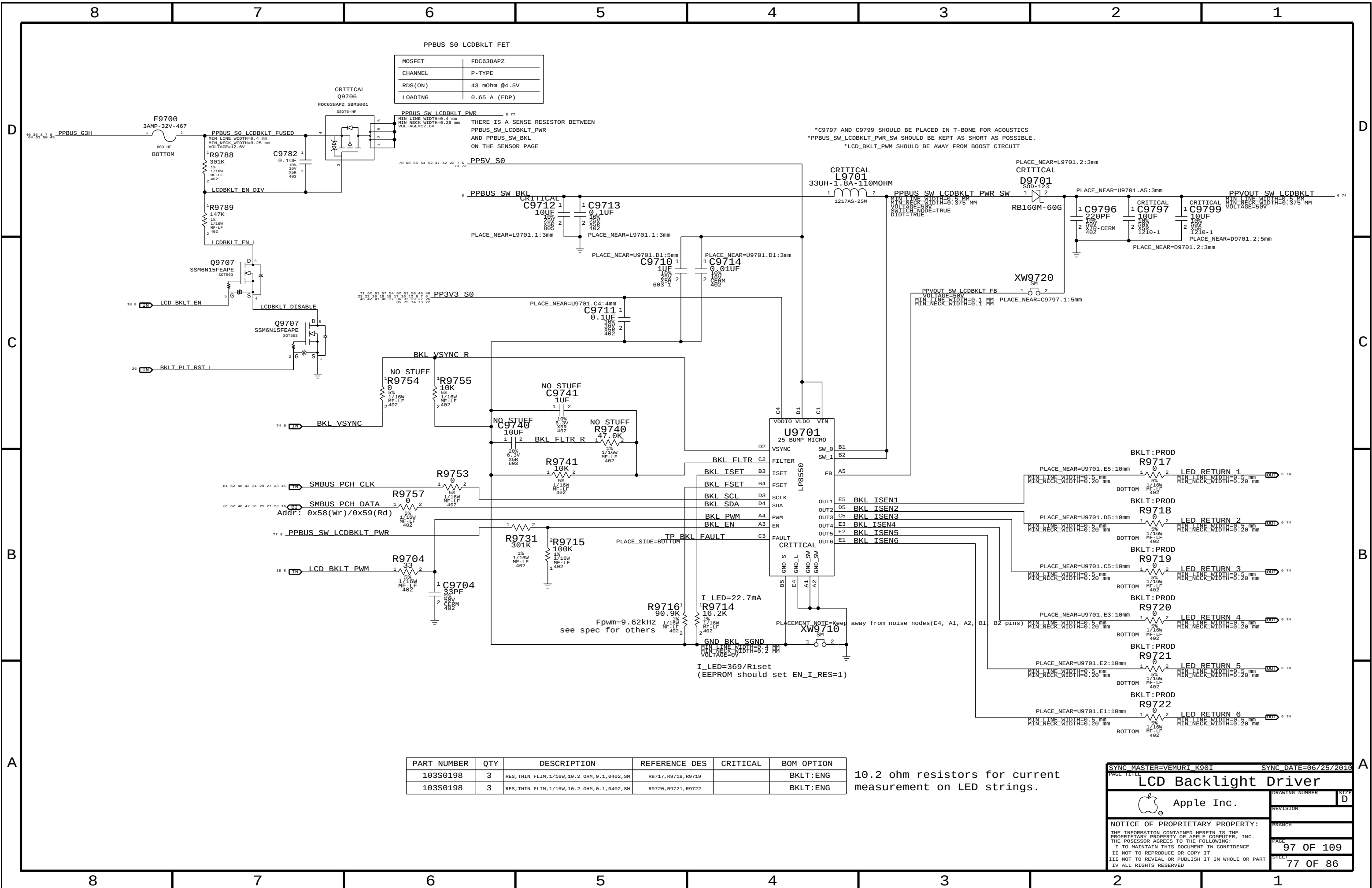
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719		BKLT:ENG
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9720, R9721, R9722		BKLT:ENG

10.2 ohm resistors for current measurement on LED strings.

SYNC MASTER=VEMURI K901		SYNC DATE=06/25/2016	
PAGE TITLE		LCD Backlight Driver	
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Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_900	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?	LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMD5 intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMD5 traces: 12 inches.
SOURCE: MCP79 LVDS Data DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=4x_DIELECTRIC	?
SATA_ICOMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.7.1.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.8

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	DP_H1	DP_850	DISPLAYPORT	DP IG ML P<3..0>	8
	DP_H1	DP_850	DISPLAYPORT	DP IG ML N<3..0>	8
	DP_EXTA_AUXCH	DP_850	DISPLAYPORT	DP_EXTA_AUXCH C P	8 17 75 81
	DP_EXTA_AUXCH	DP_850	DISPLAYPORT	DP_EXTA_AUXCH C N	8 17 75 81
	LVDS_IG_A_CLK	LVDS_900	LVDS	LVDS_IG_A_CLK P	18 74
	LVDS_IG_A_CLK	LVDS_900	LVDS	LVDS_IG_A_CLK N	18 74
	LVDS_IG_A_DATA	LVDS_900	LVDS	LVDS_IG_A_DATA P<2..0>	6 18 74
	LVDS_IG_A_DATA	LVDS_900	LVDS	LVDS_IG_A_DATA N<2..0>	6 18 74
		LVDS_900	LVDS	NC LVDS_IG_A_DATAP<3>	8 18
		LVDS_900	LVDS	NC LVDS_IG_A_DATAN<3>	8 18
		LVDS_900	LVDS	LVDS_IG_B_DATA P<3..0>	8
		LVDS_900	LVDS	LVDS_IG_B_DATA N<3..0>	8
		LVDS_900	LVDS	TP LVDS_IG_B_CLKP	6 8 18
		LVDS_900	LVDS	TP LVDS_IG_B_CLKN	6 8 18
	SATA_HDD_R2D	SATA_900	SATA	SATA_HDD_R2D C P	16 42
	SATA_HDD_R2D	SATA_900	SATA	SATA_HDD_R2D C N	16 42
	SATA_HDD_R2D_CONN	SATA_900	SATA	SATA_HDD_R2D P	6 42
	SATA_HDD_R2D_CONN	SATA_900	SATA	SATA_HDD_R2D N	6 42
	SATA_HDD_D2R	SATA_900	SATA	SATA_HDD_D2R P	16 42
	SATA_HDD_D2R	SATA_900	SATA	SATA_HDD_D2R N	16 42
	SATA_HDD_D2R_CONN	SATA_900	SATA	SATA_HDD_D2R C P	6 42
	SATA_HDD_D2R_CONN	SATA_900	SATA	SATA_HDD_D2R C N	6 42
	SATA_ODD_R2D	SATA_900	SATA	SATA_ODD_R2D C P	16 42
	SATA_ODD_R2D	SATA_900	SATA	SATA_ODD_R2D C N	16 42
	SATA_ODD_R2D	SATA_900	SATA	SATA_ODD_R2D P	6 42
	SATA_ODD_R2D	SATA_900	SATA	SATA_ODD_R2D N	6 42
	SATA_ODD_D2R	SATA_900	SATA	SATA_ODD_D2R P	16 42
	SATA_ODD_D2R	SATA_900	SATA	SATA_ODD_D2R N	16 42
	SATA_HDD_R2D_CONN	SATA_900	SATA	SATA_HDD_R2D RC P	42
	SATA_HDD_R2D_CONN	SATA_900	SATA	SATA_HDD_R2D RC N	42
	SATA_HDD_D2R_CONN	SATA_900	SATA	SATA_HDD_D2R RC P	42
	SATA_HDD_D2R_CONN	SATA_900	SATA	SATA_HDD_D2R RC N	42
	PCH_SATA_ICOMP		SATA_ICOMP	PCH_SATAICOMP	16
	USB_HUB1_UP	USB_850	USB	USB_HUB1_UP P	18 24
		USB_850	USB	USB_HUB1_UP N	18 24
	USB_HUB2_UP	USB_850	USB	USB_HUB2_UP P	18 24
		USB_850	USB	USB_HUB2_UP N	18 24
	USB_EXTA	USB_850	USB	USB_EXTA P	24 43
	USB_EXTA	USB_850	USB	USB_EXTA N	24 43
	USB_EXTB	USB_850	USB	USB_EXTB P	24 43
		USB_850	USB	USB_EXTB N	24 43
	USB_EXTC	USB_850	USB	USB_EXTC P	8 24
		USB_850	USB	USB_EXTC N	8 24
	USB_EXTD	USB_850	USB	USB_T29A P	8 24
		USB_850	USB	USB_T29A N	8 24
		USB_850	USB	T29_A_RSVD P	8 75
		USB_850	USB	T29_A_RSVD N	8 75
	USB_CAMERA	USB_850	USB	USB_CAMERA P	18 32
		USB_850	USB	USB_CAMERA N	18 32
	USB_CAMERA	USB_850	USB	USB_CAMERA_CONN P	6 32
		USB_850	USB	USB_CAMERA_CONN N	6 32
	USB_BT	USB_850	USB	USB_BT P	6 24 32
	USB_BT	USB_850	USB	USB_BT N	6 24 32
	USB_TP4D	USB_850	USB	USB_TP4D P	24 53
		USB_850	USB	USB_TP4D N	24 53
	USB_IR	USB_850	USB	USB_IR P	24 44
		USB_850	USB	USB_IR N	24 44
	USB_SDCARD	USB_850	USB	USB_SDCARD P	
		USB_850	USB	USB_SDCARD N	
	USB_BRCRYPT	USB_850	USB	USB_BRCRYPT P	
		USB_850	USB	USB_BRCRYPT N	
	PCH_USB_RBIAS	PCH_USB_RBIAS		PCH_USB_RBIAS	18
	PCH_DIEFCLK_UNUSED	CLK_PCTE_900	CLK_PCTE	PCIE_CLK100M_PCH_P	16 25
	PCH_DIEFCLK_UNUSED	CLK_PCTE_900	CLK_PCTE	PCIE_CLK100M_PCH_N	16 25
		CLK_PCTE_900	CLK_PCTE	NC_FSB_CLK133M_PCH_P	8
		CLK_PCTE_900	CLK_PCTE	NC_FSB_CLK133M_PCH_N	8
	PCH_DIEFCLK_UNUSED	CLK_PCTE_900	CLK_PCTE	PCH_CLK90M_DOT_P	16 25
	PCH_DIEFCLK_UNUSED	CLK_PCTE_900	CLK_PCTE	PCH_CLK90M_DOT_N	16 25
	PCH_DIEFCLK_UNUSED	CLK_PCTE_900	CLK_PCTE	PCH_CLK100M_SATA_P	16 25
	PCH_DIEFCLK_UNUSED	CLK_PCTE_900	CLK_PCTE	PCH_CLK100M_SATA_N	16 25
		CDU_50S	CLK_PCTE	PCH_CLK14P3M_REFCLK	16 25
		CDU_50S	CLK_PCTE	PCH_CLK33M_PCIIN	16 26
	GFX_CLK_DPLLSS	CLK_PCTE_900	CLK_PCTE	GFX_CLK120M_DPLLSS_P	
	GFX_CLK_DPLLSS	CLK_PCTE_900	CLK_PCTE	GFX_CLK120M_DPLLSS_N	

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PCH Constraints 1			
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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

DisplayPort Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_850	*	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?

PCI-Express Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_98D	*	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	LPC_AD	LPC_50S	LPC	LPC_AD<3..0>	6 16 45 47
	LPC_FRAME_L	LPC_50S	LPC	LPC_FRAME_L	6 16 45 47
	LPC_RESET_L	LPC_50S	LPC	LPCPLUS_RESET_L	6 26 47
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R	16 26
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC	26 45
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS	6 26 47
	SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS_PCH_CLK	16 23 27 29 31 42 48 62 77
	SMBUS_PCH_DATA	SMB_50S	SMB	SMBUS_PCH_DATA	16 23 27 29 31 42 48 62 77
	SMBUS_PCH_0_CLK	SMB_50S	SMB	SML_PCH_0_CLK	16 48
	SMBUS_PCH_0_DATA	SMB_50S	SMB	SML_PCH_0_DATA	16 48
	SMBUS_PCH_1_CLK	SMB_50S	SMB	SML_PCH_1_CLK	16 48
	SMBUS_PCH_1_DATA	SMB_50S	SMB	SML_PCH_1_DATA	16 48
	HDA_BIT_CLK	HDA_50S	HDA	HDA_BIT_CLK	16 57
	HDA_BIT_CLK_R	HDA_50S	HDA	HDA_BIT_CLK_R	16
	HDA_SYNC	HDA_50S	HDA	HDA_SYNC	16 57
	HDA_SYNC_R	HDA_50S	HDA	HDA_SYNC_R	16
	HDA_RST_L	HDA_50S	HDA	HDA_RST_R_L	16
	HDA_RST_L	HDA_50S	HDA	HDA_RST_L	16 57
	HDA_SDIO0	HDA_50S	HDA	HDA_SDIO0	16 57
	HDA_SDIO1	HDA_50S	HDA	AUD_SDI_R	57
	HDA_SDOOUT	HDA_50S	HDA	HDA_SDOOUT	16 57
	HDA_SDOOUT_R	HDA_50S	HDA	HDA_SDOOUT_R	16
	PM_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK	
	SPT_CLK	SPT_55S	SPT	SPT_CLK_R	16 47
	SPT_CLK	SPT_55S	SPT	SPT_CLK	47
	SPT_MOST	SPT_55S	SPT	SPT_MOST_R	16 47
	SPT_MOST	SPT_55S	SPT	SPT_MOST	47
	SPT_MISO	SPT_55S	SPT	SPT_MISO	16 47
	SPT_CS0	SPT_55S	SPT	SPT_CS0_R_L	16 47
	SPT_CS0	SPT_55S	SPT	SPT_CS0_L	47
	PCIE_R50	PCIE	PCIE	PCIE_ENET_R2D_P	37
	PCIE_R50	PCIE	PCIE	PCIE_ENET_R2D_N	37
	PCIE_ENET_R2D	PCIE_R50	PCIE	PCIE_ENET_R2D_C_P	16 37
	PCIE_ENET_R2D	PCIE_R50	PCIE	PCIE_ENET_R2D_C_N	16 37
	PCIE_ENET_D2R	PCIE_R50	PCIE	PCIE_ENET_D2R_P	16 37
	PCIE_ENET_D2R	PCIE_R50	PCIE	PCIE_ENET_D2R_N	16 37
	PCIE_ENET_D2R	PCIE_R50	PCIE	PCIE_ENET_D2R_C_P	37
	PCIE_ENET_D2R	PCIE_R50	PCIE	PCIE_ENET_D2R_C_N	37
	PCIE_R50	PCIE	PCIE	PCIE_AP_R2D_P	6 32
	PCIE_R50	PCIE	PCIE	PCIE_AP_R2D_N	6 32
	PCIE_AP_R2D	PCIE_R50	PCIE	PCIE_AP_R2D_C_P	16 32
	PCIE_AP_R2D	PCIE_R50	PCIE	PCIE_AP_R2D_C_N	16 32
	PCIE_AP_D2R	PCIE_R50	PCIE	PCIE_AP_D2R_P	16 32
	PCIE_AP_D2R	PCIE_R50	PCIE	PCIE_AP_D2R_N	16 32
	PCIE_R50	PCIE	PCIE	PCIE_FW_R2D_P	39
	PCIE_R50	PCIE	PCIE	PCIE_FW_R2D_N	39
	PCIE_FW_R2D	PCIE_R50	PCIE	PCIE_FW_R2D_C_P	16 39
	PCIE_FW_R2D	PCIE_R50	PCIE	PCIE_FW_R2D_C_N	16 39
	PCIE_FW_D2R	PCIE_R50	PCIE	PCIE_FW_D2R_P	16 39
	PCIE_FW_D2R	PCIE_R50	PCIE	PCIE_FW_D2R_N	16 39
	PCIE_FW_D2R	PCIE_R50	PCIE	PCIE_FW_D2R_C_P	39
	PCIE_FW_D2R	PCIE_R50	PCIE	PCIE_FW_D2R_C_N	39
	PCIE_AP_D2R	PCIE_R50	PCIE	PCIE_AP_D2R_PI_P	6 32
	PCIE_AP_D2R	PCIE_R50	PCIE	PCIE_AP_D2R_PI_N	6 32
	PCIE_AP_R2D	PCIE_R50	PCIE	PCIE_AP_R2D_PI_P	32
	PCIE_AP_R2D	PCIE_R50	PCIE	PCIE_AP_R2D_PI_N	32
	CLK_PCIE_90D	CLK_PCIE	CLK_PCIE	NC_PEG_CLK100MP	8 16
	CLK_PCIE_90D	CLK_PCIE	CLK_PCIE	NC_PEG_CLK100MN	8 16
	PCIE_CLK100M_ENET	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_P	16 37
	PCIE_CLK100M_ENET	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_N	16 37
	MCP_P01_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_P	16 32
	MCP_P01_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_N	16 32
	MCP_P02_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_P	16 39
	MCP_P02_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_N	16 39
	CLK_PCIE_90D	CLK_PCIE	CLK_PCIE	NC_PCIE_CLK100M_EXCARDP	8 16
	CLK_PCIE_90D	CLK_PCIE	CLK_PCIE	NC_PCIE_CLK100M_EXCARDN	8 16
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<1>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<2>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<5>	6
	CEU_27P4S	CEU_COMP		TP_PCH_VSS_NCTF<7>	
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<9>	6 81
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<9>	6 81
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<11>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<12>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<15>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<17>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<19>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<21>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<22>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<25>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<27>	6
	CEU_27P4S	CEU_COMP		PCH_VSS_NCTF<29>	6

Chipset Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
1609	DP_EXTA_ML	DP_BSD	DISPLAYPORT	DP_EXTA_ML_C P<3...0>	8 75
1620	DP_EXTA_ML	DP_BSD	DISPLAYPORT	DP_EXTA_ML_C N<3...0>	8 75
1620	DP_EXTA_ML	DP_BSD	DISPLAYPORT	DP_EXTA_ML_P<3...0>	75
1620	DP_EXTA_ML	DP_BSD	DISPLAYPORT	DP_EXTA_ML_N<3...0>	75
1620	DP_EXTA_AUXCH	DP_BSD	DISPLAYPORT	DP_EXTA_AUXCH_C P	8 17 75
1620	DP_EXTA_AUXCH	DP_BSD	DISPLAYPORT	DP_EXTA_AUXCH_C N	8 17 75
1620	DP_EXTA_AUXCH	DP_BSD	DISPLAYPORT	DP_EXTA_AUXCH_P	75
1620	DP_EXTA_AUXCH	DP_BSD	DISPLAYPORT	DP_EXTA_AUXCH_N	75
1620	DP_INT_ML	DP_BSD	DISPLAYPORT	DP_INT_ML_C P<3...0>	
1620	DP_INT_ML	DP_BSD	DISPLAYPORT	DP_INT_ML_C N<3...0>	
1620	DP_INT_AUXCH	DP_BSD	DISPLAYPORT	DP_INT_AUXCH_C P	
1620	DP_INT_AUXCH	DP_BSD	DISPLAYPORT	DP_INT_AUXCH_C N	
1620	PCIE_T29_R2D	PCIE_BSD	PCIE	PCIE_T29_R2D_C P<3...0>	8 34
1620	PCIE_T29_R2D	PCIE_BSD	PCIE	PCIE_T29_R2D_C N<3...0>	8 34
1620	PCIE_T29_R2D	PCIE_BSD	PCIE	PCIE_T29_R2D_P<3...0>	34
1620	PCIE_T29_R2D	PCIE_BSD	PCIE	PCIE_T29_R2D_N<3...0>	34
1620	PCIE_T29_D2R	PCIE_BSD	PCIE	PCIE_T29_D2R_P<3...0>	8 34
1620	PCIE_T29_D2R	PCIE_BSD	PCIE	PCIE_T29_D2R_N<3...0>	8 34
1620	PCIE_T29_D2R	PCIE_BSD	PCIE	PCIE_T29_D2R_C P<3...0>	34
1620	PCIE_T29_D2R	PCIE_BSD	PCIE	PCIE_T29_D2R_C N<3...0>	34
1620	PCIE_CLK100M_T29	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_T29_P	16 34
1620	PCIE_CLK100M_T29	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_T29_N	16 34

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
[22B]	SYSLCK_CLK32K_RTC	CLK 510M 55S	CLK 510M	SYSLCK_CLK32K_RTC	16 26
[42B]	SYSLCK_CLK25M_SB	CLK 25M 55S	CLK 25M	SYSLCK_CLK25M_SB	16 26
[42B]		CLK 25M 55S	CLK 25M	SYSLCK_CLK25M_SB R	16
[22B]		CLK 25M 55S	CLK 25M	SYSLCK_CLK25M_ENET	26 37
[42B]		CLK 25M 55S	CLK 25M	SYSLCK_CLK25M_ENET R	
[22B]	SYSLCK_CLK25M_T29	CLK 25M 55S	CLK 25M	SYSLCK_CLK25M_T29	26 34
[22B]		CLK 25M 55S	CLK 25M	SYSLCK_CLK25M_T29 R	34

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

T29 I2C Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_I2C	*	=2x_DIELECTRIC	?

T29 SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	=2x_DIELECTRIC	?

T29/DP Connector Signal Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29DP	*	=5x_DIELECTRIC	?	T29DP	TOP, BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

T29 IC Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
REGD		DP_85D	DTSP1AYPORT	DP T29SNK0 ML C P<3..0>	8 34
REGD		DP_85D	DTSP1AYPORT	DP T29SNK0 ML C N<3..0>	8 34
REGD	DP T29SNK0_MI	DP_85D	DTSP1AYPORT	DP T29SNK0 ML P<3..0>	34
REGD	DP T29SNK0_MI	DP_85D	DTSP1AYPORT	DP T29SNK0 ML N<3..0>	34
REGD		DP_85D	DTSP1AYPORT	DP T29SNK0 AUXCH C P	8 17 34
REGD		DP_85D	DTSP1AYPORT	DP T29SNK0 AUXCH C N	8 17 34
REGD	DP T29SNK0_AUXCH	DP_85D	DTSP1AYPORT	DP T29SNK0 AUXCH P	34
REGD	DP T29SNK0_AUXCH	DP_85D	DTSP1AYPORT	DP T29SNK0 AUXCH N	34
REGD		DP_85D	DTSP1AYPORT	DP T29SNK1 ML C P<3..0>	8 34
REGD		DP_85D	DTSP1AYPORT	DP T29SNK1 ML C N<3..0>	8 34
REGD	DP T29SNK1_MI	DP_85D	DTSP1AYPORT	DP T29SNK1 ML P<3..0>	34
REGD	DP T29SNK1_MI	DP_85D	DTSP1AYPORT	DP T29SNK1 ML N<3..0>	34
REGD		DP_85D	DTSP1AYPORT	DP T29SNK1 AUXCH C P	8 17 34
REGD		DP_85D	DTSP1AYPORT	DP T29SNK1 AUXCH C N	8 17 34
REGD	DP T29SNK1_AUXCH	DP_85D	DTSP1AYPORT	DP T29SNK1 AUXCH P	34
REGD	DP T29SNK1_AUXCH	DP_85D	DTSP1AYPORT	DP T29SNK1 AUXCH N	34
REGD		DP_85D	DTSP1AYPORT	DP T29SRC ML C P<3..0>	
REGD		DP_85D	DTSP1AYPORT	DP T29SRC ML C N<3..0>	
REGD		DP_85D	DTSP1AYPORT	DP T29SRC AUXCH C P	
REGD		DP_85D	DTSP1AYPORT	DP T29SRC AUXCH C N	
REGD		T29_T2C_55S	T29_T2C	I2C T29_SCL	34 48 75
REGD		T29_T2C_55S	T29_T2C	I2C T29_SDA	34 48 75
REGD	T29_SPT_CLK	T29_SPT_55S	T29_SPT	T29_SPI_CLK	34
REGD	T29_SPT_M0ST	T29_SPT_55S	T29_SPT	T29_SPI_M0SI	34
REGD	T29_SPT_MISO	T29_SPT_55S	T29_SPT	T29_SPI_MISO	34
REGD	T29_SPT_CS_L	T29_SPT_55S	T29_SPT	T29_SPI_CS_L	34
REGD		T29DP_80D	T29DP	T29_R2D C P<3..0>	8 34 75
REGD		T29DP_80D	T29DP	T29_R2D C N<3..0>	8 34 75
REGD		T29DP_100D	T29DP	T29_D2R P<3..0>	8 34 75
REGD		T29DP_100D	T29DP	T29_D2R N<3..0>	8 34 75

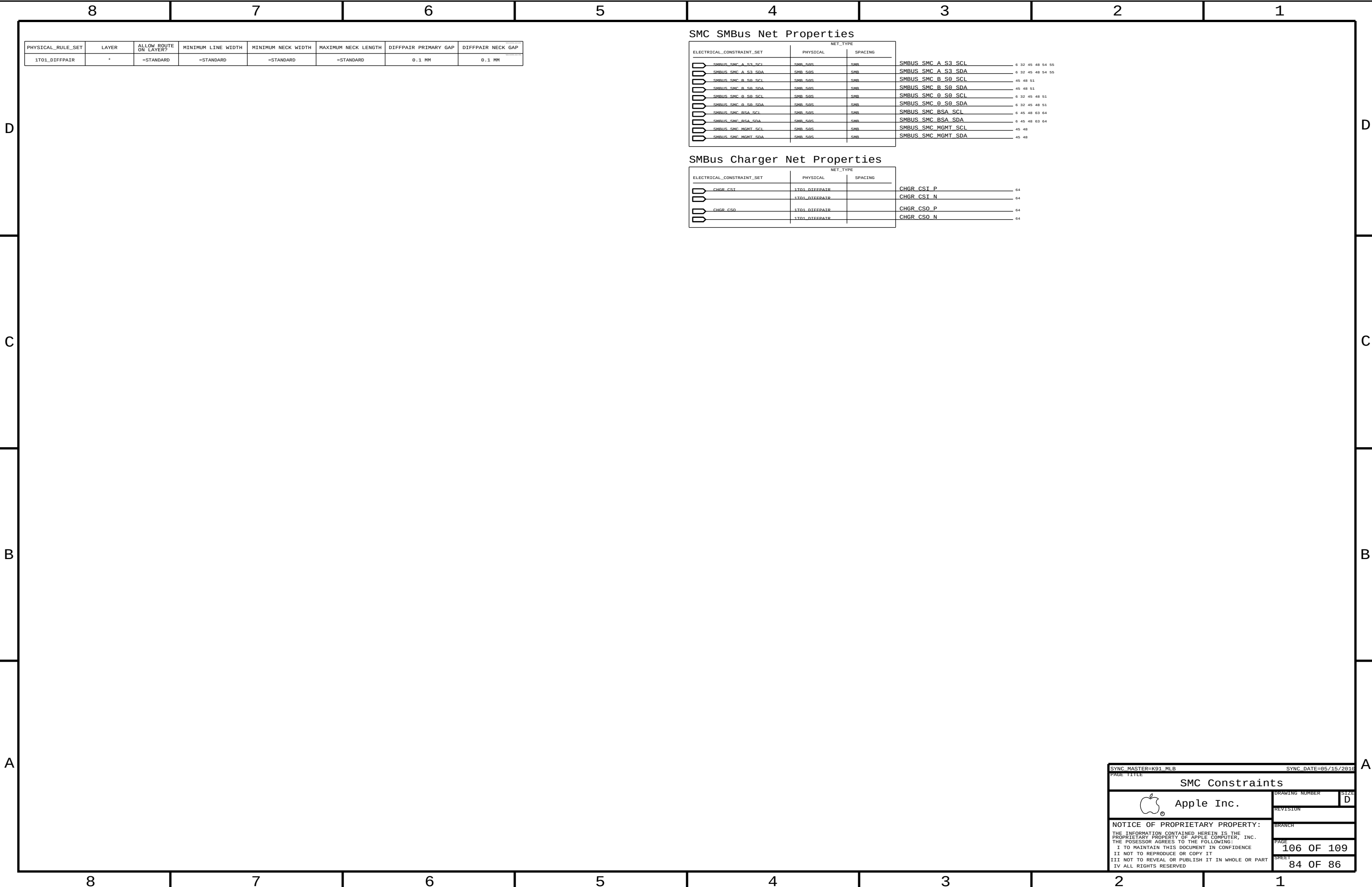
Only used on hosts supporting T29 video-in

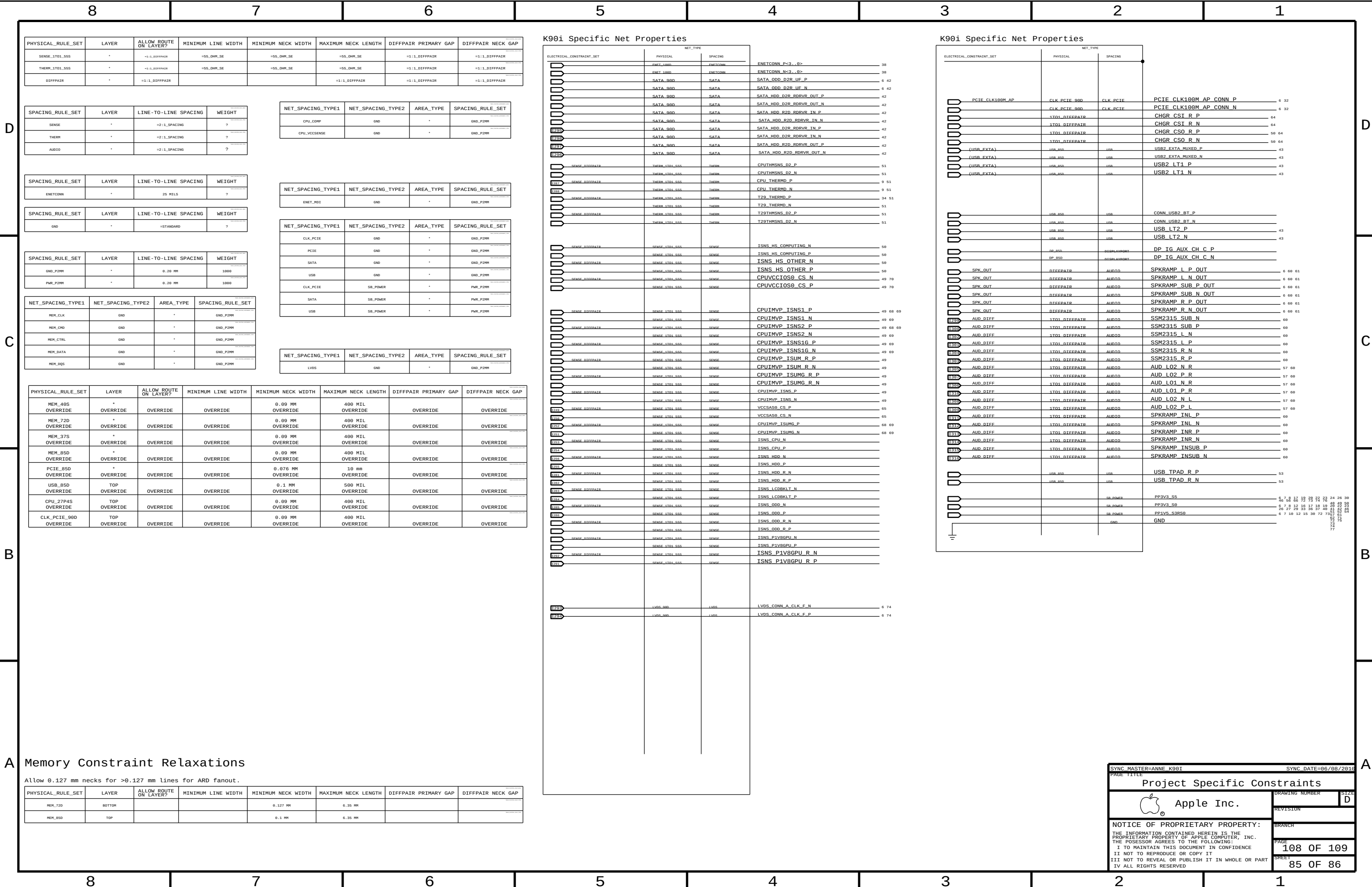
T29/DP Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
EE00	T29_R2D0	T29DP_80D	T29DP	T29_R2D P<0>
EE01	T29_R2D0	T29DP_80D	T29DP	T29_R2D N<0>
EE02	T29_R2D1	T29DP_80D	T29DP	T29_R2D P<1>
EE03	T29_R2D1	T29DP_80D	T29DP	T29_R2D N<1>
EE04		T29DP_80D	T29DP	T29_R2D C F P<1..0>
EE05		T29DP_80D	T29DP	T29_R2D C F N<1..0>
EE06	T29_D2R0	T29DP_100D	T29DP	T29_D2R C P<0>
EE07	T29_D2R0	T29DP_100D	T29DP	T29_D2R C N<0>
EE08	T29_D2R1	T29DP_100D	T29DP	T29_D2R C P<1>
EE09	T29_D2R1	T29DP_100D	T29DP	T29_D2R C N<1>
EE10		T29DP_100D	T29DP	T29DPA_D2R1_AUXCH_P
EE11		T29DP_100D	T29DP	T29DPA_D2R1_AUXCH_N
EE12		T29DP_80D	T29DP	DP_SDRVA_ML_C_P<3..0>
EE13		T29DP_80D	T29DP	DP_SDRVA_ML_C_N<3..0>
EE14		T29DP_80D	T29DP	DP_SDRVA_ML_R_P<3..0>
EE15		T29DP_80D	T29DP	DP_SDRVA_ML_R_N<3..0>
EE16	DP_SDRVA_ML_EVEN	T29DP_80D	T29DP	DP_SDRVA_ML_P<2..0:2>
EE17	DP_SDRVA_ML_EVEN	T29DP_80D	T29DP	DP_SDRVA_ML_N<2..0:2>
EE18	DP_SDRVA_ML_ODD	T29DP_80D	T29DP	DP_SDRVA_ML_P<3..1:2>
EE19	DP_SDRVA_ML_ODD	T29DP_80D	T29DP	DP_SDRVA_ML_N<3..1:2>
EE20	DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_P
EE21	DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_N
EE22		T29DP_80D	T29DP	DP_SDRVA_AUXCH_C_P
EE23		T29DP_80D	T29DP	DP_SDRVA_AUXCH_C_N
EE24		T29DP_80D	T29DP	T29DPA_ML_P<3..0>
EE25		T29DP_80D	T29DP	T29DPA_ML_N<3..0>
EE26		T29DP_80D	T29DP	T29DPA_ML_C_P<3..0>
EE27		T29DP_80D	T29DP	T29DPA_ML_C_N<3..0>
EE28		T29DP_80D	T29DP	DP_A_EXT_AUXCH_P
EE29		T29DP_80D	T29DP	DP_A_EXT_AUXCH_N
EE30	T29_R2D2	T29DP_80D	T29DP	T29_R2D P<2>
EE31	T29_R2D2	T29DP_80D	T29DP	T29_R2D N<2>
EE32	T29_R2D3	T29DP_80D	T29DP	T29_R2D P<3>
EE33	T29_R2D3	T29DP_80D	T29DP	T29_R2D N<3>
EE34		T29DP_80D	T29DP	T29_R2D C F P<3..2>
EE35		T29DP_80D	T29DP	T29_R2D C F N<3..2>
EE36	T29_D2R2	T29DP_100D	T29DP	T29_D2R C P<2>
EE37	T29_D2R2	T29DP_100D	T29DP	T29_D2R C N<2>
EE38	T29_D2R3	T29DP_100D	T29DP	T29_D2R C P<3>
EE39	T29_D2R3	T29DP_100D	T29DP	T29_D2R C N<3>
EE40		T29DP_100D	T29DP	T29DPB_D2R3_AUXCH_P
EE41		T29DP_100D	T29DP	T29DPB_D2R3_AUXCH_N
EE42		T29DP_80D	T29DP	DP_SDRVB_ML_C_P<3..0>
EE43		T29DP_80D	T29DP	DP_SDRVB_ML_C_N<3..0>
EE44		T29DP_80D	T29DP	DP_SDRVB_ML_R_P<3..0>
EE45		T29DP_80D	T29DP	DP_SDRVB_ML_R_N<3..0>
EE46	DP_SDRVB_ML_EVEN	T29DP_80D	T29DP	DP_SDRVB_ML_P<2..0:2>
EE47	DP_SDRVB_ML_EVEN	T29DP_80D	T29DP	DP_SDRVB_ML_N<2..0:2>
EE48	DP_SDRVB_ML_ODD	T29DP_80D	T29DP	DP_SDRVB_ML_P<3..1:2>
EE49	DP_SDRVB_ML_ODD	T29DP_80D	T29DP	DP_SDRVB_ML_N<3..1:2>
EE50	DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_P
EE51	DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_N
EE52		T29DP_80D	T29DP	DP_SDRVB_AUXCH_C_P
EE53		T29DP_80D	T29DP	DP_SDRVB_AUXCH_C_N
EE54		T29DP_80D	T29DP	T29DPB_ML_P<3..0>
EE55		T29DP_80D	T29DP	T29DPB_ML_N<3..0>
EE56		T29DP_80D	T29DP	T29DPB_ML_C_P<3..0>
EE57		T29DP_80D	T29DP	T29DPB_ML_C_N<3..0>
EE58		T29DP_80D	T29DP	DP_B_EXT_AUXCH_P
EE59		T29DP_80D	T29DP	DP_B_EXT_AUXCH_N

Only used on dual-port hosts.

SYNC MASTER=Master		SYNC DATE=06/21/2010	
PAGE TITLE			
T29 Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE D
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8		7		6		5		4		3		2		1	
K90i Board-Specific Spacing & Physical Constraints															
BOARD LAYERS								BOARD AREAS				BOARD UNITS (MIL or MM)		ALLEGRO VERSION	
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM								NO_TYPE, BGA				MM		15.5.1	
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
DEFAULT		*	Y	=50_OHM_SE		=50_OHM_SE		10 MM		0 MM		0 MM			
STANDARD		*	Y	=DEFAULT		=DEFAULT		10 MM		=DEFAULT		=DEFAULT			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
50_OHM_SE		TOP, BOTTOM	Y	0.110 MM		0.090 MM									
50_OHM_SE		*	Y	0.080 MM		0.080 MM		=STANDARD		=STANDARD		=STANDARD			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
40_OHM_SE		TOP, BOTTOM	Y	0.105 MM		0.105 MM									
40_OHM_SE		ISL10	N	0.120 MM		0.120 MM		=STANDARD		=STANDARD		=STANDARD			
40_OHM_SE		ISL3, ISL4, ISL9	Y	0.120 MM		0.120 MM		=STANDARD		=STANDARD		=STANDARD			
40_OHM_SE		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
37_OHM_SE		TOP, BOTTOM	Y	0.100 MM		0.1 MM									
37_OHM_SE		ISL10	N	0.145 MM		0.1 MM		=STANDARD		=STANDARD		=STANDARD			
37_OHM_SE		ISL3, ISL4, ISL9	Y	0.145 MM		0.1 MM		=STANDARD		=STANDARD		=STANDARD			
37_OHM_SE		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
27P4_OHM_SE		TOP, BOTTOM	Y	0.310 MM		0.2 MM									
27P4_OHM_SE		*	Y	0.235 MM		0.2 MM		=STANDARD		=STANDARD		=STANDARD			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
55_OHM_SE		TOP, BOTTOM	Y	0.090 MM		0.090 MM									
55_OHM_SE		*	Y	0.070 MM		0.070 MM		=STANDARD		=STANDARD		=STANDARD			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
72_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
72_OHM_DIFF		ISL3, ISL4, ISL9	Y	0.140 MM		0.140 MM				0.190 MM		0.190 MM			
72_OHM_DIFF		ISL10	N	0.140MM		0.140 MM				0.190 MM		0.190 MM			
72_OHM_DIFF		TOP, BOTTOM	Y	0.175 MM		0.175 MM				0.200 MM		0.200 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
85_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
85_OHM_DIFF		ISL3, ISL4	Y	0.101 MM		0.1 MM				0.170 MM		0.170 MM			
85_OHM_DIFF		ISL9, ISL10	Y	0.101 MM		0.1 MM				0.170 MM		0.170 MM			
85_OHM_DIFF		TOP, BOTTOM	Y	0.125 MM		0.1 MM				0.190 MM		0.190 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
90_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
90_OHM_DIFF		ISL3, ISL4	Y	0.091 MM		0.091 MM				0.180 MM		0.180 MM			
90_OHM_DIFF		ISL9, ISL10	Y	0.091 MM		0.091 MM				0.180 MM		0.180 MM			
90_OHM_DIFF		TOP, BOTTOM	Y	0.111 MM		0.111 MM				0.200 MM		0.200 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
100_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
100_OHM_DIFF		ISL3, ISL4	Y	0.076 MM		0.076 MM				0.250 MM		0.250 MM			
100_OHM_DIFF		ISL9, ISL10	Y	0.076 MM		0.076 MM				0.250 MM		0.250 MM			
100_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.200 MM		0.200 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
110_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
110_OHM_DIFF		ISL3, ISL4	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		ISL9, ISL10	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.250 MM		0.250 MM			
NOTE: These are Intel recommended impedances for PEG, unused on K90i.															
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
40_OHM_SE		TOP, BOTTOM	Y	0.105 MM		0.105 MM									
40_OHM_SE		*	Y	0.090 MM		0.090 MM		=STANDARD		=STANDARD		=STANDARD			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
80_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
80_OHM_DIFF		ISL3, ISL4	Y	0.115 MM		0.115 MM				0.180 MM		0.180 MM			
80_OHM_DIFF		ISL9, ISL10	Y	0.115 MM		0.115 MM				0.180 MM		0.180 MM			
80_OHM_DIFF		TOP, BOTTOM	Y	0.140 MM		0.140 MM				0.190 MM		0.190 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
100_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
100_OHM_DIFF		ISL3, ISL4	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
100_OHM_DIFF		ISL9, ISL10	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
100_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.200 MM		0.200 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
110_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
110_OHM_DIFF		ISL3, ISL4	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		ISL9, ISL10	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.250 MM		0.250 MM			
NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers.															
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
85_DIFF_BGA		*	=85_OHM_DIFF	=85_OHM_DIFF		=85_OHM_DIFF		=85_OHM_DIFF		=85_OHM_DIFF		=85_OHM_DIFF			
85_DIFF_BGA		ISL3, ISL4	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
85_DIFF_BGA		ISL9, ISL10	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
NOTE: 85_DIFF_BGA is 85-ohms differential impedance on outer layers and 80-ohms on inner layers.															
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
90_DIFF_BGA		*	=90_OHM_DIFF	=90_OHM_DIFF		=90_OHM_DIFF		=90_OHM_DIFF		=90_OHM_DIFF		=90_OHM_DIFF			
90_DIFF_BGA		ISL3, ISL4	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
90_DIFF_BGA		ISL9, ISL10	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
NOTE: 90_DIFF_BGA is 90-ohms differential impedance on outer layers and 85-ohms on inner layers.															
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
100_DIFF_BGA		*	=100_OHM_DIFF	=100_OHM_DIFF		=100_OHM_DIFF		=100_OHM_DIFF		=100_OHM_DIFF		=100_OHM_DIFF			
100_DIFF_BGA		ISL3, ISL4	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
100_DIFF_BGA		ISL9, ISL10	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.															
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
100_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
100_OHM_DIFF		ISL3, ISL4	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
100_OHM_DIFF		ISL9, ISL10	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
100_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.200 MM		0.200 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
110_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
110_OHM_DIFF		ISL3, ISL4	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		ISL9, ISL10	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.250 MM		0.250 MM			
NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers.															
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
40_OHM_SE		TOP, BOTTOM	Y	0.105 MM		0.105 MM									
40_OHM_SE		*	Y	0.090 MM		0.090 MM		=STANDARD		=STANDARD		=STANDARD			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
80_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
80_OHM_DIFF		ISL3, ISL4	Y	0.115 MM		0.115 MM				0.180 MM		0.180 MM			
80_OHM_DIFF		ISL9, ISL10	Y	0.115 MM		0.115 MM				0.180 MM		0.180 MM			
80_OHM_DIFF		TOP, BOTTOM	Y	0.140 MM		0.140 MM				0.190 MM		0.190 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
100_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
100_OHM_DIFF		ISL3, ISL4	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
100_OHM_DIFF		ISL9, ISL10	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM			
100_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.200 MM		0.200 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
110_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
110_OHM_DIFF		ISL3, ISL4	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		ISL9, ISL10	Y	0.070 MM		0.070 MM				0.330 MM		0.330 MM			
110_OHM_DIFF		TOP, BOTTOM	Y	0.085 MM		0.085 MM				0.250 MM		0.250 MM			
NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers.															
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
85_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
85_OHM_DIFF		ISL3, ISL4	Y	0.101 MM		0.1 MM				0.170 MM		0.170 MM			
85_OHM_DIFF		ISL9, ISL10	Y	0.101 MM		0.1 MM				0.170 MM		0.170 MM			
85_OHM_DIFF		TOP, BOTTOM	Y	0.125 MM		0.1 MM				0.190 MM		0.190 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
90_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
90_OHM_DIFF		ISL3, ISL4	Y	0.091 MM		0.091 MM				0.180 MM		0.180 MM			
90_OHM_DIFF		ISL9, ISL10	Y	0.091 MM		0.091 MM				0.180 MM		0.180 MM			
90_OHM_DIFF		TOP, BOTTOM	Y	0.111 MM		0.111 MM				0.200 MM		0.200 MM			
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP			
100_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD			
100_OHM_DIFF		ISL3, ISL4	Y	0.076 MM		0.076 MM				0.250 MM					